

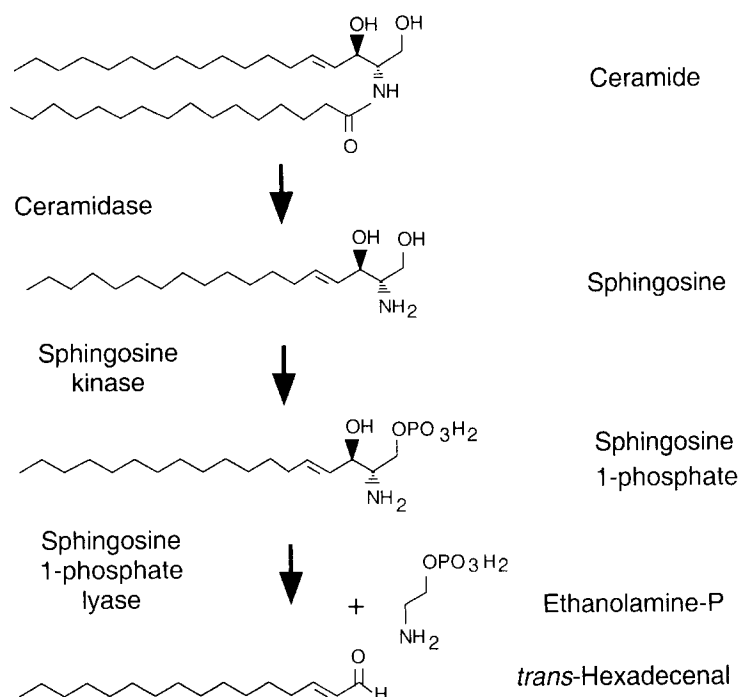


Fig. 15. Catabolism of ceramide and sphingosine [24].

ceramidases with neutral and alkaline pH optima have been found in various cell compartments (including mitochondria) and appear to be involved in signaling.

4.4. Sphingosine

Sphingosine undergoes reacylation by ceramide synthase (described in Section 3.2.2) or phosphorylation by sphingosine kinase. This family of ATP-dependent kinases is both cytosolic and membrane associated, and phosphorylates a wide range of sphingoid bases, although unnatural stereoisomers (e.g., *L-threo*-sphinganine) are inhibitors. The product can be dephosphorylated by specific sphingosine 1-phosphate phosphatases as well as more general lipid phosphatases, or cleaved to ethanolamine phosphate and *trans*-2-hexadecenal (Fig. 15) by a pyridoxal 5'-phosphate-dependent lyase (J. Zhou, 1998; P.P. Van Veldhoven, 2000). As shown first in the 1970s by W. Stoffel and co-workers, the phosphoethanolamine can be utilized for the synthesis of phosphatidylethanolamine, and *trans*-2-hexadecenal reduced to the alcohol and incorporated into alkyl ether lipids (Chapters 8 and 9). Under certain conditions, degradation of sphingoid bases can account for as much as one third of the phosphoethanolamine in phosphatidylethanolamine (E. Smith, 1995). In addition to being an intermediate in the degradation of sphingoid bases, sphingosine 1-phosphate is packaged in some cells (e.g., platelets) for secretion as discussed in Section 6.2.3.

5. Regulation of sphingolipid metabolism

Sphingolipid metabolism is regulated at a number of levels: (1) the control of ceramide biosynthesis *de novo* and by the recycling of existing sphingolipids; (2) the partitioning of ceramide toward the major classes of sphingolipids (e.g., sphingomyelin versus GlcCer and GalCer); (3) the partitioning of intermediates to determine the complex glycolipid profiles of different cell types; (4) the trafficking of sphingolipids to the appropriate cellular membranes as well as to specialized regions of a given membrane; (5) the secretion (and shedding) of some categories of sphingolipids; (6) the internalization of sphingolipids during endocytosis and other membrane functions; (7) the turnover of sphingolipids for cell signaling; and (8) degradation.

These events vary in different cell types, and in a given cell type at different stages of development and in response to varying environmental conditions [29]. These issues are obviously too complex to deal with fully in this chapter; therefore, we present only a few examples.

5.1. *De novo sphingolipid biosynthesis versus turnover in generating bioactive (signaling) metabolites*

The traditional ‘signaling’ paradigm for sphingolipids is that agonists trigger the turnover of sphingomyelin to ceramide, sphingosine and sphingosine 1-phosphate (Fig. 2); nonetheless, the *de novo* biosynthetic pathway also produces bioactive products such as sphinganine, ceramide, and sometimes sphinganine 1-phosphate. That these intermediates of *de novo* synthesis also affect cell behavior was first shown by the role of sphinganine in the toxicity of fumonisins (E. Wang, 1991) and the involvement of *de novo* synthesized ceramide in daunorubicin-induced apoptosis (R. Bose, 1995). There are now a large number of natural agonists, drugs, toxins and toxicants, and even intermediates of common metabolic pathways (such as palmitoyl-CoA) that can alter cell behavior at least in part by affecting sphingolipid biosynthesis, as illustrated in Fig. 16 (Y.A. Hannun, 2001; S.C. Linn, 2001).

5.2. *Complex sphingolipid formation in tissue development*

Glycosphingolipids undergo quantitative and qualitative changes with development [2,28], such as the stage-specific expression of the antigen SSEA-1, which appears at the 8-cell stage, is maximally expressed at the morulae stage, and disappears at the blastocyte stage. This antigenic determinant is actually present on several glycosphingolipids, all of which have a terminal Gal β 1–4[Fuc α 1–3]GlcNAc β 1–3Gal structure, as well as on cell surface glycoproteins. Globoside and Forssman antigen (IV³- α -GalNAc-Gb₄Cer) also appear at the morulae and blastocyte stages, respectively. It is likely that some of these changes contribute to cell–cell and cell–matrix interactions that affect the migration of cells to target locations within the developing embryo, and cell differentiation when the correct location(s) are found.

The requirement for glycosphingolipids in development has been established by targeted disruption of the gene encoding glucosylceramide synthase (T. Yamashita,

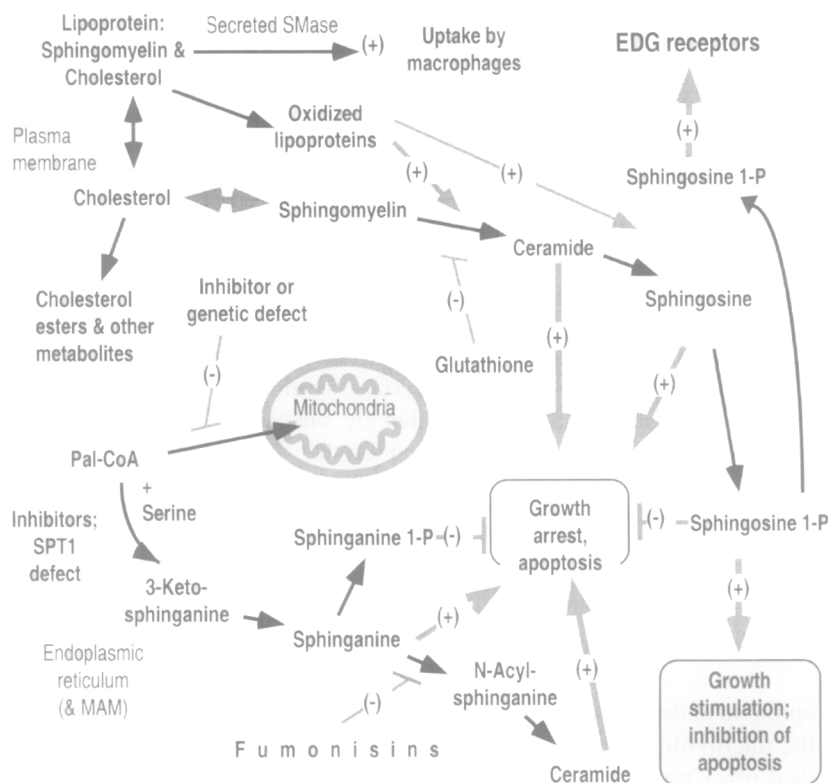


Fig. 16. Some of the factors that can modulate sphingolipid biosynthesis and turnover to impact cell behavior [23–25]. Agonist-induced changes in sphingolipid turnover were illustrated in Fig. 2; the additional pathways shown here are as follows (proceeding counter-clockwise from top left): (1) hydrolysis of lipoprotein sphingomyelin by secreted sphingomyelinase (SMase) to form particles that are taken up by macrophages; (2) induction of sphingomyelin turnover by oxidized lipoproteins to induce smooth muscle cell growth or death via sphingosine 1-phosphate versus ceramide, respectively; (3) associations between sphingomyelin and cholesterol that affect lipoprotein and membrane structure (as well as the efflux and metabolism of cholesterol, not shown); (4) perturbation of sphingolipid biosynthesis by factors that alter the amounts of the precursors (e.g., serine and palmitoyl-CoA) (Pal-CoA) including inhibitors and genetic defects of mitochondrial uptake of Pal-CoA; (5) inhibition of, or genetic defects in, serine palmitoyltransferase (SPT1); (6) inhibition of ceramide synthase (by fumonisins) to block complex sphingolipid formation and elevate sphinganine and sphinganine 1-phosphate; (7) induction of de novo sphingolipid biosynthesis at serine palmitoyltransferase and/or ceramide synthase by other stresses (irradiation, UV, cytokines, and heat shock, inter alia) (not shown); (8) the triggering of sphingomyelin hydrolysis by depletion of cytosolic glutathione, an inhibitor of neutral sphingomyelinase(s). More information about these factors is presented in the text.

1999). In the absence of glycosphingolipid synthesis, embryogenesis proceeded well into gastrulation with differentiation into primitive germ layers and patterning of the embryo, but was abruptly halted by a major apoptotic process. Embryonic stem cells deficient in glycosphingolipid synthesis were able to differentiate into endodermal, mesodermal, and ectodermal derivatives but were strikingly deficient in their ability to form well differentiated tissues.

5.3. Neural development and function

The ganglioside content of neuronal and glial cells changes quantitatively and qualitatively with development and aging (R.K. Yu, 1994) and it has been assumed that gangliosides help neuronal cells follow the appropriate path in development of neuronal networks, mediate cell–cell communication, and help regulate receptors and ion channels on neural cells, presumably by interactions with specific binding proteins (R.L. Schnaar, 1994).

These hypotheses have been confirmed by recent studies with mice lacking specific glycosyltransferases. Mice engineered to lack G_{M2}/G_{D2} synthase express only the simple gangliosides (G_{M3} and G_{D3}) and develop significant and progressive behavioral neuropathies (S. Chiavegatto, 2000). G_{M2}/G_{D2} synthase knockout mice lack a calcium regulatory mechanism that is modulated by one or more of the deleted gangliosides (G. Wu, 2001). Mice with disruption of the gene encoding G_{D3} synthase appear to undergo normal development and have a normal life span; however, when these mice are crossbred with mice carrying a disrupted gene encoding $\beta 1,4$ -N-acetylgalactosaminyltransferase, the double mutants display a sudden death phenotype and are extremely susceptible to induction of lethal seizures by sound stimulus.

5.4. Physiology (and pathophysiology) of the intestinal tract

The intestinal epithelia undergo constant renewal through cell proliferation in the crypts at the base of the microvilli, differentiation and migration up the villi, and death as the cells are sloughed off. Crypt cells have more ceramide and Gb_3 Cer, and less GlcCer and G_{M3} , than villus cells (J.-F. Bouhours, 1993), and differ also in their ceramide composition, with the crypt GlcCer having less hydroxy-fatty acids. All regions of the intestine contain relatively high levels of free ceramide (ca. 17% of the total sphingolipids). The glycosphingolipids of intestinal cells are part of the epithelial barrier, but are also used as attachment sites by microorganisms, viruses, and microbial toxins [23]; for example, *Candida albicans* binds to asialo- G_{M1} , HIV gp 120 to GalCer, cholera toxin to G_{M1} , and Shiga toxin and verotoxins to Gb_3 .

5.5. Oncogenic transformation, tumor antigens, and immunomodulation

Changes in the glycosphingolipid composition of cells have long been associated with oncogenic transformation [2]. There are multiple strategies that might utilize sphingolipids to prevent or treat neoplasia, such as (a) administration of sphingolipids or sphingolipid analogs via diet or other means (E.M. Schmelz, 2001), (b) modulation of sphingolipid metabolism to trigger apoptosis via endogenous ceramide (S. Selzner, 2001), (c) altering ceramide metabolism to overcome drug resistance (A. Senchenkov, 2001), (d) controlling the immunomodulatory effects of shed glycolipids (S. Ladisch, 1995), and (e) using abnormally expressed glycolipids as tumor markers and/or to target chemotherapeutic agents more effectively [2].

6. *Sphingolipids and signal transduction*

Findings over the last decade have provided evidence that sphingolipids are involved in essentially all aspects of cell regulation (many of which are summarized in Figs. 2 and 16). (1) Sphingolipids (particularly sphingomyelin in association with cholesterol) help define 'microdomains' of the membrane that influence receptor signaling as well as the functions of important transport proteins. (2) Complex sphingolipids serve as ligands for receptors on neighboring cells or the extracellular matrix, and mediate changes in cell behavior in response to a cell's environment. (3) Complex sphingolipids modulate the properties of receptors on the same cell, thereby controlling the responsiveness of the cell to external factors. (4) Sphingolipids are extensively involved in membrane trafficking and, therefore, influence receptor internalization, sorting and recycling, as well as the movement and fusion of secretory vesicles in response to stimuli. (5) Ceramide, sphingosine and sphingosine 1-phosphate (and possibly others) are produced from agonist- or stress-induced turnover of sphingomyelin (and de novo synthesis) to participate in cell signaling. (6) Sphingosine 1-phosphate is released from some cells to serve as an extracellular agonist for SIP receptors. Moreover, because diverse factors can affect sphingolipid metabolism (availability of precursor substrates, genetic mutations in related pathways, naturally occurring inhibitors, etc.), these pathways may play a central role in many aspects of cell physiology and pathophysiology (Fig. 16).

6.1. *Interactions between gangliosides and growth factor receptors*

Gangliosides are bimodal regulators of cell growth: both inhibitory (often as the complex species) and stimulatory (as catabolites). This is exemplified by ganglioside G_{M3} , which inhibits growth through extension of the G_1 phase of the cell cycle, and makes cells refractory to stimulation by epidermal growth factor [7]. Upon removal of the fatty acid, lyso- G_{M3} strongly inhibits cell proliferation whereas removal of the *N*-acetyl group from the sialic acid (producing de-*N*-acetyl- G_{M3}) enhances growth and epidermal growth factor receptor tyrosine kinase activity. The neutral glycolipid cores of G_{M3} , LacCer and GlcCer, also affect growth (K. Ogura, 1992).

6.2. *Hydrolysis to bioactive lipid backbones*

The finding that sphingosine is a potent inhibitor of protein kinase C (Y.A. Hannun, 1986) introduced the paradigm that the cellular functions of sphingolipids may reside not only in the complex species, but also in the lipid backbones. Soon thereafter, ceramide was found to be released from sphingomyelin in GH3 pituitary cells treated with diacylglycerol (R.N. Kolesnick, 1989) and HL-60 cells treated with $1\alpha,25$ -dihydroxyvitamin D_3 (which induces these cells to differentiate) (T. Okazaki, 1989). Furthermore, treatment of HL-60 cells with a short-chain ceramide, or exogenous sphingomyelinase, could mimic the effects of $1\alpha,25$ -dihydroxyvitamin D_3 , which linked the agonist-induced turnover of a sphingolipid to a metabolite that met many of the criteria for an intracellular mediator. A large number of studies have now explored how the lipid backbones of sphingolipids serve as second messengers, but caution must be exercised

in reading this literature because some of the effects of exogenous sphingolipids have not yet been linked to changes in endogenous mediators.

6.2.1. Ceramide

Sphingomyelin turnover to ceramide is now thought to mediate, at least in part, cellular responses to a wide spectrum of agents (cytokines, ionizing irradiation, corticosteroids, *inter alia*), many of which deal with some type of stress (R.N. Kolesnick, 1998; Y.A. Hannun, 2000; A. Huwiler, 2000). Both neutral and acidic (including a secreted acidic) sphingomyelinases are involved. The regulatory mechanisms for these enzymes are involved, and their mechanisms of activation and inhibition are still being uncovered. As an example, CD95-triggers translocation of acid sphingomyelinase to the plasma membrane outer surface, where it releases ceramide from sphingomyelin and enables the clustering of CD95 in sphingolipid-rich membrane rafts triggering apoptosis (H. Grassme, 2001).

At least five direct targets for ceramide have been identified: ceramide-activated protein phosphatases, which are both type 1 and 2A phosphoprotein phosphatases (C.E. Chalfant, 2000); ceramide-activated protein kinase, which has been identified as the kinase suppressor of ras (Y.H. Zhang, 1997; D.B. Polk, 2001); cathepsin D (M. Heinrich, 2000); protein kinase C zeta (N.A. Bourbon, 1998); and cytosolic phospholipase A₂ (A. Huwiler, 2001). In general, dihydroceramides (*N*-acylsphinganine) are ineffective in activating these targets, which makes them useful as controls. Elevations in cellular ceramide are usually associated with growth inhibition, differentiation and, in many cells, induction of apoptosis.

6.2.2. Sphingoid bases

A remarkable number of cellular systems are activated or inhibited by sphingoid bases, with the systems that are most likely to reflect direct activation being a 14-3-3 kinase (T. Megidish, 2000), and inhibition being protein kinase C (E.R. Smith, 2000), phosphatidic acid phosphohydrolase (D. Perry, 1992) and Akt kinase (H.C. Chang, 2001). There are also strong indications that sphingosine can affect ion transporters (C. Mathes, 1998; Y. Shin, 2000).

Relatively little is known about agonist-induced turnover of sphingolipids to sphingosine and the coupling of the sphingosine to intracellular responses. When added exogenously to cells, sphingoid bases are usually growth inhibitory and pro-apoptotic (V.L. Stevens, 1989; T. Shirahama, 1997). Agents affecting cellular levels of sphingosine include dexamethasone (R. Ricciolini, 1994), lipoproteins and phorbol esters (E. Wilson, 1988), platelet-derived growth factor (A. Olivera, 1993; E. Coroneos, 1995), Fas-induced apoptosis of type II Jurkat T cells (O. Cuvillier, 2000) and doxorubicin in MCF7 breast adenocarcinoma cells (O. Cuvillier, 2001). The *in vivo* correlates are few, but it has been recently reported that apoptosis in the skeletal muscle of rats with heart failure is associated with increased serum levels of tumor necrosis factor- α and sphingosine (L. Dalla Libera, 2001).

In addition, free sphingoid bases, sphingoid base 1-phosphates, and ceramides can be elevated by disruption of *de novo* sphingolipid biosynthesis by, for example, heat shock in yeast (R.C. Dickson, 1997; G.M. Jenkins, 1997) and fumonisins [4,15]. It should also

be borne in mind when studying cells in culture that many cell lines undergo a ‘burst’ of sphinganine and sphingosine formation from both increased de novo sphingolipid biosynthesis and turnover when ‘conditioned’ medium is removed (E.R. Smith, 1995). This may have a physiologic significance because cells produce a novel factor (named betrachamine) that suppresses the burst (L. Warden, 1999).

6.2.3. *Sphingosine 1-phosphate*

Platelet-derived growth factor induces rapid increases in cellular sphingosine and sphingosine 1-phosphate, and several lines of evidence indicate that sphingosine 1-phosphate is a mediator of growth stimulation by this growth factor (A. Olivera, 1993; Y. Su, 1994). The downstream responses are release of calcium from intracellular compartment(s) (S. Kim, 1995) as well as activation of AP1 transcription factor. Sphingosine 1-phosphate is also mitogenic when added to cells exogenously, and is thought to signal through a pertussis toxin-sensitive G protein (K.A. Goodemote, 1995) that activates the mitogen-activated protein kinase pathway (J. Wu, 1995). Progress in characterizing these receptors, members of the EDG family (now called S1P) receptors, has been rapid [10], and sphingosine 1-phosphate clearly qualifies as both an extracellular and an intracellular messenger.

Whereas ceramide and sphingosine are often growth inhibitory and pro-apoptotic, sphingosine 1-phosphate is mitogenic and anti-apoptotic, which has been likened to a rheostat that determines cell survival versus death pathways (S. Spiegel, 1999). Ceramides and sphingosine induce release of cytochrome *c* from mitochondria (an important event in apoptosis), and recent studies suggest that sphingosine 1-phosphate antagonizes apoptosis of human leukemia cells by inhibiting release of cytochrome *c* and Smac/DIABLO from mitochondria (O. Cuvillier, 2001). A major determinant of the type(s) of sphingolipids that are formed is whether the agonists activate only sphingomyelinase or a combination of sphingomyelinase and ceramidase with or without sphingosine kinase (as shown in Fig. 2) (A. Olivera, 1993; E. Coroneos, 1995; M.N. Nikolova-Karakashian, 1997; N. Augé, 1999).

6.2.3.1. Other bioactive lysosphingolipids. Considerable progress has been made in characterizing the occurrence and biological activities of other lysosphingolipids, but much work remains to elucidate their roles in cell regulation. For example, sphingosylphosphorylcholine (lysosphingomyelin) is a potent mitogen (T. Seufferlein, 1995), ceramide 1-phosphate is a potent calcium-mobilizing agent (S. Gijsbers, 1999), and psychosine (lyso-GlcCer or -GalCer), which has long been known to be highly cytotoxic, is the agonist for an orphan G protein-coupled receptor, T cell death-associated gene 8 (D.S. Im, 2001).

7. *Bioactive sphingolipids appear to be at the heart of numerous aspects of cell regulation in normal and pathologic conditions*

Fig. 16 illustrates interrelationships between sphingolipid metabolism and some of the factors that can alter the amounts of key intermediates and/or products to impact cell

behavior. Beginning with the more familiar signaling paradigm:

- agonists, stress, oxidized lipoproteins and depletion of glutathione can induce sphingomyelinase activation to produce ceramide; and if ceramidase is also activated, sphingosine and/or sphingosine 1-phosphate (when sphingosine kinase is also active) (Section 6.2, and Fig. 2);
- a secreted acid sphingomyelinase hydrolyzes sphingomyelin associated with lipoproteins to form particles that are taken up by macrophages and promote foam cell formation (S. Marathe, 1998);
- hydrolysis of cellular sphingomyelin alters the structure of rafts and other membrane microdomains, as well as 'frees' cholesterol for efflux (Chapter 20), metabolism and down-regulation of de novo cholesterol synthesis (Chapter 15) [5,6];
- a number of factors can perturb sphingolipid biosynthesis by altering the amounts of the precursors (e.g., serine and palmitoyl-CoA) including inhibitors and genetic defects of mitochondrial uptake of palmitoyl-CoA (H. Vesper, 1999);
- genetic defects in serine palmitoyltransferase can, in an as yet to be elucidated way, result in sensory neuropathy (J.L. Dawkins, 2001; K. Bejaoui, 2001);
- a host of factors up-regulate de novo sphingolipid biosynthesis by transcriptional or post-translational activation of serine palmitoyltransferase and/or activation of ceramide synthase, often resulting in cell death due to elevations in pro-apoptotic intermediates (Section 5.1) [25];
- inhibition of ceramide synthase (by fumonisins) blocks complex sphingolipid formation and elevates sphinganine (to induce toxicity) and sometimes sphinganine 1-phosphate (which may induce a mitogenic response), whereas other factors (such as irradiation) can increase ceramide synthase and induce cell death (Section 3).

8. *Future directions*

Knowledge about basic 'sphingolipidology' — the physical properties of sphingolipids, sphingolipid metabolizing enzymes and targets regulated by sphingolipids — has at least doubled since the 1996 edition of this book. New ideas have also emerged about how sphingolipids are involved in disease etiology, and how naturally occurring and/or synthetic sphingolipids may be useful in disease prevention and treatment. Like other areas of biologic research, 'sphingolipidology' is rapidly evolving into 'sphingolipidomics', that is, a field where a rigorous evaluation of any one part requires a comprehensive analysis also of essentially all of the other components. For example, the 'sphingomyelin ceramide signaling pathway' was initially envisioned to involve agonist- (or stress-) induced turnover of sphingomyelin to ceramide to activate (or inhibit) intracellular target(s). As it is now understood, sphingomyelin turnover affects the organization (and intracellular trafficking) of important membrane microdomains (rafts and caveolae) and their associated receptors, transporters, and other bioactive lipids as well as the production of a cascade of products (ceramide, sphingosine and sphingosine 1-phosphate, etc.), each of which can interact with multiple intracellular (and sometimes extracellular) targets. Very often, a stimulus triggers not only sphingolipid turnover but also changes in de novo synthesis, including the formation of additional categories

of bioactive sphingolipids (such as GlcCer). Hence, to understand completely how sphingolipids regulate a given biological process, one must analyze essentially all of the sphingolipids in (and around!) the cells. It is likely that many of the seeming contradictions in the current research literature occur because each study has evaluated only one or a few of these factors, which may be analogous to the parable of the blind men trying to describe an elephant.

New technologies such as electrospray tandem mass spectrometry, DNA and protein microarrays, etc., are making ‘sphingolipidomic’ analyses feasible. Nonetheless, additional methods are still needed to evaluate temporal, localized changes in subcellular compartments, and to be able to study integrated systems such as tissues, organs, and intact organisms.

With better understanding comes greater utility, and sphingolipids are impacting almost every translational research field, from bioengineering to nutrition. It is intriguing that Thudichum’s interests included nutrition, and a century before ‘nutraceuticals’ and ‘functional foods’ were in vogue, he wrote in the preface to his last book, *Cookery. Its Art and Practice* (1895): “Physiologic deduction proves that . . . no cookery is rational which does not attain the utmost theoretically possible effect, namely, the production of the highest physiological force. . . . It is believed and hoped that the medical profession will find in this work many materials to assist them. . . .”. He surely had sphingolipids in mind.

Abbreviations

Cer	ceramide
ESI-MS/MS	electrospray tandem mass spectrometry
Fuc	fucose
Gal	galactose
G, with subscript for the subclass	ganglioside
Gb, with subscript for the number of carbohydrates	globoside
Glc	glucose
GlcA	glucuronic acid
GM2-AP	GM ₂ activator protein
Hex A or B	hexosaminidase A or B
Lac	lactose
Man	mannose
GalNAc	N-acetylgalactosamine
GlcNAc	N-acetylglucosamine
Neu5Ac	N-acetylneuraminic acid
Neu5Gc	N-glycolylneuraminic acid
Pal-CoA	palmitoyl-CoA
Ser	serine
SPT	serine palmitoyltransferase
SAP	sphingolipid activator protein

SM	sphingomyelin
SMase	sphingomyelinase
TNF α	tumor necrosis factor- α
UDP-sugar	uridine dinucleotide phosphate sugar
UV	ultraviolet

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Cholesterol biosynthesis

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1. Introduction

Cholesterol's structure, biosynthetic pathway and metabolic regulation have tested the ingenuity of chemists, biochemists and cell biologists for over 100 years. The last century began with the pioneering work of Heinrich Wieland, who deduced the structure of cholesterol and bile acids, for which Wieland was awarded the Nobel Prize in Chemistry in 1926. How was such a complex molecule synthesized by the cell? Investigation into the cholesterol biosynthetic pathway required the development of isotopic tracer methods in Rudi Schoenheimer's lab in the 1930s. Using these novel techniques, Konrad Bloch and David Rittenberg showed that the ring structure and side chain of cholesterol were derived from acetate, and they identified intermediates in the pathway. Subsequent work by Bloch, John Cornforth and George Popjak succeeded in establishing the biosynthetic origin of all 27 carbons of cholesterol. For his elegant work, Bloch was awarded the Nobel Prize in Chemistry in 1964.

By the 1980s, the cholesterol biosynthetic pathway was understood to be a complex pathway of over 40 cytosolic and membrane-bound enzymes, which was subject to feedback regulation by the end-product, cholesterol, and oxygenated forms (called oxysterols). Genes encoding the key enzymes were cloned, which subsequently revealed the transcriptional and post-translational control of these enzymes. Michael Brown and Joseph Goldstein were awarded the Nobel Prize in Physiology or Medicine in 1985 for their comprehensive work on feedback regulation of cholesterol metabolism. Today, the mechanisms of regulation have been elucidated on a molecular level, although it is still not clear how cholesterol elicits all of the regulation. Furthermore, the evidence is rapidly building that cholesterol's precursors and metabolites might serve as biologically active signaling molecules.

Fig. 1 is an overview of the metabolic and transport pathways that control cholesterol levels in mammalian cells (reviewed in Liscum and Munn [1]). Cholesterol is synthesized from acetyl-CoA via the isoprenoid pathway, and at least four enzymes in the biosynthetic pathway are regulated by cellular cholesterol levels. Essential non-steroidal isoprenoids, such as dolichol, prenylated proteins, heme A and isopentenyl adenosine-containing tRNAs are also synthesized by this pathway. In extrahepatic tissues, most cellular cholesterol is derived from *de novo* synthesis [2], whereas hepatocytes obtain most of their cholesterol via the receptor-mediated uptake of plasma lipoproteins, such as low-density lipoprotein (LDL). LDL is bound and internalized by the LDL receptor

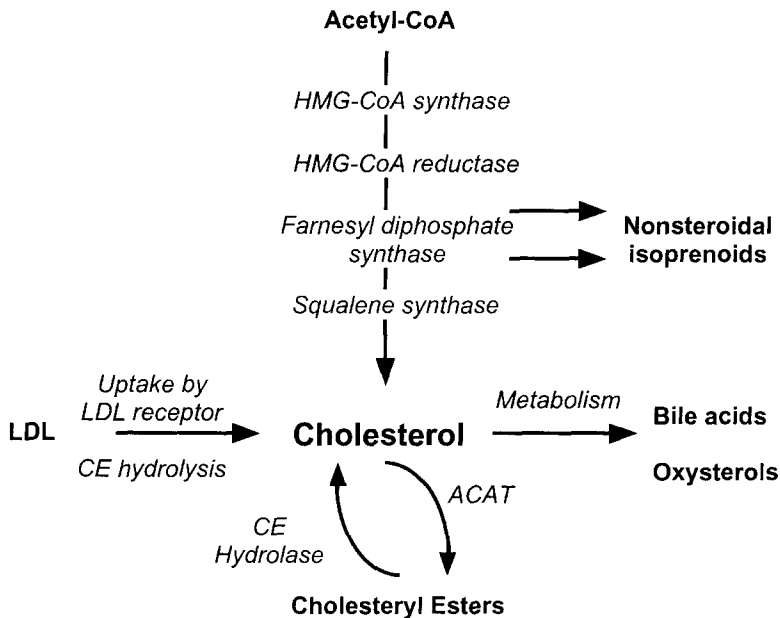


Fig. 1. Overview of the metabolic and transport pathways that control cholesterol levels in mammalian cells. Cholesterol is synthesized from acetyl-CoA and the four key enzymes that regulate cholesterol synthesis are indicated. Cells also obtain cholesterol by uptake and hydrolysis of LDL's cholesteryl esters (CE). End-products derived from cholesterol or intermediates in the pathway include bile acids, oxysterols, cholesteryl esters and non-steroidal isoprenoids. ACAT, acyl-CoA : cholesterol acyltransferase.

and delivered to the acidic late endosomes and lysosomes, where hydrolysis of the core cholesteryl esters occurs (discussed in Chapter 21). The cholesterol that is released is transported throughout the cell. Normal mammalian cells tightly regulate cholesterol synthesis and LDL uptake to maintain cellular cholesterol levels within narrow limits and supply sufficient isoprenoids to satisfy metabolic requirements of the cell. Regulation of cholesterol biosynthetic enzymes takes place at the level of gene transcription, mRNA stability, translation, enzyme phosphorylation and enzyme degradation. Cellular cholesterol levels are also modulated by a cycle of cholesterol esterification by acyl-CoA : cholesterol acyltransferase (ACAT) and hydrolysis of the cholesteryl esters, and by cholesterol metabolism to bile acids and oxysterols.

2. The cholesterol biosynthetic pathway

Fig. 2 takes a closer look at the cholesterol biosynthetic pathway, focusing on the enzymes that are regulated, sterol intermediates and the location of enzymes in the cell. Sterols are synthesized from the two-carbon building block, acetyl-CoA. The soluble enzyme acetoacetyl-CoA thiolase interconverts acetyl-CoA and acetoacetyl-CoA, which are then condensed by 3-hydroxy-3-methylglutaryl (HMG)-CoA synthase to form HMG-CoA. There are two forms of HMG-CoA synthase. A mitochondrial

form, involved in ketogenesis, predominates in the liver. In extrahepatic tissues, the most abundant form is a soluble enzyme of 53 kDa that is highly regulated by supply of cholesterol (G. Gil, 1986). Like acetoacetyl-CoA thiolase, HMG-CoA synthase has classically been described as a cytosolic enzyme because it is found in the $100,000 \times g$ supernatant of homogenized cells and tissues. However, both enzymes contain peroxisomal targeting sequences [3] and may reside in multiple cellular compartments.

HMG-CoA reductase catalyzes the reduction of HMG-CoA to mevalonate, utilizing two molecules of NADPH. HMG-CoA reductase is a 97-kDa glycoprotein of the endoplasmic reticulum (L. Liscum, 1985) and peroxisomes [3]. Analysis of the endoplasmic reticulum enzyme's domain structure revealed an N-terminal membrane domain with eight transmembrane spans (E.H. Olender, 1992), a short linker, and a C-terminal catalytic domain facing the cytosol (Fig. 3). Transmembrane spans 2–5 share a high degree of sequence similarity with several other key proteins in cholesterol metabolism; this region is termed the sterol-sensing domain (described in Section 3.5). Elucidation of the crystal structure of the HMG-CoA reductase catalytic domain indicated that the active protein is a tetramer [4], which is consistent with biochemical analysis. The monomers appear to be arranged in two dimers, with the active sites at the monomer–monomer interface. The dimer–dimer interface is predominantly hydrophobic.

HMG-CoA reductase is the rate-determining enzyme of the cholesterol biosynthetic pathway and, like HMG-CoA synthase, is highly regulated by supply of cholesterol. Thus, the enzyme has received intense scrutiny as a therapeutic target for treatment of hypercholesterolemia. The enzyme is inhibited by a class of pharmacological agents, generally called statins, which have an HMG-like moiety and a bulky hydrophobic group [5] (Fig. 4). Statins occupy the HMG-binding portion of the active site, preventing HMG-CoA from binding (E.S. Istvan, 2001). Also, the bulky hydrophobic group causes disordering of several catalytic residues. Thus, statins are potent, reversible competitive inhibitors of HMG-CoA reductase with K_i values in the nanomolar range. Elevated plasma cholesterol levels are a primary risk factor for coronary artery disease, and statin inhibition of HMG-CoA reductase effectively reduces cholesterol levels and decreases overall mortality. However, complete inhibition of HMG-CoA reductase by statins will kill cells, even if exogenous cholesterol is supplied. That is because complete inhibition deprives cells of all mevalonate-derived products, including essential non-steroidal isoprenoids. To survive, cells must produce a small amount of mevalonate that, when limiting, is used preferentially by higher affinity pathways for non-steroidal isoprenoid production (S. Mosley, 1983).

Mevalonate is metabolized to farnesyl-diphosphate (-PP) by a series of enzymes localized in peroxisomes. First, mevalonate kinase phosphorylates the 5-hydroxy group of mevalonic acid. The enzyme is a homodimer of 40 kDa that is subject to feedback inhibition by several isoprenoid intermediates [6]. Mutations in the mevalonate kinase gene lead to the human genetic disease mevalonic aciduria (discussed in Section 2.2). The product of mevalonate kinase, mevalonate-5-P, is then phosphorylated to form mevalonic acid-5-PP, which is decarboxylated and dehydrated by mevalonate-PP decarboxylase to form isopentenyl-PP. Isopentenyl-PP is in equilibrium with its isomer, dimethylallyl-PP. Farnesyl-PP synthase catalyzes the head to tail condensations of two molecules of isopentenyl-PP with dimethylallyl-PP to form farnesyl-PP. The

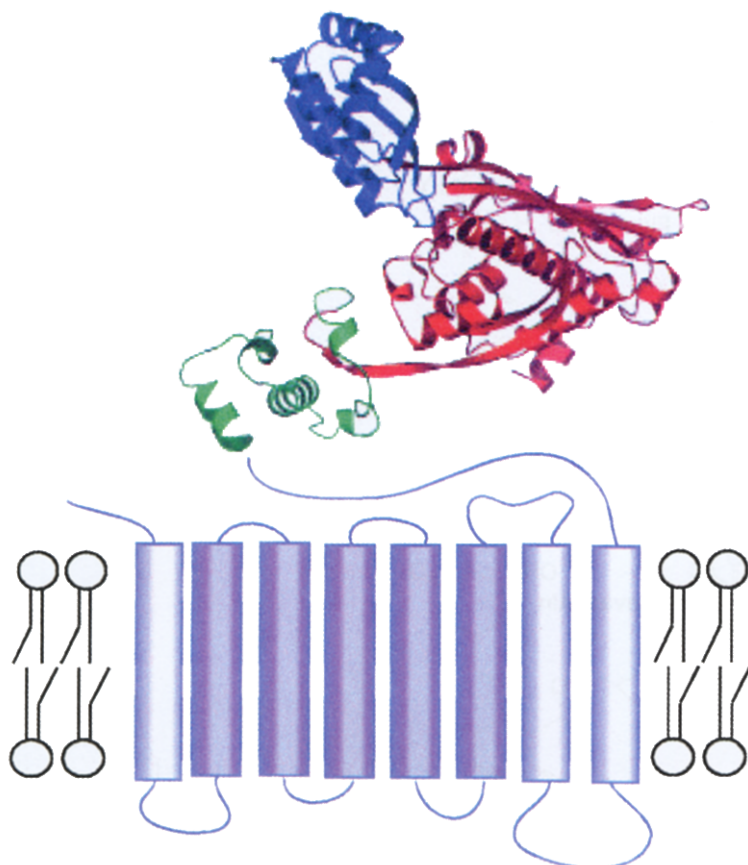


Fig. 3. Domain structure of the endoplasmic reticulum HMG-CoA reductase. The crystal structure of the catalytic domain has been determined and is depicted as a ribbon diagram (courtesy of Eva S. Istvan, Washington University School of Medicine). The catalytic domain consists of a small helical domain (green), a large central element resembling a prism (red), which contains the HMG-CoA-binding site, and a small domain to which NADPH binds (blue) [4]. The structure of the membrane domain has not been solved; however, it is known that eight transmembrane spans embed the protein into the endoplasmic reticulum membrane. Spans 2–5 (darker cylinders) are termed the sterol-sensing domain and mediate the regulated degradation of the enzyme.

enzyme is part of a large family of prenyltransferases that synthesize the backbones for all isoprenoids, including cholesterol, steroids, prenylated proteins, heme A, dolichol, ubiquinone, carotenoids, retinoids, chlorophyll and natural rubber (K.C. Wang, 2000).

Squalene synthase is a 47-kDa protein of the endoplasmic reticulum and catalyzes the first committed step in cholesterol synthesis. The enzyme condenses two molecules of farnesyl-PP and then reduces the presqualene-PP intermediate to form squalene. A large N-terminal catalytic domain faces the cytosol, anchored to the membrane by a C-terminal domain. This orientation may allow the enzyme to receive the hydrophilic substrates from the cytosol and release the hydrophobic product into the endoplasmic

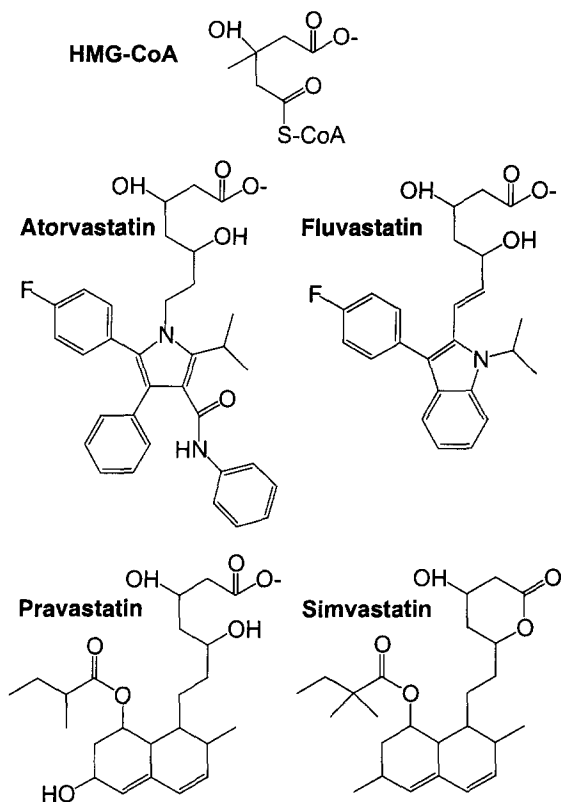


Fig. 4. Chemical structures of HMG-CoA and several statin inhibitors of HMG-CoA reductase. Atorvastatin (Lipitor), fluvastatin (Lescol), pravastatin (Pravachol) and simvastatin (Zocor) are widely prescribed cholesterol-lowering drugs.

reticulum membrane for further metabolism [7]. Squalene synthase is highly regulated by the cholesterol content of the cell. Thus, it plays an important role in directing the flow of farnesyl-PP into the sterol or non-sterol branches of the pathway (M.S. Brown, 1980) [7].

Squalene is converted into the first sterol, lanosterol, by the action of squalene epoxidase and oxidosqualene cyclase. Lanosterol is then converted to cholesterol by a series of oxidations, reductions, and demethylations. The required enzyme reactions have been defined and metabolic intermediates identified; however, the precise sequence of reactions between lanosterol and cholesterol remains to be established [8] (Fig. 5). There is evidence for two alternative pathways that differ in when the Δ^{24} double bond is reduced (discussed in Section 2.3). Both 7-dehydrocholesterol and desmosterol have been postulated to be the immediate precursor of cholesterol. One of the key enzymes in the latter part of the pathway is 7-dehydrocholesterol Δ^7 -reductase, a 55-kDa integral membrane protein. Mutations in the gene for 7-dehydrocholesterol Δ^7 -reductase cause the human genetic disease Smith–Lemli–Opitz syndrome (discussed in Section 2.3).

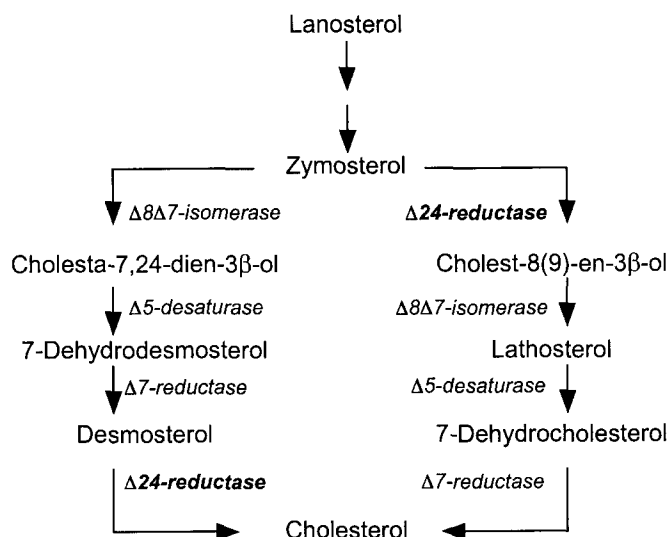


Fig. 5. Final steps in the cholesterol biosynthetic pathway. Alternate steps have been proposed for the conversion of zymosterol to cholesterol, which differ in when the $\Delta 24$ -reductase reaction occurs. Figure adapted from Waterham and Wanders [8] and Kelly and Hennekam [11].

2.1. Enzyme compartmentalization

Where does cholesterol synthesis take place? All of the enzymes that convert acetyl-CoA to farnesyl-PP have classically been thought of as cytosolic enzymes, with the exception of HMG-CoA reductase, which is typically depicted as an endoplasmic reticulum enzyme with the catalytic site facing the cytosol. Enzymes that convert farnesyl-PP to cholesterol are classically described as microsomal. However, there is now strong evidence that all but one of these enzymes is also, or exclusively, peroxisomal [3]. The molecular cloning of cDNAs encoding many of these enzymes has revealed peroxisomal targeting sequences. The availability of antibodies has allowed immunocytochemical localization to peroxisomes. Together these data suggest that peroxisomes may play an active role in all steps in the cholesterol biosynthetic pathway except the conversion of farnesyl-PP to squalene, which is catalyzed by squalene synthase found solely in the endoplasmic reticulum.

HMG-CoA reductase is the one exception to the rule. Immunocytochemistry and immunoblotting have localized HMG-CoA reductase to both the endoplasmic reticulum and peroxisomes; however, no peroxisomal targeting motif has been found in the HMG-CoA reductase protein sequence. Furthermore, the peroxisomal HMG-CoA reductase has an apparent molecular weight of 90 kDa whereas the endoplasmic reticulum enzyme is 97 kDa (W.H. Engfelt, 1997). The peroxisomal enzyme exhibits other distinct properties: it is resistant to statin inhibition, the enzyme's activity is not regulated by phosphorylation, the protein's turnover is not regulated by mevalonate. Altogether, this evidence suggests that the endoplasmic reticulum and peroxisome enzymes are functionally and structurally distinct (N. Aboushadi, 2000).

Additional evidence for the involvement of peroxisomes in cholesterol biosynthesis comes from analysis of diseases of peroxisomal deficiency. Zellweger syndrome, neonatal adrenoleukodystrophy, and infantile Refsum's disease are all diseases of peroxisome biogenesis [9]. In most of these peroxisomal disorders, the peroxisomal matrix proteins are synthesized in the cytosol as normal, but they cannot be assembled into nascent peroxisomes due to mutations in one of at least 12 different genes encoding proteins necessary for peroxisomal protein targeting and import. Fibroblasts from individuals with peroxisome biogenesis disorders show reduced enzymatic activities of cholesterol biosynthetic enzymes, reduced levels of cholesterol synthesis and lower cholesterol content [3]. These data support the hypothesis that part of the cholesterol synthesis pathway is peroxisomal.

It is not clear why cholesterol synthesis is compartmentalized and requires intermediates to cycle between peroxisomes and the cytosol. It is also unclear why some of the enzymes are found in multiple compartments and others are solely in endoplasmic reticulum or peroxisomes. As noted, cholesterol synthesis is a very complex process and compartmentalization may represent another level of regulation [3].

2.2. *Mevalonic aciduria*

Cholesterol synthesis is essential for normal development and maintenance of tissues that cannot obtain cholesterol from plasma lipoproteins, such as brain. Furthermore, the biosynthetic pathway supplies non-steroidal isoprenoids that are required by all cells. Thus, it is not surprising that metabolic defects in the cholesterol biosynthetic pathway have devastating consequences.

The first recognized human metabolic defect in the biosynthesis of cholesterol and isoprenoids was mevalonic aciduria [10]. Mevalonic aciduria is an autosomal recessive disorder that is quite rare, with only 19 known patients. In normal individuals, a small amount of mevalonic acid diffuses into the plasma at levels proportional to the rate of cellular cholesterol formation. Patients with mild mevalonic aciduria excrete 3000–6000 times the normal amount of mevalonic acid and patients with the severe form of the disease excrete 10,000–200,000 times the normal amount. Enzyme assays using cell lysates showed that mevalonate kinase activity was markedly deficient in patient samples and genetic analysis has revealed nucleotide changes in the mevalonate kinase gene that lead to amino acid substitutions. Because of this enzyme deficiency, there is little to no feedback inhibition of HMG-CoA reductase and, thus, mevalonate is overproduced.

Clinical features of mevalonic aciduria include failure to thrive, anemia, gastroenteropathy, hepatosplenomegaly, psychomotor retardation, hypotonia, ataxia, cataracts, and dysmorphic features [10]. Surprisingly, patients with severe deficiencies in mevalonate kinase show normal plasma cholesterol levels and cultured mevalonic aciduria fibroblasts show rates of cholesterol synthesis half that of normal cells. Close examination of cholesterologenic enzymes in mevalonic aciduria fibroblasts has revealed a 6-fold increase in HMG-CoA reductase activity, which is postulated to compensate for the low mevalonate kinase activity.

2.3. *Smith–Lemli–Opitz syndrome*

A second metabolic defect in cholesterol synthesis leads to Smith–Lemli–Opitz syndrome (SLOS) (B.U. Fitzkey, 1999) [11]. SLOS is a relatively common autosomal recessive disorder, with estimates of incidence ranging from 1 in 10,000 to 1 in 60,000. Four lines of evidence pointed to the metabolic defect in SLOS patients. (1) Individuals with SLOS were found to have markedly elevated levels of plasma 7-dehydrocholesterol and low plasma cholesterol levels. (2) 7-Dehydrocholesterol $\Delta 7$ -reductase activity was deficient in SLOS patient samples and the amount of residual activity could be correlated with severity of the disease. (3) Rodents treated with AY-9944, an inhibitor of 7-dehydrocholesterol $\Delta 7$ -reductase, developed SLOS-like malformations [11]. (4) Cloning of the gene for 7-dehydrocholesterol $\Delta 7$ -reductase led to identification of a splice-site mutation and amino acid substitutions in SLOS patients.

Severely reduced cholesterol synthesis is predicted to have severe consequences on development of the fetus because cholesterol is only obtained from the maternal circulation during the first trimester [11]. In addition, the brain is predicted to be severely affected because plasma lipoproteins cannot cross the blood–brain barrier and most, if not all, cholesterol needed for brain growth is synthesized locally (S.D. Turley, 1998) [2,12]. Indeed, severely affected SLOS infants who died soon after birth were found to have functionally null 7-dehydrocholesterol $\Delta 7$ -reductase alleles [12], whereas typical affected individuals likely have some residual 7-dehydrocholesterol $\Delta 7$ -reductase catalytic activity.

Patients with SLOS have mental retardation and microcephaly, which is consistent with cholesterol synthesis being required for normal brain development. Clinical features also include failure to thrive, and characteristic craniofacial, skeletal and genital anomalies. The clinical phenotype appears to be due to a lack of cholesterol rather than the cellular accumulation of 7-dehydrocholesterol (W. Gaoua, 2000). A recent multicenter clinical trial has shown that SLOS children fed a diet supplemented with cholesterol show improved growth and neurodevelopment (i.e. language and cognitive skills) (M. Irons, 1997; E.R. Elias, 1997). It is likely that the diet fulfilled the daily requirement for cholesterol and down-regulated endogenous 7-dehydrocholesterol synthesis.

What are the final steps in the cholesterol biosynthetic pathway? SLOS may provide an answer to that question. As noted above, there is evidence for two alternative pathways, which differ in when the $\Delta 24$ double bond is reduced [11]. In both pathways, lanosterol is demethylated to form zymosterol (Fig. 5). Then, zymosterol can be metabolized sequentially by a $\Delta 24$ -reductase, $\Delta 8, \Delta 7$ -isomerase, and $\Delta 5$ -desaturase to form 7-dehydrocholesterol, which is reduced at the $\Delta 7$ position to form cholesterol. Alternatively, zymosterol can be metabolized by the $\Delta 8, \Delta 7$ -isomerase and $\Delta 5$ -desaturase, to form 7-dehydridesmosterol. 7-Dehydridesmosterol is metabolized by the $\Delta 7$ -reductase to form desmosterol and then by the $\Delta 24$ -reductase to form cholesterol. The fact that the SLOS deficiency in $\Delta 7$ -reductase leads to a buildup of 7-dehydrocholesterol rather than 7-dehydridesmosterol is interpreted to mean that the former pathway is the principal one. However, the latter pathway must also be used because desmosterol is an abundant cholesterol precursor in certain tissues. It has been suggested that the final steps in the biosynthetic pathway may be tissue specific.

Table 1
Inborn errors of sterol biosynthesis

Syndrome	Metabolic defect
Mevalonic aciduria	Mevalonate kinase
Smith–Lemli–Opitz	Sterol $\Delta 7$ -reductase
Desmosterolosis	Sterol $\Delta 24$ -reductase
Rhizomelic chondrodysplasia punctata (CDP)	Pex7 peroxisomal enzyme import
CDP X-linked dominant (CDPX2)	Sterol $\Delta 8, \Delta 7$ -isomerase
CHILD syndrome (congenital hemidysplasia with ichthyosis and limb defects)	Sterol $\Delta 8, \Delta 7$ -isomerase Sterol C-4 demethylase
Greenberg skeletal dysplasia	Sterol $\Delta 14$ -reductase

These syndromes and their corresponding metabolic defects are reviewed in Kelley [13].

Perhaps, in SLOS cells, any 7-dehydrodesmosterol that accumulates is metabolized by the available $\Delta 24$ -reductase to form 7-dehydrocholesterol.

2.4. Other enzyme deficiencies

Other inborn errors of sterol biosynthesis have been reviewed by Kelley [13] and are summarized in Table 1. Rhizomelic chondrodysplasia punctata, like Zellweger syndrome, exhibits defective sterol synthesis due to the lack of key peroxisomal enzymes of cholesterol biosynthesis. CDPX2, also known as Conradi–Hünemann syndrome, and most cases of CHILD syndrome are due to mutations in the sterol $\Delta 8, \Delta 7$ -isomerase gene, which is located on the X chromosome. Mutations in a single gene may lead to different syndromes with similar, but distinct, pathologies due to the mosaicism of X-chromosome inactivation. A few cases of CHILD syndrome may be due to mutations in the sterol C-4 demethylase gene, also located on the X chromosome.

3. Regulation of cholesterol synthesis

Isoprenoid synthesis is regulated by the sterol end-product of the biosynthetic pathway, by non-sterol intermediates, and also by physiological factors. The cholesterol content of the cell controls several enzymes in the biosynthetic pathway, but the focus has been on the rate-limiting enzyme, HMG-CoA reductase. Different regulators have different mechanisms of action. For example, sterols have been shown to regulate at the level of HMG-CoA reductase transcription whereas non-sterols regulate HMG-CoA reductase mRNA translation. Both sterols and non-sterols are needed for regulation of HMG-CoA reductase protein degradation [14]. Physiological factors that influence cholesterol synthesis include diurnal rhythm, insulin and glucagon, thyroid hormone, glucocorticoids, estrogen and bile acids [15]. These factors regulate HMG-CoA reductase by transcriptional, translational and post-translational mechanisms.

3.1. Transcriptional regulation

HMG-CoA reductase is the rate-limiting enzyme in the cholesterol biosynthetic pathway and combined regulation of HMG-CoA reductase synthesis and turnover can alter steady state levels of the enzyme 200-fold. HMG-CoA reductase is regulated in parallel with at least three other enzymes in the cholesterol biosynthetic pathway, HMG-CoA synthase, farnesyl-PP synthase and squalene synthase, as well as the LDL receptor. This coordinate regulation is due to the fact that each gene has a similar sequence (*cis*-acting element) within the promoter that recognizes a common *trans*-acting transcription factor. Availability of the transcription factor to bind to the promoter sequence is influenced by the cellular cholesterol content.

Fig. 6 illustrates the current model of cholesterol-mediated transcriptional regulation [16–19]. The 5' flanking regions of cholesterol-regulated genes have one to three copies of a 10-bp non-palindromic nucleotide sequence termed the sterol regulatory element (SRE). SREs are conditional positive elements that are required for gene transcription in cholesterol-depleted cells. The SRE sequence found in the LDL receptor gene is 5'-ATCACCCCAC-3'. SREs have been identified in the HMG-CoA synthase, HMG-CoA reductase, farnesyl-PP synthase and squalene synthase genes, as well as genes of fatty acid synthesis. However, there is not a strict SRE consensus sequence and identifying functional SREs has been difficult [17].

The transcription factor that binds the SRE is termed the SRE-binding protein (SREBP) [16,20,21]. The first SREBP to be identified was the protein that bound to the LDL receptor promoter (M.R. Briggs, 1990; X. Wang, 1990). Cloning of SREBP cDNAs (C. Yokoyama, 1993; X. Hua, 1993) revealed that there are two SREBP genes that produce three distinct proteins. SREBP-1a and -1c are derived from one gene that contains two promoters and differ in the length of the N-terminal transactivation domain. SREBP-2 is derived from a second gene and is 45% identical to SREBP-1a. SREBP-1c is the predominant isoform in liver and adipocytes. It was isolated independently and called adipocyte determination and differentiation-dependent factor 1 (P. Tontonoz, 1993). Here when the term SREBP is used, the information is relevant for all three isoforms.

SREBPs are cytosolic 68-kDa proteins with a canonical basic helix–loop–helix leucine zipper (bHLH-Zip) motif that is present in other transcription factors. Unlike other bHLH-Zip transcription factors, SREBPs have a tyrosine in place of a conserved arginine, which allows them to bind to the inverted E-box motif (5'-CANNTG-3') in addition to SREs (J.B. Kim, 1995). By binding to SREs, SREBPs coordinately regulate multiple enzymes involved in fatty acid synthesis and lipogenesis [20,21].

An additional feature that distinguishes SREBPs from other bHLH-Zip transcription factors is that SREBP genes encode 125-kDa membrane proteins that are inserted into the endoplasmic reticulum and serve as precursors for the active transcription factors. SREBPs have three functional domains: an N-terminal 68-kDa fragment containing the bHLH-Zip transcription factor, two membrane-spanning segments, and a C-terminal regulatory domain. It is the sequential two-step cleavage of the full-length precursor SREBP and release of the 68-kDa N-terminal bHLH-Zip domain that is influenced by the cellular cholesterol content.

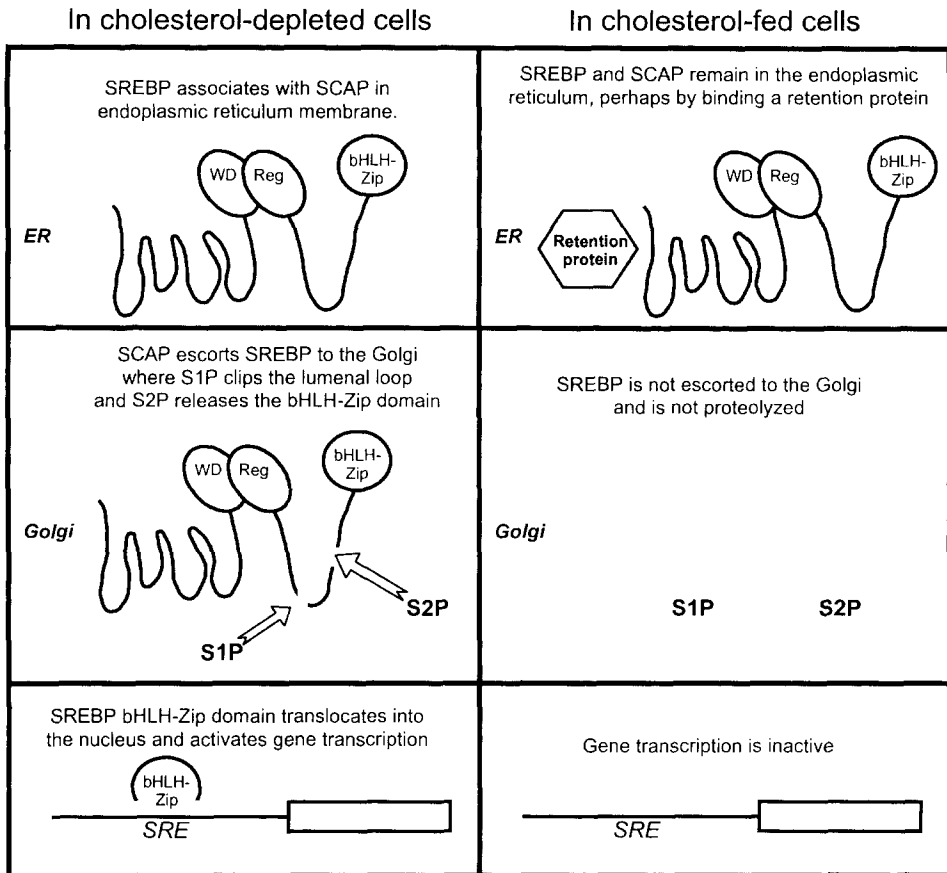


Fig. 6. Current model of cholesterol regulation of SREBP proteolysis. The sterol regulatory element-binding protein (SREBP) precursor is inserted into the endoplasmic reticulum (ER) membrane. The SREBP regulatory domain (Reg) interacts with the SREBP cleavage-activating protein (SCAP), likely through SCAP's WD repeats. When cholesterol levels are low, SCAP escorts SREBP to the Golgi where the bHLH-ZIP domain is released by site-1 protease (S1P) cleavage of a luminal loop followed by site-2 protease (S2P) cleavage within a transmembrane span. The mature SREBP translocates into the nucleus and activates gene transcription. In cholesterol replete cells, the SREBP precursor and SCAP remain in the endoplasmic reticulum and the SREBP precursor is not proteolyzed to release the bHLH-Zip transcription factor.

Identification of the proteins required for regulated SREBP cleavage was accomplished using somatic cell genetic approaches. Mutant Chinese hamster ovary cells with abnormal regulation of cholesterol and fatty acid metabolism, which were selected over the past 20 years, proved invaluable for this goal [22]. Using expression cloning strategies, genes were isolated that restored SREBP-mediated transcription in each mutant. This work led to identification of two proteases and an escort protein required for SREBP precursor cleavage. Cleavage of the SREBP precursor at site 1 requires a subtilisin-like serine protease (J. Sakai, 1998), whereas cleavage at site 2 requires a zinc metalloprotease (R.B. Rawson, 1997). Transcription factor release is controlled

by a chaperone, SREBP cleavage-activating protein (SCAP), which escorts the SREBP precursor to the Golgi where the proteases reside (X. Hua, 1996).

How is SREBP proteolysis controlled by cholesterol? A hint that this event may involve vesicle trafficking came from the finding that the SREBP precursor's *N*-linked carbohydrates are endoglycosidase-H-resistant (trimmed by Golgi mannosidases) when cellular cholesterol levels are low, and endoglycosidase-H-sensitive when cellular cholesterol levels are high (A. Narturfft, 1998, 1999). Thus, release of the mature SREBP transcription factor appeared to coincide with transport to the Golgi. The current model is as follows (Fig. 6). When cellular cholesterol levels are low, the SREBP precursor is synthesized and inserted into the endoplasmic reticulum membrane. The C-terminal regulatory domain of the SREBP precursor interacts with the C-terminus of SCAP, likely through SCAP's four WD repeats (J. Sakai, 1998). SCAP and the SREBP precursor are then transported to the Golgi where the luminal loop of the SREBP precursor is clipped by the site-1 protease; however, the two halves of the protein remain membrane-anchored. Then the site-2 protease clips the N-terminal SREBP intermediate within the first membrane-spanning segment, releasing the soluble transcription factor, mature SREBP. Upon translocation to the nucleus, the mature SREBP binds SRE sequences within the promoters of target genes and enhances their transcription.

When cellular cholesterol levels rise to a threshold level, SCAP and the SREBP precursor no longer travel to the Golgi and the SREBP precursor is not proteolyzed to produce the mature SREBP. As a result, transcription of target genes declines to basal levels. Evidence for SCAP-SREBP transport is two-fold. The sterol-dependent movement of SCAP has been directly visualized in cultured cells transfected with a green fluorescent protein-SCAP fusion protein (A. Narturfft, 2000). Furthermore, *in vitro* vesicle-budding assays have demonstrated that oxysterols suppress SREBP-SCAP complexes from entering into vesicles budding from the endoplasmic reticulum.

The mechanism by which SCAP senses cellular cholesterol is not known. Sensing is postulated to involve a five-transmembrane segment of SCAP with sequence similarity to a five-transmembrane segment of HMG-CoA reductase, called the sterol-sensing domain (discussed in Section 3.5). Consistent with this hypothesis, several constitutively active SCAP mutants have been isolated that have point mutations in the sterol-sensing domain (X. Hua, 1996). Also, there is recent evidence that SCAP interacts with a protein that is retained in the endoplasmic reticulum (T. Yang, 2000). Therefore, control of SCAP and SREBP transit to the Golgi could depend upon sterol-dependent binding of SCAP's sterol-sensing domain to a retention protein.

SREBPs control not only cholesterol synthesis, but also the synthesis of fatty acids (via fatty acid synthase, acetyl-CoA carboxylase, and stearoyl-CoA desaturase 2), triacylglycerols and phospholipids (glycerol-3-phosphate acyltransferase). This was illustrated most dramatically in transgenic mice overexpressing the nuclear forms of SREBP-1a, -1c and -2 in liver [20]. Overexpression of each isoform resulted in activation of a full spectrum of cholesterol and fatty acid biosynthetic enzymes; however, absolute levels of induction of each enzyme and the subsequent liver phenotype varied according to the isoform expressed. From these data, the following conclusions can be drawn. SREBP-1a is a strong activator of cholesterol and fatty acid synthesis and is likely to be important in rapidly dividing cells that require lipid for membrane production.

SREBP-1c predominates in the liver, where it primarily activates genes of fatty acid synthesis. SREBP-1c appears to be important for maintaining basal transcription levels of fatty acid and cholesterol biosynthetic enzymes during periods of fasting. It also plays a role in the insulin response [21]. SREBP-2 selectively activates cholesterol biosynthetic genes and the LDL receptor, and primarily responds when the liver's demand for cholesterol rises.

3.2. mRNA translation

HMG-CoA reductase is also subject to translational control by a mevalonate-derived non-sterol regulator (D. Peffley, 1985; M. Nakanishi, 1988). This component of the regulatory mechanism can only be observed when cultured cells are acutely incubated with statins, which block mevalonate formation. Under those conditions, sterols have no effect on HMG-CoA reductase mRNA translation; however, mevalonate reduces the HMG-CoA mRNA translation by 80% with no change in mRNA levels. Translational control of hepatic HMG-CoA reductase by dietary cholesterol was shown in an animal model in which polysome-associated HMG-CoA reductase mRNA was analyzed in cholesterol-fed rats (C.M. Chambers, 1997). It was found that cholesterol feeding increased the portion of mRNA associated with translationally inactive monosomes and decreased the portion of mRNA associated with translationally active polysomes. The mechanism of HMG-CoA reductase translational control has not been elucidated.

3.3. Phosphorylation

Many key metabolic enzymes are modulated by phosphorylation–dephosphorylation and it has long been known that HMG-CoA reductase catalytic activity is inhibited by phosphorylation (Z.H. Beg, 1973). Rodent HMG-CoA reductase is phosphorylated on Ser 871 by an AMP-activated protein kinase that uses ATP as a phosphate donor (P.R. Clarke, 1990). However, examination of HMG-CoA reductase activity in rat liver showed that phosphorylation–dephosphorylation could not account for the long-term regulation that occurred with diurnal light cycling, fasting, or cholesterol-supplemented diet (M.S. Brown, 1979). Approximately 75–90% of HMG-CoA reductase enzyme was found to be phosphorylated (inactive) under all physiological conditions. This reservoir of inactive enzyme may allow cells to respond transiently to short-term cholesterol needs.

The AMP-activated kinase that phosphorylates and inactivates HMG-CoA reductase also phosphorylates and inactivates acetyl-CoA carboxylase. It has been suggested that, when cellular ATP levels are depleted causing AMP levels to increase, the resultant activation of the kinase would inhibit cholesterol and fatty acid biosynthetic pathways, thus conserving energy (D.G. Hardie, 1992). Consistent with this hypothesis, cholesterol synthesis was reduced when ATP levels were depleted by incubation with 2-deoxy-D-glucose (R. Sato, 1993). However, cholesterol synthesis was not reduced when ATP levels declined in cells expressing a Ser 871 to Ala mutant form of HMG-CoA reductase, which is not phosphorylated (R. Sato, 1993). Therefore, HMG-CoA reductase phosphorylation appears to be important for preserving cellular energy stores rather than end-product feedback regulation.

3.4. Proteolysis

Raising the cellular cholesterol content not only stops transcription of the genes encoding the cholesterol biosynthetic enzymes, but it also leads to accelerated degradation of the rate-limiting enzyme, HMG-CoA reductase. In cholesterol-depleted cells, HMG-CoA reductase is a stable protein that is degraded slowly ($t_{1/2} = 13$ h) (J.R. Faust, 1982). If cholesterol repletion simply stopped transcription of the HMG-CoA reductase gene, that would lead to a slow decline in HMG-CoA reductase enzyme activity owing to stability of the protein; however, in the presence of excess sterols or mevalonate there is rapid ($t_{1/2} = 3.6$ h) (J.R. Faust, 1982) and selective degradation of the enzyme, which results in more precise control of cellular sterol synthesis.

The HMG-CoA reductase membrane domain is necessary and sufficient for regulated degradation. Expression of the cytosolic catalytic domain results in a stable protein that is not subject to regulated degradation (G. Gil, 1985), whereas expression of the N-terminal membrane domain linked to a reporter protein results in regulated degradation of the reporter (D. Skalnik, 1988). Despite the complex topology of HMG-CoA reductase, no proteolytic intermediates have ever been detected. How is HMG-CoA reductase proteolyzed? Possibilities include vesicular translocation to lysosomes or autophagy of HMG-CoA reductase-containing endoplasmic reticulum membranes; however, the degradation appears to be rapid and selective for HMG-CoA reductase. It is also unaffected by inhibitors of protein transport through the Golgi (K. Chun, 1990). Another possibility is HMG-CoA reductase digestion by resident endoplasmic reticulum proteases. Subcellular fractionation and use of specific protease inhibitors has revealed that HMG-CoA reductase degradation occurs in purified endoplasmic reticulum membranes and is inhibited by lactacystin, an inhibitor of the proteasome (T. McGee, 1996). Additional evidence that the proteasome is involved comes from study of the yeast ortholog, Hmg2p, which is subject to regulated degradation like the mammalian enzyme. A yeast genetic approach has led to the identification of three proteins that are involved in the reverse translocation of endoplasmic reticulum proteins and disposal by the proteasome (R.Y. Hampton, 1996) [23]. Evidence that HMG-CoA reductase is polyubiquitinated prior to proteolysis has been provided for the yeast (R.Y. Hampton, 1997) and mammalian (T. Ravid, 2000) enzymes.

What is the signal for accelerated HMG-CoA reductase degradation? In mammalian cells, both mevalonate-derived non-sterols and sterols are required. That is, in cholesterol-depleted cells, the addition of sterols leads to accelerated HMG-CoA reductase degradation only when the sterols are accompanied by a mevalonate-derived non-sterol signal (M. Nakanishi, 1988; J. Roitelman, 1992; T.E. Meigs, 1997). Regulated degradation has also been demonstrated through *in vitro* experiments. HMG-CoA reductase degradation is more rapid in endoplasmic reticulum membranes isolated from mevalonate- or sterol-treated cells (T. McGee, 1996) and in hepatic microsomes prepared from mevalonate-treated rats (C. Correll, 1994). Both the sterol and non-sterol signals are blocked by cycloheximide, indicating that regulated turnover of HMG-CoA reductase requires ongoing protein synthesis (K. Chun, 1990; J. Roitelman, 1992; T. Ravid, 2000).

There is strong evidence that the non-sterol isoprenoid signal for HMG-CoA reduc-

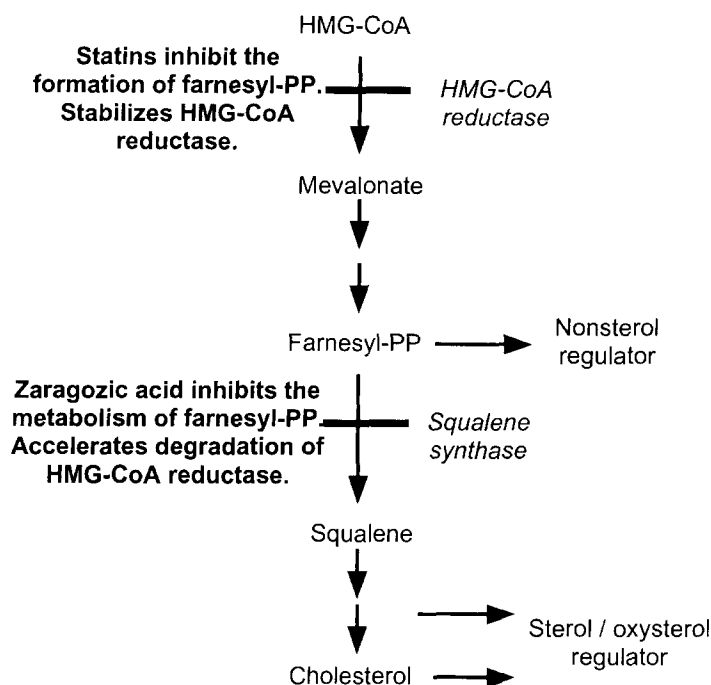


Fig. 7. Evidence that a farnesyl-PP-derived signal modulates HMG-CoA reductase degradation. In yeast, conditions that decrease farnesyl-PP levels stabilize HMG-CoA reductase levels whereas conditions that increase farnesyl-PP levels accelerate HMG-CoA reductase degradation.

tase degradation is derived from farnesyl-PP. This evidence has been recently reviewed (S.F. Petras, 2001) and includes the following. Intracellular farnesol levels increase significantly after mevalonate addition to cells (T.E. Meigs, 1996). Accelerated HMG-CoA reductase degradation can be induced in cells incubated with mevalonate, farnesyl-PP, or farnesol but not with a non-hydrolyzable analog of farnesyl-PP (C.C. Correll, 1994; M.D. Giron, 1994; T.E. Meigs, 1997). Furthermore, inhibition of the enzyme farnesyl pyrophosphatase blocks the mevalonate-dependent, sterol-accelerated degradation of the enzyme (T.E. Meigs, 1997).

Many, but not all, aspects of regulated HMG-CoA reductase degradation are conserved among eukaryotes. In yeast, modulation of Hmg2p stability by a farnesyl-PP-derived signal has been shown using pharmacologic and genetic approaches (R.G. Gardner, 1999). Conditions chosen to increase farnesyl-PP levels (inhibition of squalene synthase by zaragozic acid or down-regulation of squalene synthase) accelerated Hmg2p ubiquitination and degradation, whereas conditions chosen to decrease farnesyl-PP levels (inhibition of HMG-CoA reductase by statins, farnesyl-PP synthase down-regulation, or squalene synthase overexpression) stabilized Hmg2p (Fig. 7). One difference between the yeast and mammalian systems is the requirement for ongoing protein synthesis, which has not been shown in yeast (R.G. Gardner, 1999).

As mentioned above, regulation of mammalian HMG-CoA reductase turnover re-

quires both sterols and non-sterol isoprenoids. In contrast, the yeast isozyme can be suppressed by farnesyl-PP alone, although sterols enhance the non-sterol regulation. Pharmacologic and genetic approaches have again been extremely informative, showing that endogenously produced oxysterols serve as a positive signal for Hmg2P degradation in yeast (R.G. Gardner, 2001). Oxysterols have long been known to accelerate HMG-CoA reductase degradation when added to the medium of cultured mammalian cells; however, an endogenously produced oxysterol regulator of mammalian HMG-CoA reductase degradation has not yet been identified.

How do sterol or non-sterol regulators influence HMG-CoA reductase degradation? They could have a direct effect on the enzyme itself, as an allosteric regulator or by binding to the membrane domain. Alternatively, they could have an effect on the biophysical properties of the endoplasmic reticulum membrane (i.e. fluidity, membrane organization) or affect the proteolytic machinery. These possible effects could be direct or through interaction with an effector protein (R.G. Gardner, 1999; S.F. Petras, 2001).

The half-life of HMG-CoA reductase is also influenced by the enzyme's oligomerization state and expression level (H.H. Cheng, 1999). Oligomerization of HMG-CoA reductase through its cytosolic domain appears to stabilize the protein, as does a higher expression level. Analysis of the crystal structure of the catalytic portion of human HMG-CoA reductase revealed that this domain forms tight tetramers (E.S. Istvan, 2000). It was suggested that oligomerization of the catalytic domain may induce association of the membrane domains, which would decrease accessibility of the enzyme to proteases. Sterols may induce dissociation of the HMG-CoA reductase tetramer, resulting in accelerated proteolysis.

3.5. Sterol-sensing domain

How do HMG-CoA reductase and SCAP detect rising cellular cholesterol levels? A key feature of these proteins was revealed when SCAP was cloned and the deduced protein sequence compared with that of HMG-CoA reductase (X. Hua, 1996). A five transmembrane domain was found in SCAP with 25% identity and 55% similarity with a corresponding region in HMG-CoA reductase. This domain, termed the sterol-sensing domain, appeared to be critical for SCAP's cholesterol-regulated SREBP escort function since certain amino acid substitutions in the sterol-sensing domain led to constitutive activity (X. Hua, 1996; A. Nohturfft, 1996). In HMG-CoA reductase, transmembrane span 2 (which is within the sterol-sensing domain) was shown to be necessary for regulated degradation (H. Kumagai, 1995). Chimeric mutants of HMG-CoA reductase were constructed that combined membrane-spanning domains from the hamster enzyme (which is subject to regulated degradation) with membrane-spanning domains from the sea urchin enzyme (which shares 62% amino acid sequence identity with the hamster enzyme in the membrane domain, but is not subject to regulated degradation). Analysis of regulated turnover of the chimeric molecules showed that hamster transmembrane span 2 was sufficient to confer regulated degradation upon the sea urchin enzyme (H. Kumagai, 1995).

Sterol-sensing domains with a high degree of sequence similarity are found in several other proteins with obvious connections to cholesterol homeostasis. One is the

biosynthetic enzyme 7-dehydrocholesterol Δ^7 -reductase. The function of the sterol-sensing domain in 7-dehydrocholesterol Δ^7 -reductase is not clear; however, amino acid substitutions in that region cause the loss of 90% of catalytic activity (S.H. Bae, 1999). Another sterol-sensing domain-containing protein is NPC1, a 1278-amino-acid glycoprotein found in late endosomes and lysosomes (E.D. Carstea, 1997). NPC1 is hypothesized to play a role in trafficking of cholesterol, gangliosides and other cargo from late endosomes to destinations throughout the cell [24] (L. Liscum, 2000). Mutations in NPC1 lead to the predominant form of Niemann–Pick C disease, a human genetic disease characterized by progressive neurodegeneration. The biological function of NPC1 is still not clear, but structure/function analysis indicates that the sterol-sensing domain is important. Mutations that cause amino acid substitutions within the sterol-sensing domain lead to a rapidly progressing, infantile form of the disease, whereas amino acid substitutions throughout the rest of the protein cause the classical juvenile presentation [24]. NPC1L1 is a Golgi protein with 42% identity and 51% similarity with NPC1 (J.P. Davies, 2000). NPC1L1 has a sterol-sensing domain, but the function of this NPC1-like protein is unknown.

Other proteins with a sterol-sensing domain have a more tenuous link to cholesterol homeostasis. Two proteins, Patched and dispatched, are involved in developmental patterning (A.P. McMahon, 2000). Patched is the receptor for the morphogen, Sonic Hedgehog, which is the only known protein with a covalently attached cholesterol moiety. Dispatched is the plasma membrane protein required for secretion of cholesterol-modified Hedgehog. Hedgehog binding to Patched leads to a signal transduction cascade that activates transcription of specific genes. Mutations in Patched cause basal cell nevus syndrome, which is characterized by developmental abnormalities and basal cell carcinomas (R.L. Johnson, 1996). A role for Patched and dispatched in cholesterol metabolism has not been established; however, it is intriguing that exposure of embryos to inhibitors of cholesterol biosynthesis, such as Triparanol, AY-9944 or BM 15.766, cause profound developmental defects that resemble those in Sonic Hedgehog mutant embryos (J.A. Porter, 1996).

4. Metabolism of cholesterol

Cellular cholesterol levels are regulated, not only by feedback inhibition of cholesterol synthesis, but also by feedforward regulation of cholesterol metabolism. Excess cholesterol is metabolized to oxysterols. In addition to blocking SCAP-facilitated proteolysis of SREBP and thereby down-regulating endogenous cholesterol synthesis and LDL receptor levels, oxysterols also activate bile acid synthesis (discussed in Chapter 16) and cholesterol esterification, which further reduces the cellular content of unesterified cholesterol.

4.1. Oxysterols

Oxysterols are potent suppressors of cholesterol synthesis (A.A. Kandutsch, 1973, 1974). Their effectiveness has been attributed to their ability to diffuse into and through

cells to activate regulatory processes, thus bypassing the need for receptor-mediated entry. It was long assumed that cholesterol was the natural regulator and that oxysterols were contaminants found in commercial supplies of cholesterol or formed upon storage of stock cholesterol solutions. Now it is known that there are many naturally occurring oxysterols that have diverse actions on cellular lipid metabolism (G.J. Schroepfer, Jr., 2000). 25-Hydroxycholesterol is the most studied oxysterol; however, other oxysterols are as, or more, physiologically important.

Oxysterols can be formed by the action of at least three distinct hydroxylases [25]. The mitochondrial sterol 27-hydroxylase participates in an alternative pathway of bile acid biosynthesis, hydroxylating cholesterol and several other intermediates in the bile acid synthetic pathway. 27-Hydroxycholesterol formed in peripheral tissues is a potent inhibitor of endogenous cholesterol synthesis. It is also thought to be secreted into the bloodstream and transported to the liver, where 27-hydroxycholesterol binds the liver X receptor (LXR) nuclear hormone receptor (described in Chapter 16). LXR forms a heterodimer with the retinoid X receptor (RXR) and activates transcription of genes encoding bile acid biosynthetic enzymes. The physiological significance of sterol 27-hydroxylase is illustrated by the genetic disease cerebrotendinous xanthomatosis, which is caused by mutations in the sterol 27-hydroxylase gene [26]. The absence of this critical hydroxylase activity precludes the mobilization of excess cholesterol from peripheral tissues and leads to cholesterol deposition and xanthoma development.

24-Hydroxylase is an endoplasmic reticulum enzyme predominantly expressed in brain. Bjorkhem and colleagues have provided strong evidence that 24-hydroxylase maintains cholesterol homeostasis in the brain, which cannot participate in high-density lipoprotein-mediated reverse cholesterol transport (I. Bjorkhem, 1999). 24-Hydroxycholesterol is secreted from the brain into the circulation, taken up by the liver and metabolized into bile acids.

25-Hydroxylase is an endoplasmic reticulum and Golgi enzyme with low-level expression in most tissues [25]. 25-Hydroxycholesterol is a potent regulator of SREBP proteolytic processing. Given that the enzyme resides in the same subcellular compartment as SREBP and SCAP, 25-hydroxycholesterol may be a physiological regulator of cholesterol synthesis.

Oxysterol binding to LXR activates transcription of several genes that play key roles in maintaining bodily cholesterol homeostasis. One is the gene encoding the ATP-binding cassette transporter ABCA1, a 2201-amino-acid plasma membrane protein that stimulates cholesterol and phospholipid efflux. Cholesterol effluxed from peripheral cells by the action of ABCA1 is transferred by plasma high-density lipoproteins to the liver in a process called reverse cholesterol transport [27] (discussed in Chapter 20). A second example is LXR-activated transcription of the gene encoding cholesteryl ester transfer protein, which promotes transfer of cholesteryl esters from high-density lipoproteins to very low-density lipoproteins for clearance by the liver. Finally, oxysterol binding to LXR also stimulates expression of SREBP-1c, but not SREBP-1a or SREBP-2 (J.J. Repa, 2000). Therefore, increased cellular cholesterol should lead to oxysterol formation, which would increase expression of SREBP-1c and increase fatty acid synthesis. Indeed, administration of an LXR selective agonist to mice led to increased lipogenesis and higher plasma triacylglycerol and phospholipid levels (J.R. Schultz, 2000).

Another protein that binds oxysterols with high affinity was first reported by A.A. Kandutsch et al. (1977) and called oxysterol-binding protein (OSBP). At the time the protein was purified and cDNA-cloned (F.R. Taylor, 1989; P.A. Dawson, 1989), OSBP was expected to be a cytosolic protein that translocated into the nucleus and repressed transcription of cholesterologenic genes when oxysterols were present. Given our current knowledge of transcriptional control, we might expect OSBP to bind to the SREBP precursor or SCAP in the endoplasmic reticulum or site-1 protease or site-2 protease in the Golgi to interfere with SREBP proteolytic processing. However, a direct role for OSBP in transcriptional control has not been demonstrated. OSBP is a high-affinity 25-hydroxycholesterol-binding protein (K_d 10 nM) that translocates from cytosol and vesicles to the Golgi when ligand is bound [28].

How might OSBP transduce signals? One reasonable hypothesis is that when cellular cholesterol levels are high, cholesterol hydroxylation occurs. The resultant oxysterol then binds to OSBP, which translocates to the Golgi and signals suppression of cholesterol synthesis. However, OSBP responds paradoxically to cholesterol rather than oxysterols [28]. When cells are cholesterol replete, OSBP moves to the cytosol and vesicles, not to the Golgi. OSBP moves to the Golgi when cells are cholesterol-depleted. Thus, it has been difficult to establish the identity of OSBP's endogenous ligand and which downstream events are mediated by OSBP. The story is made more complex by the finding that the human OSBP family has at least five members, with sequence similarity in the C-terminal ligand-binding domain, whereas *Saccharomyces cerevisiae* has six related proteins [28].

4.2. Cholesteryl ester synthesis

Excess cholesterol can also be metabolized to cholesteryl esters. ACAT is the endoplasmic reticulum enzyme that catalyzes the esterification of cellular sterols with fatty acids. In vivo, ACAT plays an important physiological role in intestinal absorption of dietary cholesterol, in intestinal and hepatic lipoprotein assembly, in transformation of macrophages into cholesteryl ester laden foam cells, and in control of the cellular free cholesterol pool that serves as substrate for bile acid and steroid hormone formation. ACAT is an allosteric enzyme, thought to be regulated by an endoplasmic reticulum cholesterol pool that is in equilibrium with the pool that regulates cholesterol biosynthesis. ACAT is activated more effectively by oxysterols than by cholesterol itself, likely due to differences in their solubility. As the fatty acyl donor, ACAT prefers endogenously synthesized, monounsaturated fatty acyl-CoA.

The cloning of the human ACAT gene and its orthologs, as well as the subsequent generation of ACAT-deficient mice, led to the realization that two ACAT isozymes must contribute to the enzyme activity (reviewed in Farese [29], Rudel et al. [30] and Chang et al. [31]). Human ACAT-1 was cloned using an expression cloning strategy (C.C.Y. Chang, 1993). The gene encodes an integral membrane protein of 550 amino acids that is present in almost all cells and tissues examined. Orthologs were identified in other mammalian species, as well as *Drosophila melanogaster* and *Caenorhabditis elegans*. The first indication of multiple ACATs came from the cloning of two ACAT-related enzymes (ARE1 and ARE2) from *S. cerevisiae* (H. Yang, 1996). The inactivation of both

yeast genes was required to eliminate sterol esterification. In addition, ACAT-1-deficient mice showed the expected depletion of cholesteryl esters in adrenals, ovaries, testes and macrophages, but no changes in intestinal cholesterol absorption or hepatic cholesterol esterification (V.L. Meiner, 1996). This result indicated that a second ACAT must be present in those mouse tissues.

The cloning of ACAT-2 (R.A. Anderson, 1998; S. Cases, 1998; P. Oelkers, 1998) revealed a protein of similar size to ACAT-1, with a novel N-terminus but a C-terminus highly similar to ACAT-1. In adult humans, ACAT-2 is confined to the apical region of intestinal enterocytes, with low levels also expressed in hepatocytes. Disruption of the ACAT-2 gene in mice led to dramatic reduction in cholesterol absorption and prevention of hypercholesterolemia (A.K.K. Buhman, 2000). The data suggest that, in humans, ACAT-1 plays a critical role in foam-cell formation and cholesterol homeostasis in extrahepatic tissues, whereas ACAT-2 has an important role in absorption of dietary cholesterol [31]. ACAT-1 is the major isozyme in hepatocytes, although the total pool of cholesteryl esters produced by both enzymes regulates very low-density lipoprotein synthesis and assembly [31].

5. Future directions

Fifty years ago, it was recognized that hepatic cholesterol synthesis was subject to feedback regulation by dietary cholesterol (R.G. Gould, 1950). Only in the last decade have the mechanisms been elucidated for transcriptional and degradative regulation of the rate-limiting enzyme, HMG-CoA reductase. Both forms of regulation require that proteins sense the local cholesterol concentration. Rising cholesterol levels cause HMG-CoA reductase to be ubiquitinated and degraded by the proteasome. They cause SCAP to remain localized to the endoplasmic reticulum rather than translocating to the Golgi. The challenge ahead is to determine how HMG-CoA reductase and SCAP transduce the signal of increased cellular cholesterol content into action, i.e. protein degradation or movement to Golgi.

HMG-CoA reductase and SCAP are not the only cellular proteins equipped with a sterol-sensing domain. Does the sterol-sensing domain in 7-dehydrocholesterol Δ^7 -reductase confer cholesterol-mediated feedback regulation upon this last step in the biosynthetic pathway? What is the function of the sterol-sensing domain in NPC1, NPC1L1, Patched and dispatched? Is their subcellular location or their binding to another protein altered by cholesterol? Finally, is sterol-sensing domain-mediated regulation due to the action of cholesterol itself or another biologically active sterol?

Abbreviations

ACAT	acyl-CoA : cholesterol acyltransferase
bHLH-Zip	basic helix-loop-helix leucine zipper
CE	cholesteryl ester
DHC	dehydrocholesterol

ER	endoplasmic reticulum
HMG	3-hydroxy-3-methylglutaryl
LDL	low-density lipoprotein
LXR	liver X receptor
OSBP	oxysterol-binding protein
PP	diphosphate
RXR	retinoid X receptor
S1P	site-1 protease
S2P	site-2 protease
SCAP	SREBP cleavage-activating protein
SLOS	Smith–Lemli–Opitz syndrome
SRE	sterol regulatory element
SREBP	SRE-binding protein

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Metabolism and function of bile acids

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1. Introduction

Bile acids make up a group of sterol-derived compounds that act as detergents in the intestine to facilitate the digestion and absorption of fats and fat-soluble molecules. In mammalian species, the cholesterol side chain is trimmed to yield C24-sterol derivatives. In other vertebrate species, the hydroxylation of the side chain does not lead to its removal and the products of the biosynthetic pathway are referred to as bile alcohols. Invertebrate species do not synthesize sterol bile acids. Over the last few years, much information has been gained about the function of bile acids and the mechanisms that regulate their synthesis. The focus of this chapter is to provide a general overview of bile acid biochemistry and to review recent discoveries that have advanced our understanding of bile acid metabolism and function in mammals.

The concept of bile was developed around the late 1600s to mid 1700s. It was early in the 1800s when bile solutes were crudely isolated. Among the components identified were the amino acid taurine (identified in ox bile, hence its name), cholesterol and a nitrogenous acid. The term 'cholic acid' was initially applied to the acidic component but this was changed to the generic term 'bile acid' shortly after. By the mid 1800s, taurine- and glycine-conjugated bile acids could be distinguished and it was also around this time that the idea that bile acids were responsible for solubilizing cholesterol in bile emerged. Nearly half a century ago, it became evident that bile acids are synthesized from cholesterol [1]. Bile acids are the major solutes in bile. The typical mammalian bile is comprised of about 82% water, 12% bile acids, 4% phospholipids (mostly phosphatidylcholines), 1% unesterified cholesterol and the remaining 1% as assorted solutes (including proteins).

2. Bile acid structure

The structure of bile acids holds the key for their ability to act as efficient detergents. In general, cholesterol is modified by epimerization of the pre-existing 3 β -hydroxyl group, saturation and hydroxylation of the steroid nucleus and trimming of the side chain [2]. Fig. 1 shows the positions of the carbons in the steroid nucleus that are modified during bile acid biosynthesis. Under normal physiological conditions, one or two hydroxyl groups are added to the steroid nucleus. This modification renders the sterol less hydrophobic, enabling it to interact with an aqueous environment more efficiently. The hydroxyl groups of many bile acids are oriented towards one face of the

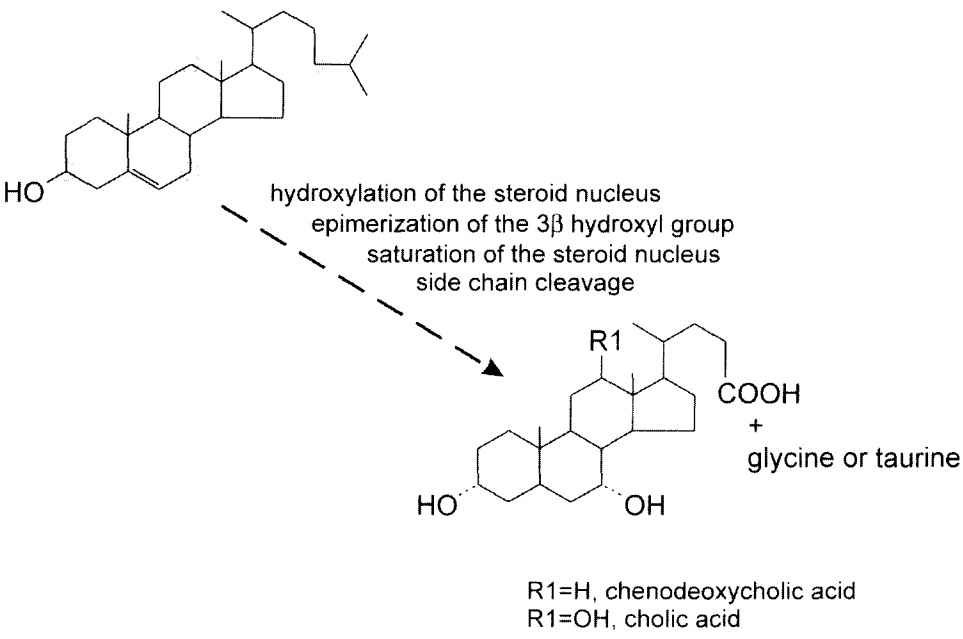


Fig. 1. Conversion of cholesterol into bile acids. The carbons in the cholesterol molecule that are modified during the conversion process are circled.

steroid nucleus giving the molecule an amphipathic character. After trimming 3 carbons from the side chain, the ‘free’ bile acids are covalently linked to one of two amino acids (either taurine or glycine) to form ‘conjugated’ bile acids. Conjugated bile acids readily ionize, allowing these polar molecules to efficiently interact with both hydrophobic and hydrophilic substances (Fig. 2).

The number and specific orientation of the hydroxyl groups added to the steroid nucleus vary according to animal species. Table 1 lists the bile acids that are commonly found in the bile of different mammalian species. Some bile acids, such as cholic and

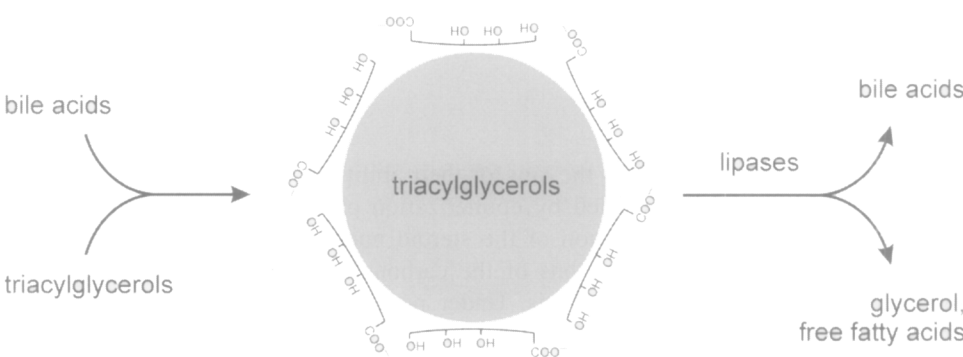


Fig. 2. Interaction of bile acids with triacylglycerols. Lipid-soluble nutrients may be present in the triacylglycerol droplet. Lipases hydrolyze the triacylglycerols to liberate free fatty acids and glycerol.

Table 1
Abundant bile acids found in the bile of selected mammalian species

Bile acid	Position and orientation of hydroxyl groups	Species
Chenodeoxycholic acid	3 α , 7 α	bear, hamster, human, pig
Ursodeoxycholic acid	3 α , 7 β	bear
Deoxycholic acid	3 α , 12 α	cat, human, rabbit
Hyocholic acid	3 α , 6 α , 7 α	pig
β -muricholic acid	3 α , 6 β , 7 β	mouse, rat
Cholic acid	3 α , 7 α , 12 α	bear, cat, hamster, human, mouse, pig, rabbit, rat

chenodeoxycholic acids, are common to many mammalian species whereas others are unique to certain species. Ursodeoxycholic acid, which is abundant in bear bile, has been found to be therapeutically useful for treating primary biliary cirrhosis and dissolving gallstones. It is chemically and biologically distinct from its isomer chenodeoxycholic acid, which differs only in the orientation of the hydroxyl group attached to carbon 7 of the steroid nucleus. Among the mammalian species that are commonly studied in the laboratory, the hamster is the only one that shows a biliary bile acid composition that is comparable to that of humans.

Fig. 3 shows the synthesis of taurine from cysteine via oxidation and decarboxylation reactions. Taurine is very rare in plants but is abundant in animal tissue, particularly in the brain. The bile acids of carnivores are mostly conjugated to taurine whereas those of herbivores are conjugated to glycine. Both taurine- and glycine-conjugated bile acids are found in the bile of omnivores. The bile of cats contains taurine-conjugated

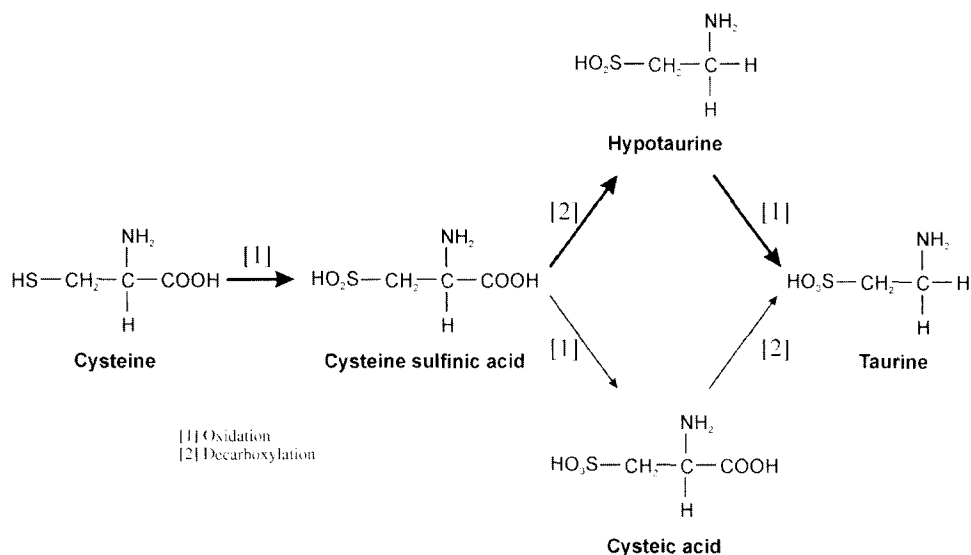


Fig. 3. Synthesis of taurine from cysteine. The major pathway for the formation of taurine is via hypotauroine.

bile acids exclusively. Cats appear to have a requirement for taurine as withdrawal of dietary taurine causes the degeneration of the retina leading ultimately to blindness. Interestingly, taurine deficiency does appear to have any significant consequences in other mammalian species. Conjugated bile acids are more acidic than unconjugated bile acids due to the additional carboxyl group contributed by the amino acid. Consequently, conjugated bile acids readily ionize and exist mainly as bile salts at physiological pH. The functional significance of the choice of amino acid used for conjugation is not clear. Cultured rat hepatoma cells show differential sensitivity to taurine- and glycine-conjugated bile acids [3]. In these cells, glycine-conjugated bile acids are toxic and induce cell death by apoptosis whereas taurine-conjugated bile acids are well-tolerated and even promote cell survival.

3. Biosynthesis of bile acids

Classical studies elucidated the major steps in the bile acid biosynthetic pathway mainly by analyzing the metabolites formed from labeled cholesterol and oxysterols [4]. At least 18 distinct reactions occurring in various subcellular compartments (cytosol, endoplasmic reticulum, mitochondria, and peroxisomes) are necessary to transform cholesterol into bile acids. Reactions involving modifications of the steroid nucleus occur in the endoplasmic reticulum and mitochondria. The removal of the cholesterol side chain involves peroxisomes. Many of the enzymes that catalyze these reactions have been purified, their cDNAs cloned and ectopically expressed in a variety of cultured cell lines. In addition, the impact of overactivity and deficiency of some of these enzymes in the formation of bile acids *in vivo* has been studied through the use of gene therapy, transgenic, and targeted gene disruption techniques.

3.1. The classical and alternative bile acid biosynthetic pathways

The classical pathway operates entirely in the liver (Fig. 4). It begins by α -hydroxylation of carbon 7 of the cholesterol steroid nucleus. This reaction is catalyzed by the microsomal cytochrome *P*-450 monooxygenase referred to as cholesterol 7 α -hydroxylase (cyp7a) and is the rate-limiting step of the classical pathway. Several of the enzymes that participate in the transformation of cholesterol into bile acids belong to the cytochrome *P*-450 family. In general, this class of enzymes catalyzes the hydroxylation of various organic compounds using molecular oxygen as a cosubstrate. The heme-containing monooxygenases recognize specific molecules, or a group of related compounds, and work in concert with NADPH:cytochrome *P*-450 oxidoreductase which supplies electrons for the reactions. Cyp7a shows a high degree of selectivity towards cholesterol. Bile acid output from the liver is correlated with the cyp7a activity, and it is generally considered that the classical pathway is the source of the bulk of the bile acids made by the liver.

The existence of an alternate pathway for the synthesis of bile acids was suspected because it was possible for oxysterols to be converted into bile acids (N. Wachtel, 1968). It is now recognized that a variety of oxysterols produced by an assortment of cell types

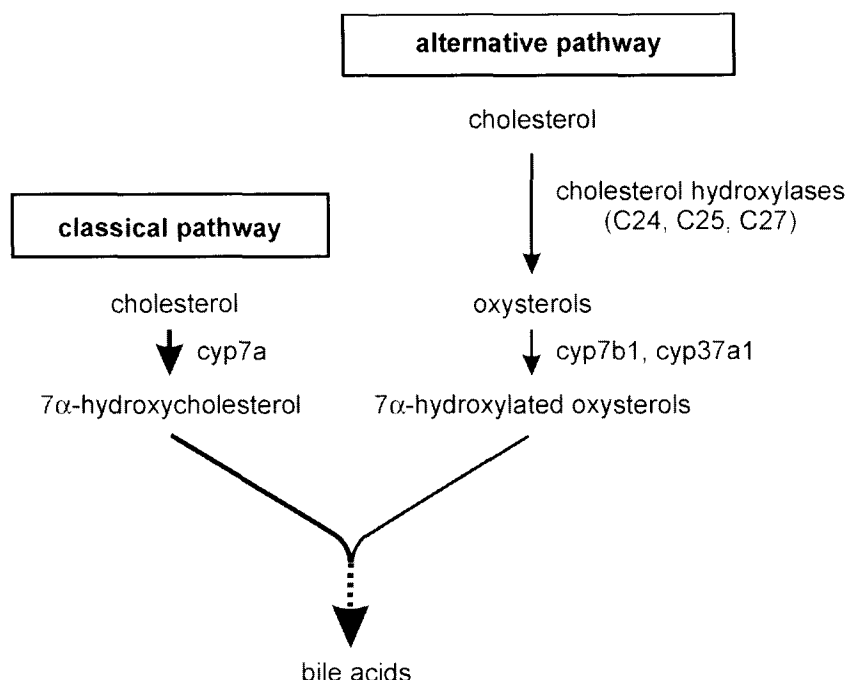


Fig. 4. The bile acid biosynthetic pathways. The classical pathway operates entirely in the liver. In other tissues, the entry of cholesterol into the alternate pathways is facilitated by cholesterol hydroxylases. The oxysterols generated by these enzymes are 7 α -hydroxylated by oxysterol hydroxylases and the products enter the latter steps of the classical pathway.

can be converted into bile acids. The production of these oxysterols is catalyzed by several sterol hydroxylases: sterol 27-hydroxylase (cyp27) (J.J. Cali, 1991), cholesterol 25-hydroxylase (E.G. Lund, 1998) and cholesterol 24-hydroxylase (cyp46) (E.G. Lund, 1999). Cholesterol 25-hydroxylase is not a cytochrome *P*-450 monooxygenase, unlike the two other enzymes. Almost all of the 24-hydroxycholesterol that ends up in the liver originates from the brain, and it has been suggested that the production of this oxysterol is the major mechanism responsible for eliminating excess cholesterol from this organ (I. Bjorkhem, 2001). Cyp27 is also important in the latter stages of bile acid synthesis in the liver, as it is the major enzyme that catalyzes the hydroxylation of the side chain to facilitate the cleavage of the sterol side chain. The oxysterols generated outside the liver are 7 α -hydroxylated, mainly by oxysterol hydroxylases distinct from cyp7a. The cyp7b1 oxysterol 7 α -hydroxylase prefers 25-hydroxycholesterol and 27-hydroxycholesterol, while the cyp37a1 oxysterol 7 α -hydroxylase is selective for 24-hydroxycholesterol (J. Li-Hawkins, 2000). Cyp7a can accept some oxysterols as a substrate, although it largely prefers cholesterol.

The latter steps required to complete the synthesis of bile acids occur only in the liver and are common to both the classical and alternative pathways. Consequently, the liver is the only organ in the body capable of producing bile acids. The isomerization of the 3 β hydroxyl group and the saturation of the steroid nucleus involve 3 β -hydroxy-

Δ^5 -C₂₇-steroid oxidoreductase (K. Wikvall, 1981), 3-oxo- Δ^4 -steroid 5 β -reductase (O. Berseus, 1967) and 3 α -hydroxysteroid dehydrogenase (A. Stolz, 1987). The activities of these enzymes are necessary for the formation of normal bile acids. The enzyme sterol 12 α -hydroxylase (cyp8b1) catalyzes the addition of a hydroxyl group to carbon 12 of the steroid nucleus, and therefore controls the production of the cholic and chenodeoxycholic acids. Changes in the ratio of cholic to chenodeoxycholic acid affect the overall hydrophobicity of the bile acid pool.

The cDNA for an enzyme capable of catalyzing the 6 α -hydroxylation of the steroid nucleus was cloned from pig liver (K. Lundell, 2001). This enzyme, named cyp4a21, is believed to be responsible for the formation of hyocholic acid, a bile acid typically found in porcine bile. The steps leading to the synthesis of β -muricholic acid (a 6 β -hydroxylated bile acid) are less understood [5]. This bile acid appears only in rat and mouse bile. The conversion of lithocholic acid (a 3 α -monohydroxylated bile acid) and chenodeoxycholic acid into β -muricholic acid has been observed, but the identities of the enzymes catalyzing the reactions are not known. Intestinal bacteria are thought to be responsible for modifying the steroid nucleus to form 7 β -hydroxylated bile acids. However it was recently noted that bear liver has the capacity to produce ursodeoxycholic acid, indicating the existence of hepatic enzymes that can catalyze the direct 7 β -hydroxylation of the steroid nucleus or epimerization of the 7 α -hydroxyl group (L.R. Hagey, 1993).

3.2. Mutations affecting key enzymes involved in bile acid biosynthesis

Bile acid synthesis represents a major pathway for cholesterol catabolism. In humans, bile acid excretion can account for the disposal of up to ~ 0.5 g of cholesterol per day. In animal studies, direct stimulation of bile acid synthesis by increasing the abundance of cyp7a enzyme in the liver through gene therapy, reduces the concentration of cholesterol in the plasma (D.K. Spady, 1995, 1998; L.B. Agellon, 1997). It was reasonably expected that inhibiting bile acid synthesis by repression of cyp7a would impair cholesterol catabolism and lead to an increased concentration of plasma cholesterol. In mice, the complete loss of cyp7a function results in the high incidence of neonatal lethality due mainly to inefficient absorption of fats and fat-soluble vitamins [6]. Cyp7a-deficient mice that manage to survive beyond the weaning period synthesize bile acids via the alternative pathway [7] but these mice do not develop hypercholesterolemia [8]. In contrast, a recently discovered mutation in the human *CYP7A1* gene that causes cyp7a deficiency appears to cause hypercholesterolemia (J.P. Kane, 2002). It is not yet known if the loss of cyp7a activity has an effect on human neonatal survival.

Mutations in human cyp27 cause cerebrotendinous xanthomatosis (CTX) (J.J. Cali, 1991). This disorder, which is characterized by neurological defects and premature atherosclerosis, may well be the consequence of sterol accumulation in neural and other tissues [9]. CTX patients have reduced capacity for normal bile acid synthesis but produce large amounts of bile alcohols. This is consistent with the importance of cyp27 in the removal of the cholesterol side chain. Interestingly, deficiency of cyp27 in mice does not elaborate the full complement of defects observed in humans with CTX [10]. The basis for the difference is not completely understood. It has been suggested

that other cytochrome *P*-450 enzymes, cyp3a4 in particular, partially compensate for the missing functions supplied by cyp27 in the murine species (A. Honda, 2001). Indeed, cyp27-deficient mice are still capable of producing normal C24 bile acids but overall bile acid synthesis is markedly diminished. This finding confirms that cyp27 activity is quantitatively important in side chain cleavage, but that hydroxylation of another carbon in the side chain can permit some side chain cleavage to proceed. McArdle RH-7777 rat hepatoma cells are deficient in both cyp7a and cyp27, and no longer possess the capacity to synthesize bile acids. Reinstatement of cyp7a activity enables these cells to synthesize C24 bile acids despite the absence of cyp27 activity (E.D. Labonté, 2000). Cyp27 deficiency causes hypertriglyceridemia and hepatomegaly in mice, indicating that cyp27 function affects other metabolic processes in this species (J.J. Repa, 2000).

The importance of the cyp7b1 oxysterol 7 α -hydroxylase in bile acid synthesis has also been studied in mice. Mice lacking this microsomal enzyme are viable and do not exhibit obvious defects in cholesterol or bile acid metabolism [11]. The notable feature in cyp7b1-deficient mice is the accumulation of 25- and 27-hydroxycholesterol in plasma and cells, suggesting that cyp7b1 is important in the catabolism of these oxysterols into bile acids. In humans, mutations in cyp7b1 results in severe neonatal liver disease characterized by cholestasis (arrest of bile flow) and cirrhosis (damage and scarring of liver tissue resulting from chronic impaired liver function) [12]. Mutations in 3 β -hydroxy- Δ^5 -C₂₇-steroid oxidoreductase and 3-oxo- Δ^4 -steroid 5 β -reductase are also known to cause progressive intrahepatic cholestasis [13–15].

The importance of peroxisomes in the cleavage of the cholesterol side chain during bile acid synthesis is well illustrated in Zellweger syndrome (Chapter 9). This genetic disorder is characterized by peroxisome deficiency and accumulation of large amounts of bile alcohols in the plasma of afflicted patients (R.J. Wanders, 1987). Mice carrying an induced mutation in the *Scp2* gene also accumulate bile alcohols similar to those seen in Zellweger patients [16]. The *Scp2* gene codes for two proteins: the cytosolic sterol carrier protein-2 (SCP2) and the peroxisomal sterol carrier protein-x (SCPx) (Chapter 17). SCPx contains the entire SCP2 sequence plus an N-terminal domain that has a β -ketothiolase activity (U. Seedorf, 1994). The basis for the accumulation of bile alcohols in mice homozygous for a mutant *Scp2* gene is the deficiency in peroxisomal β -ketothiolase activity supplied by SCPx.

4. Transport of bile acids

4.1. Enterohepatic circulation

Bile acids circulate between the liver and intestines via bile and portal blood. The path traced by bile acids between these two organs is depicted in Fig. 5, and is referred to as the enterohepatic circulation [17]. A number of transporters involved in the transport of bile acids have been described [18]. Hepatocytes recover bile acids from portal blood by an active process involving sodium/taurocholate co-transporting polypeptide (ntcp) (B. Hagenbuch, 1991). The recovered bile acids, along with newly synthesized bile

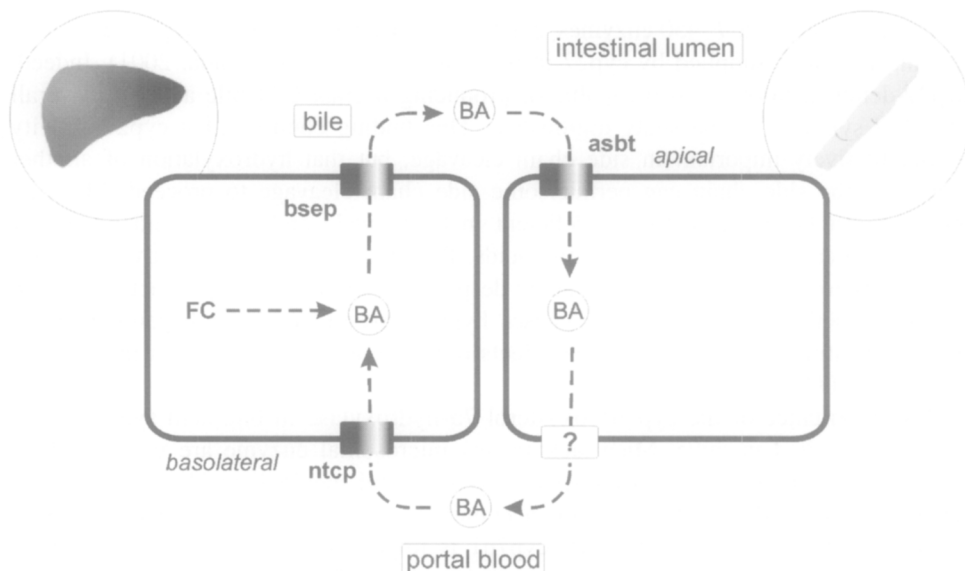


Fig. 5. Transport of bile acids in the enterohepatic circulation. The left and right sides of the figure depict a liver and intestinal cell, respectively. Note that the movement of bile acids in the enterohepatic circulation is vectorial. Abbreviations: asbt, apical/sodium bile acid cotransporter; BA, bile acids; bsep, bile salt export pump; FC, unesterified cholesterol; ntcp, sodium/taurocholate cotransporting polypeptide. The identity of the protein (depicted by '?') mediating the exit of bile acids from the basolateral pole of enterocytes is not yet known.

acids, are secreted into bile via the bile salt export pump (bsep, also known as sister of *p*-glycoprotein) (T. Gerloff, 1998). This protein belongs to the adenosine triphosphate binding (ABC) cassette family of transporters (M. Dean, 2001). Mutations in human bsep are known to cause progressive familial intrahepatic cholestasis type 2 [19]. However, targeted disruption of the murine gene encoding the bsep does not reproduce the human disease phenotype in mice [20]. This finding is another example illustrating a difference between human and murine bile acid metabolism.

The secreted bile acids are stored in the gallbladder prior to being released into the small intestine. An exception occurs in the rat (but not in the mouse), which lacks a gallbladder and thus continuously releases bile into the intestine. The primary bile acids (the products of bile acid biosynthesis in the liver) are metabolized by enteric bacteria to produce deconjugated (i.e., lacking taurine or glycine), and/or dehydroxylated derivatives referred to as secondary bile acids. The secondary bile acids may be further modified by sulfation and/or glucuronidation, but these modifications are not significant under normal physiological conditions. The deconjugated bile acids ('free' bile acids) are absorbed along the entire axis of the intestines. The majority of the conjugated bile acids are recovered in the terminal ileum via an active process involving the apical/sodium bile acid transporter (asbt) (M.H. Wong, 1994). In humans, mutations in asbt cause primary bile acid malabsorption (P. Oelkers, 1997). The identity of the intestinal bile acid exporter is not yet established.

As already mentioned, the major lipids found in bile are bile acids, phospholipids (mainly phosphatidylcholines, PC) and unesterified cholesterol. The solubility of cholesterol in bile is dependent upon the ratio of these lipids. The acyl chain composition of biliary PC (predominantly C16:0 at the *sn*-1 position and either C18:1 or C18:2 at the *sn*-2 position) differs from that normally found in bulk cell membranes (predominantly C18:0 at the *sn*-1 position and C20:4 at the *sn*-2 position). It is now known that the secretion of PC into bile requires a canalicular membrane protein referred to as *mdr2* (an ABC-type transporter encoded by the *Abcb4* gene in mice). Mice that are deficient in *mdr2* have a very low concentration of PC in bile (J.J.M. Smit, 1993). The secretion of bile acids into bile is not affected by *mdr2* deficiency. However, cholesterol concentration in the bile of *mdr2*-deficient mice is diminished, indicating that the secretion of cholesterol into bile is dependent on biliary PC. Mutations in MDR3 (the human equivalent of *mdr2*) cause progressive familial intrahepatic cholestasis type 3 (J.M. De Vree, 1998). It was recently proposed that *abca1* (Chapter 20), another ABC-type transporter, is involved in the cellular efflux of cholesterol. However, it is not yet clear if this transporter is directly responsible for mediating the transport cholesterol. *Abca1* is found in a variety of organs including the liver. It remains to be determined if this protein is found in canalicular membrane of hepatocytes.

4.2. Intracellular transport

The mechanism for the intracellular transport of bile acids is less understood than the uptake and secretion of bile acids by liver and intestinal cells. Several intracellular proteins capable of binding bile acids have been identified but it is not yet clear if these proteins are involved in the transcellular transport of bile acids [18]. The best candidate protein in the intestine is the ileal lipid binding protein (ilbp). This protein, a member of the intracellular lipid binding protein family (A.V. Hertz, 2000), is abundantly expressed in the distal portion of the small intestine where *asbt* is found. It has been suggested that ilbp and *asbt* interact to form a macromolecular bile acid transport system in intestinal cells [21]. The protein providing the equivalent function in liver cells is not known. The liver-fatty acid binding protein does not bind bile acids efficiently. High level expression of the human bile acid binder in hepatoma cells capable of active bile acid uptake does not appear to influence bile acid transport [3].

5. Molecular regulation of key enzymes in the bile acid biosynthetic pathways

Bile acid synthesis is modulated by a variety of hormonal and nutrient factors. Alterations in bile acid metabolism have been documented in response to thyroid hormones, glucocorticoids and insulin. It is also well known that cholesterol and bile acids have opposite effects on the activity of the bile acid biosynthetic pathway (Fig. 6). A major advance into the understanding of the mechanisms that regulate bile acid synthesis came with the cloning of the rat *cyp7a* cDNA, which permitted the expression of the *cyp7a*

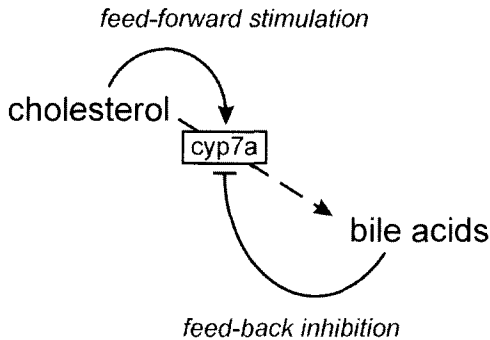


Fig. 6. Regulation of bile acid synthesis.

gene to be monitored at the molecular level [22]. Many of the details relating to the molecular mechanisms involved in regulating bile acid synthesis have been elucidated using both cultured cells and genetically modified mouse strains.

Feeding rats with a cholesterol-enriched diet induces bile acid synthesis. This increase is attributable to the rise in cyp7a activity, which catalyzes the rate-limiting step of the classical pathway (Fig. 6). Interrupting the return of bile acids to the liver, by diverting bile or by feeding a bile acid-binding resin, also stimulates the synthesis of bile acids. In contrast, reintroduction of bile acids into bile-diverted rats reverses the stimulatory effect, indicating that bile acid synthesis is subject to end-product inhibition. It was later discovered that cyp7a enzyme activity is closely correlated with cyp7a mRNA abundance, indicating that cyp7a gene transcription is the major determinant of cyp7a activity.

In many of the early studies, crystalline cholesterol was added directly to the standard rodent chow and fed to the animals. Although this experimental condition was useful in illustrating the stimulation of the cyp7a gene in response to dietary cholesterol, it does not normally exist in nature. The use of semi-purified diets has revealed that the composition of the fat in which cholesterol is presented has a marked influence on the ability of cholesterol to regulate cyp7a gene expression [23]. It has also become apparent that the fat component of the diet is capable of stimulating murine *Cyp7a1* gene expression, independent of exogenous cholesterol.

The regulation of the classical and alternative bile acid biosynthetic pathways has been studied mostly in mice and rats. The data indicate that the classical pathway is under stringent regulation, with much of the control exerted on the cyp7a gene. In contrast, the alternative pathway appears to operate constitutively. There is also emerging evidence indicating that the synthesis of bile acids in humans is only moderately regulated, unlike that in mice and rats.

5.1. Transcriptional control

Several transcription factor binding sites have been mapped in the cyp7a gene promoter, and many of these bind transcription factors that are members of the nuclear receptor superfamily (Table 2). Some of these receptors, notably the liver x receptor α (LXR α)

Table 2
Transcription factors shown to have functional interaction with the *cyp7a* gene promoter

Transcription factor	Cyp7a gene
BTEB	rat
C/EBP β	rat
COUP-TFII (ARP-1)	rat
DBP	rat
HNF-1	human
HNF-3	human, hamster, rat
HNF-4	human, hamster, rat
LRH-1 (also known as CPF, FTF)	human, rat
LXR α	rat
PPAR α	mouse
TR α and TR β	human

The data in this table are compiled from studies published by academic (L.B. Agellon, J.Y. Chiang, A.D. Cooper, E. De Fabiani, G. Gil, D.J. Waxman, U. Schibler) and pharmaceutical (Tularik Inc., Glaxo-Wellcome Research and Development) laboratories.

[NR1H3]) and peroxisome proliferator-activated receptor α (PPAR α [NR1C1]), bind to their target elements as heterodimers with retinoid x receptor α (RXR α [NR2B1]).

The stimulation of *cyp7a* gene expression by cholesterol involves LXR α (Fig. 7),

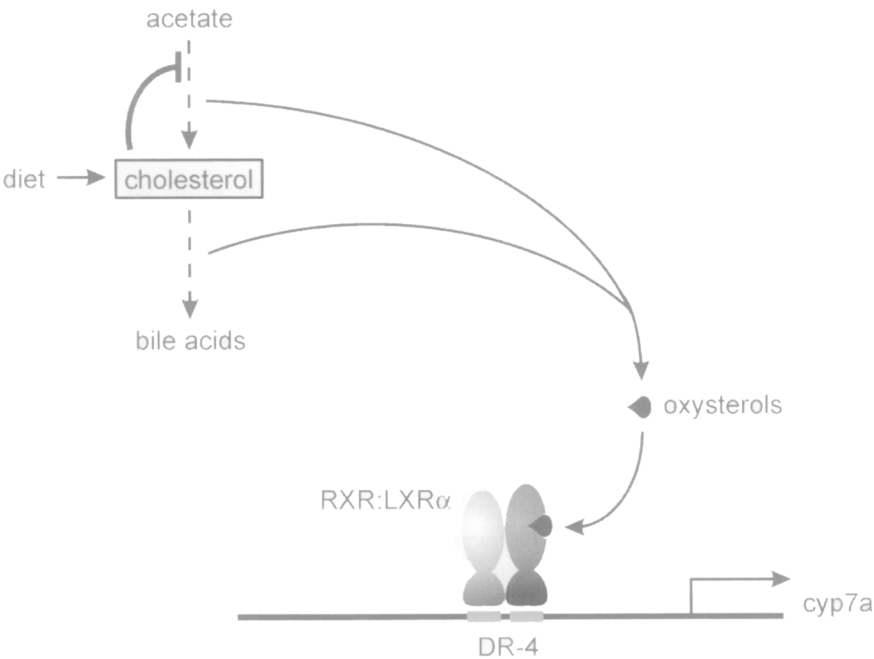


Fig. 7. Induction of *cyp7a* gene expression by oxysterol-activated LXR α :RXR α . Both the bile acid and cholesterol biosynthetic pathways generate oxysterols. The binding site of LXR α :RXR α in the *cyp7a* gene promoter is a DR-1 (a direct repeat of the hexanucleotide hormone response element separated by 4 nt).

an oxysterol-activated transcription factor (D.J. Mangelsdorf, 1996). In cultured cells, induction of the rat *Cyp7a1* gene promoter by oxysterols is dependent on LXR α (J.M. Lehman, 1997). In LXR α -deficient mice, the *Cyp7a1* gene is no longer induced by cholesterol feeding [24]. The oxysterols that serve as potent ligands for LXR α are likely generated by the early steps in the alternative bile acid biosynthetic pathway (i.e., 25-hydroxycholesterol and 27-hydroxycholesterol), and by the cholesterol biosynthetic pathway (i.e., 24(S),25-epoxycholesterol). It is notable that 24(S),25-epoxycholesterol is also capable of repressing 3-hydroxy-3-methyl-glutaryl coenzyme A reductase activity (T.A. Spencer, 1985).

Fatty acids and their metabolites can stimulate the murine *Cyp7a1* gene promoter via PPAR α : RXR α in hepatoma cells [25]. Interestingly, LXR α : RXR α and PPAR α : RXR α heterodimers bind to overlapping regions in the murine *CYP7a1* gene promoter. It is currently not known how these transcription factors interact with the *cyp7a* gene promoter when both are simultaneously activated, or whether this is relevant to the earlier finding that the type of fat in the diet influences the response of the murine *Cyp7a1* gene to dietary cholesterol. The corresponding region of the human *CYP7A1* gene promoter does not interact with either PPAR α : RXR α or LXR α : RXR α . In transgenic mice, the human *CYP7A1* gene is not stimulated by cholesterol feeding (L.B. Agellon, 2002). A differential interaction of thyroid hormone receptors (TR α [NR1A1] and TR β [NR1A2]) with the *cyp7a* gene promoters of different species has also been documented [26]. The human *CYP7A1* gene promoter binds and is inhibited by the thyroid hormone receptor. In contrast, the murine *Cyp7a1* gene promoter does not interact with the thyroid hormone receptor. These differences may indicate that the *cyp7a* gene promoters of different organisms are configured to respond to regulatory cues relevant to each species.

An indirect mechanism for the inhibition of *cyp7a* gene expression by bile acids has been proposed (Fig. 8). The liver receptor homolog protein-1 (LRH-1 [NR5A2]) is a monomeric orphan nuclear receptor bound to *cyp7a* gene promoter to enable expression in the liver (M. Nitta, 1999). It was recently discovered that bile acids are the physiological ligands of the farnesoid x receptor (FXR [NR1H4]), another transcription factor belonging to the nuclear receptor superfamily (M. Makashima, 1999; B.M. Forman, 1999). In the liver, FXR stimulates the expression of the gene encoding the nuclear factor known as small heterodimer partner (SHP; NR0B2) [27,28]. The interaction of SHP with LRH-1 renders the *cyp7a* gene promoter insensitive to stimulation by other transcription factors. However the proposed model (Fig. 8) cannot account for some observations. For example, the *Cyp7a1* gene is resistant to inhibition by bile acids in FXR-deficient mice (C.J. Sinal, 2000) even though it has been suggested that bile acids can stimulate SHP gene expression via an alternative mechanism involving the JNK/c-Jun pathway (J.H. Miyake, 2000; S. Gupta, 2001). Furthermore, feeding a diet containing both cholesterol and bile acids does not abolish *cyp7a* gene expression [29].

Bile acids inhibit the expression of the *cyp8b1* gene (encodes the sterol 12 α -hydroxylase) in parallel with the *cyp7a* gene (Z.R. Vlahcevic, 2000). Suppression of *cyp8b1* gene expression is likely mediated through SHP, as LRH-1 is also required for *cyp8b1* promoter activity (A. del Castillo-Olivares, 2000). The expression of the

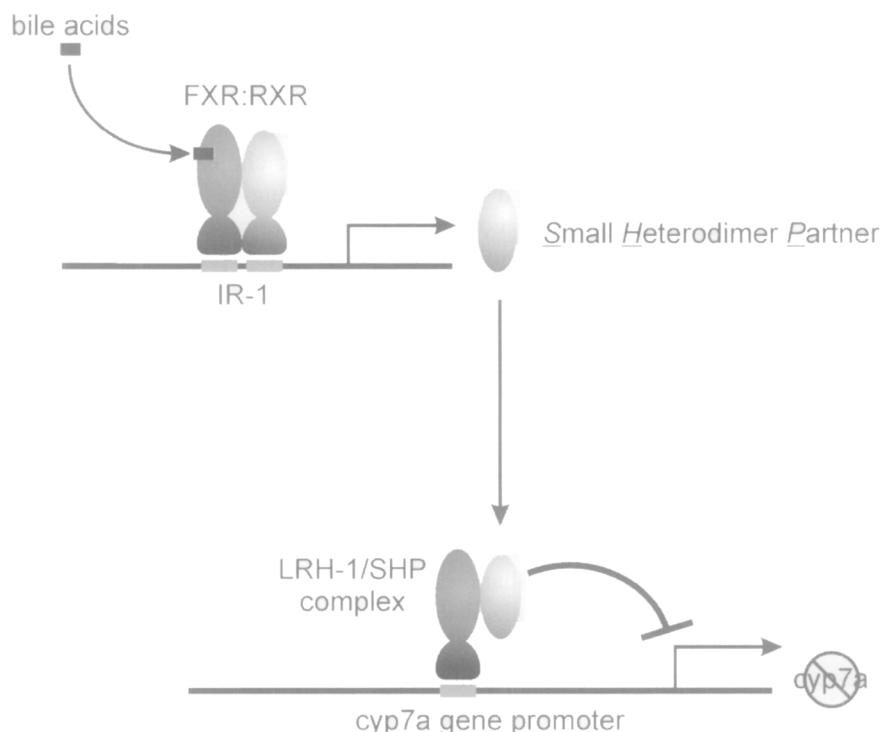


Fig. 8. Repression of *cyp7a* gene expression by bile acids. The liver receptor homolog-1 (LRH-1) binds to the *cyp7a* gene promoter to enable expression in the liver. A bile acid-activated FXR:RXR heterodimer binds to an IR-1 (an inverted repeat of the hexanucleotide hormone response element separated by 1 nt) in the promoter of the gene that encodes the small heterodimer partner (SHP) and stimulates its expression. Binding of SHP to LRH-1 arrests the expression of the *cyp7a* gene.

rat *Cyp8b1* gene is also inhibited by thyroid hormone, but it is not clear whether this effect involves the interaction of the thyroid hormone receptor with the rat *cyp8b1* gene promoter (U. Andersson, 1999). Moreover thyroid hormones, unlike bile acids, exhibit opposite effects on the expression of *Cyp8b1* and *Cyp7a1* genes in rats.

5.2. Post-transcriptional control

The majority of the studies described in the literature dealing with the regulation of bile acid synthesis have focused on *cyp7a* and it is apparent that most of the control is exerted at the level of gene transcription. The *cyp7a* mRNA has a short half-life, and this is attributable to the existence of multiple copies of the AUUUA motif in its 3'-untranslated region. However, some bile acids can further accelerate the decay of chimeric mRNAs containing the 3'-untranslated region of the murine *cyp7a* mRNA in hepatoma cells, and this effect is independent of the AUUUA element [30]. The regulation of *cyp7a* enzyme activity by phosphorylation/dephosphorylation has been suggested. However the results obtained by several studies are conflicting and the topic remains controversial.

6. *Future directions*

The well-known function of bile acids is to aid in the digestion and absorption of lipids and lipid-soluble nutrients in the intestine. Since bile acids are synthesized from cholesterol, they are also regarded as the terminal products of cholesterol catabolism. Research in the past decade has provided a new understanding of the physiological importance of bile acids. Bile acids are now known to be active regulators of cellular processes, such as signal transduction, by influencing the activity of proteins involved in signaling cascades, and gene expression by influencing the turnover of specific mRNA species as well as by serving as the natural ligand for the nuclear receptor FXR.

The synthesis of bile acids is under tight control, with both substrate and end-product actively participating in the regulatory process. The enzymes involved in bile acid synthesis are coordinately regulated with the proteins that transport bile acids. Much of the control is exerted at the level of transcription. It will be of interest to determine how bile acid metabolism is integrated into other processes, such as reverse cholesterol transport and perhaps fat metabolism. The use of targeted gene disruption technology to generate specific mutations in the bile acid biosynthetic pathway has been highly useful in uncovering new components and evaluation of their relative importance. It should be realized that there are already recognized differences in the way standard mouse strains metabolize bile acids. Nevertheless, surprising new information has emerged from studies employing engineered mouse strains. It has become apparent that gender dimorphism exists with respect to several components of bile acid metabolism in the mouse. In addition, these studies reveal that the metabolism of bile acids in humans and mice may be more dissimilar than previously believed. It will be important to explain how these differences arise. Significant advances have also been gained in the area of bile acid transport. Proteins that mediate the passage of bile acids across cell membranes have been identified, although some still await discovery. The transport of bile acids within cells remains poorly understood but this should become clear in the coming years.

Abbreviations

cyp7a	cholesterol 7 α -hydroxylase
cyp7b1	oxysterol 7 α -hydroxylase
cyp8b1	sterol 12 α -hydroxylase
cyp27	sterol 27-hydroxylase
FXR	farnesoid x receptor
LXR α	liver x receptor α
LRH-1	liver receptor homolog-1
PC	phosphatidylcholine
PPAR α	peroxisome proliferator-activated receptor α
SHP	small heterodimer partner
RXR	retinoid x receptor

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Lipid assembly into cell membranes

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1. Introduction

A fundamental problem of cell biology and biochemistry is the elucidation of the mechanisms by which the specific components of subcellular membranes are assembled into mature organelles. The major components of all cell membranes are lipids and proteins. The presence of discrete structural motifs contained in the primary sequence of proteins directs a large number of post-translational processes that enable their sorting among different membrane compartments [1]. The sorting process for proteins is essentially absolute such that plasma membrane proteins are never found in the mitochondria or vice versa. In contrast, lipid molecules do not contain discrete structural subdomains that exclusively direct their movement to specific membranes. The distribution of lipids among different organelles is heterogeneous, but (with a few exceptions) is not usually absolute. These observations about lipids indicate that specialized sorting and transport machinery must exist for their assembly into different membranes.

2. The diversity of lipids

A multiplicity of individual lipids can contribute to membrane formation. The biological role of this lipid heterogeneity is not completely defined and the list of significant actions continues to grow. Some of the diversity contributes to membrane fluidity. Other roles for lipid diversity are the storage of precursors that are metabolized to potent second messengers (e.g., diacylglycerol, ceramide, sphingosine, inositol trisphosphate and eicosanoids) (see Chapters 12, 13 and 14). In addition, many of the polyphosphoinositide (PI3P, PI4P, PI45P₂) function as membrane recognition and attachment sites for protein complexes involved in protein traffic and membrane fusion events [2]. Multiple anionic lipids (PS and polyphosphoinositides) can also regulate attachment of cytoskeletal proteins to membranes. Segregated domains of cholesterol and sphingolipid form microdomains or 'rafts' with unusual physical properties that contribute to protein sorting and are enriched in specific subsets of membrane proteins (see Chapter 1).

In addition to the large numbers of chemically distinct lipid species that occur within a prokaryotic or eukaryotic cell, there is another level of complexity, i.e., the asymmetric

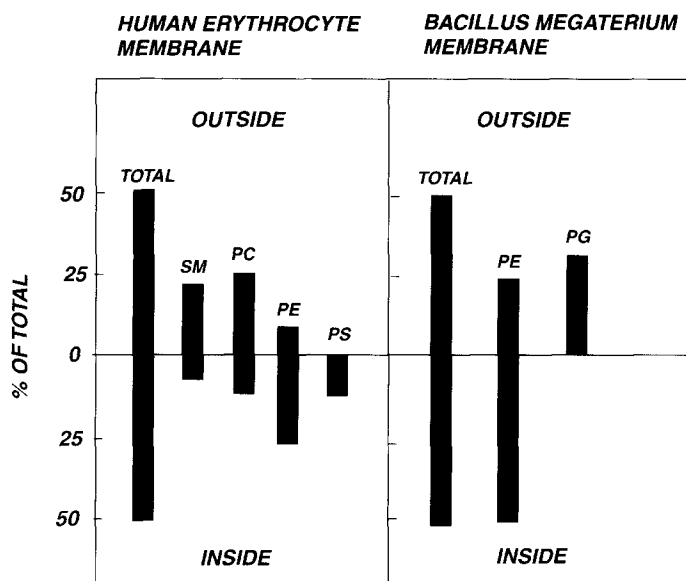


Fig. 1. The asymmetric distribution of lipids across the plane of the cell membrane of the human erythrocytes and *Bacillus megaterium*.

distribution of the lipids across the plane of the bilayer. Two striking examples of membrane lipid asymmetry are found in the red blood cell membrane [3], and the cytoplasmic membrane of *Bacillus megaterium* [4]. The data in Fig. 1 demonstrate that in the red cell membrane the outer leaflet of the lipid bilayer is composed primarily of sphingomyelin (SM) and phosphatidylcholine (PC), and the inner leaflet contains phosphatidylserine (PS) and phosphatidylethanolamine (PE), with lesser amounts of PC and SM. Relatively small amounts of phosphatidylinositol (PI) and its phosphorylated derivatives are also found in the erythrocyte membrane and these anionic lipids are distributed such that the majority is localized to the inner leaflet of the bilayer. In the prokaryote *B. megaterium* the distribution of PE has been shown to be asymmetric, with 30% of this lipid present on the outer leaflet of the bilayer and 70% on the inner leaflet. PE comprises about 70%, and phosphatidylglycerol (PG) about 30%, of the total phospholipid. Thus, nearly all the PG is in the outer leaflet of the bilayer.

Yet another level of complexity is found in cells that possess multiple membrane systems. The Gram-negative bacteria have both an inner and outer membrane system that differ in lipid composition (Chapters 1 and 3). In animal eukaryotes there are numerous membrane systems, the best characterized being endoplasmic reticulum (ER), Golgi membranes, plasma, mitochondrial, lysosomal, peroxisomal and nuclear membranes. In higher plants the eukaryotic organelle repertoire is expanded to include chloroplasts and other plastids, vacuoles and glyoxysomes (Chapter 4). Several of these membrane systems have dramatically different lipid compositions, as shown in Table 1 (A. Colbeau, 1971; T.W. Keenan, 1970). These differences in lipid content raise a variety of interesting questions: How are the different lipid compositions of different

Table 1
Lipid compositions of subcellular organelles from rat liver

Phospholipid ^a	Endoplasmic reticulum		Mitochondrial membranes		Lysosomal membrane	Nuclear membrane	Golgi membrane	Plasma membrane
	Rough	Smooth	Inner	Outer				
Lysophosphatidylcholine	2.9	2.9	0.6	–	2.9	–	5.9	1.8
Sphingomyelin	2.4	6.3	2.0	2.2	16.0	6.3	12.3	23.1
Phosphatidylcholine	59.6	54.4	40.5	49.4	41.9	52.1	45.3	43.1
Phosphatidylinositol	10.1	8.0	1.7	9.2	5.9	4.1	8.7	6.5
Phosphatidylserine	3.5	3.9	1	1	–	5.6	4.2	3.7
Phosphatidylethanolamine	20.0	22.0	38.8	34.9	20.5	25.1	17.0	20.5
Cardiolipin	1.2	2.4	17.0	4.2	–	–	–	–
Phospholipid/protein (μ mole P/mg)	0.33	0.47	0.34	0.46	0.21	–	–	0.37
Cholesterol molar ratio	0.07	0.24	0.06	0.12	0.49	–	0.152	0.76
Phospholipid								

Values for individual lipids are percentage of total phospholipid phosphorus.

organelles established? How are these differences maintained? Are the different lipid compositions essential for organelle function?

3. Methods to study intra- and inter-membrane lipid transport

3.1. Fluorescent probes

Pagano [5] and coworkers pioneered methods for the rapid insertion of fluorescent phospholipid analogs from liposomes into the plasma membranes of cultured cells. Virtually all of these analogs exhibit slight water solubility and high hydrophobic partitioning coefficients that enable them to be efficiently and reversibly transferred to cell membranes at low temperature from liposomes or albumin complexes containing the fluorescent lipid. A commonly used fluorochrome is nitrobenzoxadiazole (NBD) which is conjugated to short-chain fatty acids in the *sn*-2 position of glycerophospholipids or on the amine of sphingosine. Other analogs such as boron dipyrromethene difluoride (BODIPY) derivatized fatty acids have proved equally effective. Non-fluorescent molecules containing short-chain sphingosines or fatty acids such as diC8 SM or diC8 PC have similar physical properties to the fluorescent molecules, and have also proved to be important probes. The structures of some of these analogs are shown in Fig. 2. Subsequent to the insertion of these lipid analogs into cell membranes, the cells can be washed at low temperature to remove the donor liposomes or albumin complexes. In almost all cases this procedure results in the pulse labeling of the outer leaflet of the plasma membrane with the lipid analog. The lipid analogs can also be removed from the outer leaflet of the plasma membrane at reduced temperature by washing cells with a solution that contains liposomes (e.g., composed of dioleoyl-PC) or albumin. When the lipid analogs are fluorescent their intracellular movement can be observed

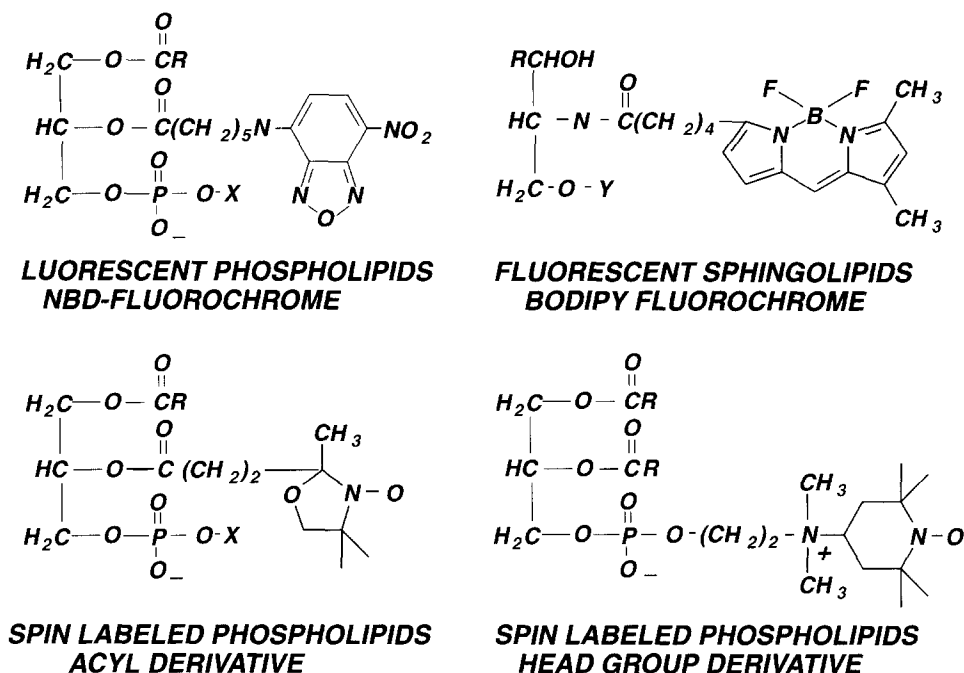


Fig. 2. General structural features of fluorescent and spin-labeled lipid analogs. The fluorescent lipids contain a short-chain fatty acid, amino-caproic acid, that is derivatized with 4-nitrobenzo-2-oxa-1,3-diazole (NBD), or valeric acid that is derivatized with a boron dipyrromethene difluoride (BODIPY) moiety. For fluorescent phospholipids, X can be hydrogen, or the esterified forms of choline, ethanolamine, serine, or inositol. For fluorescent sphingolipids, Y can be hydrogen, or the esterified forms of phosphocholine or glucose. The spin-labeled lipids modified in the fatty acid portion contain a 4 doxylpentanoyl fatty acid in the *sn*-2 position. Those modified in the polar head group contain a tempocholine moiety in place of choline. The X substituent for the acyl spin-labeled lipids can be hydrogen, or the esterified forms of choline, ethanolamine or serine.

by fluorescence microscopy. A simplified outline of the use of these fluorescent lipids is shown in Fig. 3. In addition to their utility for examining the fluorescence pattern within cells, these lipids can be extracted from cells and their chemical metabolism analyzed using thin-layer chromatography or liquid chromatography/mass spectrometry. The fluorescent sterols dehydroergosterol and NBD-cholesterol are also available for *in vivo* studies. These sterols can be delivered to cells in reconstituted lipoproteins or from complexes with methyl β -cyclodextrin (E. Kilsdonk, 1995).

3.2. Spin-labeled analogs

Paramagnetic analogs of phospholipids have also been used to investigate lipid transport phenomena in model membrane systems (R.D. Kornberg, 1971) and in biological membranes. Representative structures are shown in Fig. 2. Several of these spin-labeled lipid analogs that are modified in the fatty acid chain can be readily and reversibly transferred from the bulk aqueous phase to biological membranes, in much the same way

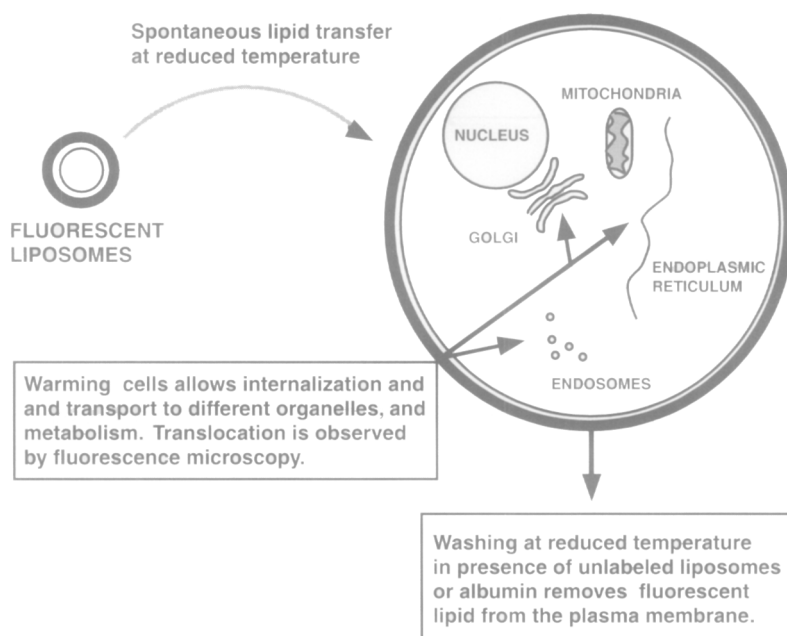


Fig. 3. Fluorescent labeling of living cells with lipid analogs. The heavy line represents the fluorescent phospholipid. Incubation of liposomes containing fluorescent lipid with eukaryotic cells at temperatures of 2–7°C results in the spontaneous transfer of fluorescence to the outer leaflet of the plasma membrane. The fluorescent lipid remains in the plasma membrane at low temperatures and can be reversibly removed by washing the cells with unlabeled liposomes or albumin solutions. Warming the cells to 37°C or intermediate temperature results in the internalization of phospholipid and subsequent labeling of intracellular organelles that can be monitored by fluorescence microscopy.

as the fluorescent lipid analogs. Since the amplitude of the ESR spectra is proportional to the amount of spin-labeled lipid present, these analogs can be used to measure the depletion or retention of the lipids (M. Seigneuret, 1984). In a typical experiment, an intact red cell is incubated with trace amounts of spin-labeled phospholipid at reduced temperature. This treatment effectively pulse labels the outer leaflet of the plasma membrane. Upon warming, the spin-labeled lipids can either remain in the outer leaflet of the plasma membrane or be internalized. If the cells are subsequently cooled and incubated in the presence of ascorbate, the ESR signal of lipid present in the outer leaflet (but not the inner leaflet) of the plasma membrane is quenched and the spectral difference can be used to determine both the rate and extent of transbilayer movement.

3.3. Asymmetric chemical modification of membranes

One method for ascertaining the distribution of lipids across the plane of the membrane bilayer is the use of membrane-impermeant reagents that react with the primary amines of PS and PE on only the external leaflet of the bilayer. Reagents such as trinitrobenzenesulfonate (TNBS) and isethionylacetimidate (IAA) (Fig. 4) are impermeant at reduced temperatures [6] and the chemically modified lipids can be readily identified

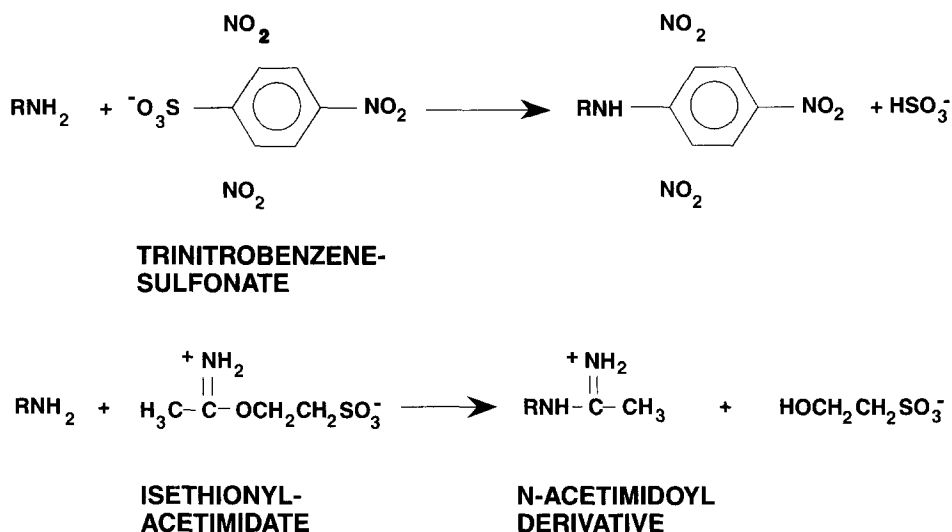


Fig. 4. Primary amine modifying reagents. Phospholipids containing the primary amines PE and PS can be modified by treatment with either trinitrobenzenesulfonate or isethionylacetimidate yielding the *N*-trinitrophenyl derivative or the *N*-acetimidoyl derivative.

by thin-layer chromatography. When such reagents are used in conjunction with *in vivo* radiolabeling of the lipid, it is possible to discern the temporal and metabolic conditions required for the newly synthesized lipids to reach the compartment that is accessible to the chemical modifying reagents. A useful variation of this approach combines chemical reduction of NBD phospholipids with dithionite to eliminate fluorescence (J.C. McIntyre, 1991). When this latter technique is employed with fluorescence microscopy or spectrofluorometry, it can be extremely informative for resolving questions about transbilayer topology.

Specific pools of lipids on the external surface of cells can also be modified by the action of enzymes such as phospholipases, sphingomyelinases [3], and cholesterol oxidase (Y. Lange, 1985). These enzymes also generate characteristic derivatives of the parental lipids that can be readily identified by thin-layer chromatography and this approach provides another technique for identifying specific pools of lipid on the external surface of the cell membrane. Another means to sample lipids at membrane interfaces relies upon specific chemical desorption. The interaction between sterols and methyl β -cyclodextrin provides a high affinity interaction that can be used to selectively remove (or deliver) cholesterol and dehydroergosterol to membranes (E. Kilsdonk, 1995).

3.4. Phospholipid transfer proteins

In 1968, K.W. Wirtz identified a soluble intracellular protein derived from rat liver that was capable of binding PC and transferring it from one population of (donor) membranes to a second population of (acceptor) membranes (K. Wirtz, 1968). Since

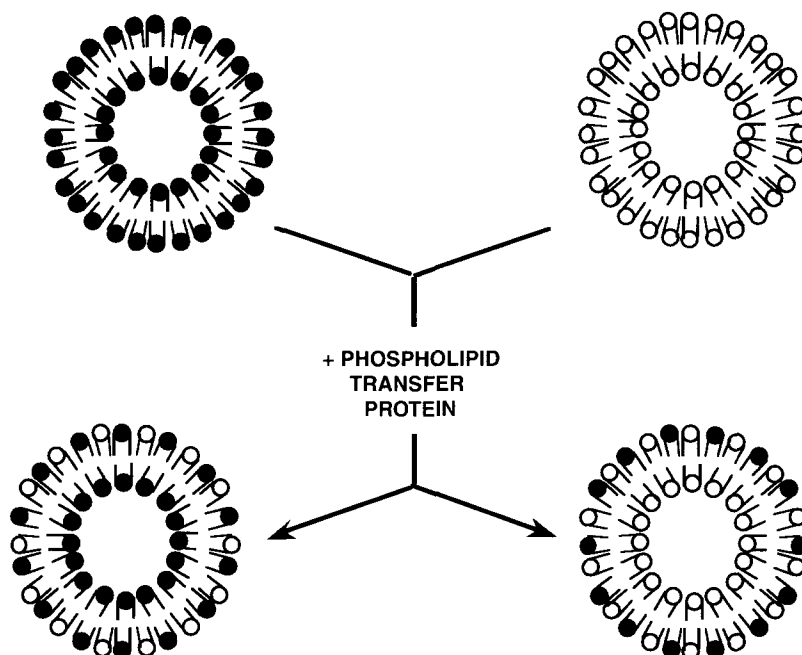


Fig. 5. The action of phospholipid transfer proteins. Mixing of equivalent populations of labeled (denoted in black) and unlabeled liposomes (denoted in white) with phospholipid transfer protein leads to the transfer of lipid between outer leaflets. In the absence of transbilayer movement of the lipid, only the outer leaflets equilibrate with each other.

this initial observation, many of these proteins have been identified in virtually all mammalian tissues, in plants, and in yeast and other microorganisms [7]. The well characterized phospholipid transfer proteins fall into three main categories: (1) those specific for PC; (2) those with high activity for PI and less, but significant, activity with PC, and in some cases SM (J. Westerman, 1995); and (3) those with transfer activity with most phospholipids and cholesterol (this latter protein is referred to as the nonspecific lipid transfer protein). In addition to the phospholipid transfer proteins, there are also intracellular proteins with high transfer activity for sphingolipids (T. Sasaki, 1990). The action of these proteins is typically a one for one exchange of lipid molecules between donor and acceptor membranes. As shown schematically in Fig. 5, the transfer proteins typically equilibrate the lipid present in the outer leaflets of liposomes. The ability of these proteins to transfer lipids from accessible membrane compartments has made them useful tools for inserting lipids into, or removing them from, membranes and probing the transbilayer movement of phospholipids. The role of these proteins in membrane biogenesis will be discussed in Section 4.3.2.6.

3.5. Rapid plasma membrane isolation

One approach to sampling the arrival of newly synthesized lipids at the cell surface utilizes a rapid plasma membrane isolation technique (R.F. DeGrella, 1982). In this

approach, the intact cells are adsorbed onto cationic beads at reduced temperature. After adsorption, the cells are lysed by brief sonication, which liberates the majority of intracellular organelles. The density of the beads containing adsorbed plasma membrane allows them to be separated from the intracellular organelles by low-speed centrifugation. Subsequent to this isolation procedure, the lipids present in the membrane can be extracted and analyzed. When this method is used in combination with radiolabeling of the intracellular pool, the characteristics of the processes required for movement of the lipid from within the cell to the plasma membrane can be determined. A related approach has employed an iron-derivatized wheat germ agglutinin to bind the external surface of the plasma membrane (D. Warnock, 1993) that can be selectively isolated using high-intensity magnetic fields.

3.6. Organelle specific lipid metabolism

For a few lipids, distinct changes in structure also serve to define the arrival at certain organelles or their subcompartments. The enzyme PS decarboxylase is located at the inner mitochondrial membrane of mammalian cells (L.M.G. van Golde, 1974). The synthesis of PS, however, occurs primarily in the ER and related membranes. Thus the decarboxylation of PS can be used as an indicator of the transport of this lipid to the inner mitochondrial membrane [8]. Yeast also contain a mitochondrial PS decarboxylase (PSD1), and in addition, a second enzyme (PSD2) is found in the Golgi and vacuoles. Mutations in the *PSD1* or *PSD2* genes of yeast make it possible to use PS metabolism to PE as an index of lipid transport to the locus of either the mitochondria or the Golgi–vacuolar compartment. In yeast, the movement of PE (derived from either PSD1 or PSD2) to the ER can also be followed by measuring its methylation to PC, since the methyltransferases are only present in the ER. Site-specific metabolism also occurs for sphingolipids (Chapter 14). Ceramide, the hydrophobic precursor for all sphingolipids, is synthesized in the ER. The formation of SM from ceramide (Cer) occurs at the luminal surface of the *cis*-medial Golgi (A.H. Futerman, 1990; D. Jeckel, 1990). Thus, SM synthesis from Cer can be used to follow ceramide transport from ER to Golgi. The synthesis of GlcCer also occurs at the Golgi but at the cytosolic side of the membrane. Subsequently, the GlcCer moves into the lumen of the organelle and is converted to LacCer and more complex glycosphingolipids. As with phospholipids, each metabolic step that occurs in a separate organelle or with different topology from the precursor, can serve as an indicator of lipid transport/translocation.

Important elements of sterol metabolism can also be used to elucidate where in the cell a particular precursor has moved [9]. The arrival of cholesteryl esters within lysosomes is revealed by cleavage of the fatty acid to yield free cholesterol. The subsequent transport of cholesterol to the ER can be monitored by the action of acyl CoA:cholesterol acyltransferase (Chapter 15) which results in the formation of new cholesteryl esters. In addition, sphingomyelinase treatment of the cell surface induces cholesterol movement from the plasma membrane to the ER where its arrival can likewise be monitored by acyl CoA:cholesterol acyltransferase action. Import of cholesterol into mitochondria (usually restricted to steroidogenic cells) can be followed by side-chain cleavage reactions that produce pregnenolone. Movement of pregnenolone

out of mitochondria can also be followed by oxidations at positions 3, 17 and 21 which occur in the ER.

4. Lipid transport processes

The movement of lipids within the cell can be divided into two different general classes of transport: intramembrane transport, which entails the transbilayer movement of the lipid molecule; and intermembrane transport which is the movement of lipid molecules from one distinct membrane domain to another. Extensive reviews of these processes have been published [8–14].

4.1. Intramembrane lipid translocation and model membranes

The observation that biological membranes can be asymmetric with respect to transbilayer disposition of lipid components (Fig. 1) initially raised basic questions about how such asymmetry was established and maintained. An important issue that needed to be resolved on theoretical grounds, was whether lipids in model membranes could undergo spontaneous transbilayer movement. A simple consideration of the events that occur in the transbilayer movement of a zwitterionic molecule such as PC suggests that at least two energetically unfavorable events must occur. The first is desolvation of the molecule and the second is movement of the charged portion of the lipid through the hydrophobic portion of the bilayer.

Direct experiments to examine the transbilayer movement of phospholipids (R.D. Kornberg, 1971) made use of spin-labeled analogs of PC in which the choline moiety was replaced with the tempocholine probe, *N,N*-dimethyl-*N*-(1'-oxyl-2',6',6'-tetramethyl-4'-piperidyl)-ethanolamine (Fig. 2). These workers found that only the ESR signal generated by molecules in the outer leaflet of unilamellar liposomes could be rapidly quenched by ascorbate. The ESR signal from lipid molecules initially residing at the inner leaflet of liposomes was accessible to ascorbate with a $t_{1/2}$ of >6.5 h, indicating slow transbilayer lipid movement (Fig. 6).

Additional evidence for slow transbilayer phospholipid movement in liposomes came from experiments using ^3H -PC-labeled liposomes and PC-transfer protein. In the presence of excess unlabeled acceptor membranes, only the PC in the outer leaflet of the liposome membrane was rapidly transferred (J.E. Rothman, 1975). The ^3H -PC initially present in the inner leaflet of the membrane moved to the outer leaflet with a $t_{1/2}$ of 11–15 days (Fig. 6). Further evidence demonstrating slow transbilayer movement of phospholipids was obtained from unilamellar liposomes containing 90% PC and 10% PE [6]. In these liposomes the PE initially residing in the outer leaflet of the membrane was rapidly modified by IAA. The PE at the inner leaflet remained refractory to modification by trinitrobenzenesulfonate (i.e., did not undergo transbilayer movement) with a $t_{1/2}$ of >80 days (Fig. 6).

In contrast to phospholipids, non-polar lipids such as diacylglycerol (DG) behave differently. B.R. Ganong (1984) synthesized a structural analog of DG in which the *sn*-3 hydroxyl group was replaced by an SH group that could be detected with dithio-

ESR LABELING OF PC

PC Analog in the outer leaflet rapidly quenched by ascorbate. Analog in the inner leaflet slowly accessible to ascorbate with a $t_{1/2} = 6.5h$

TRANSFER PROTEIN & PC

PC in outer leaflet rapidly transferred to excess acceptor vesicles. PC in inner leaflet accessible with $t_{1/2} = 10d$

MODIFICATION OF PE

PE in outer membrane rapidly modified by IAA. PE in inner membrane available for reaction with TNBS with $t_{1/2} > 80d$

MODIFICATION OF DG ANALOG

Thiol analog of DG reacts with DTNBA. Lipid from both leaflets reacts rapidly. For inner leaflet estimated $t_{1/2} = 15sec.$

MODIFICATION OF CHOLESTEROL

Cholesterol present in liposomes is oxidized by cholesterol oxidase with a $t_{1/2} = 1min$

CHANGE IN FLUORESCENCE

BODIPY analog of ceramide changes fluorescence intensity with change in concentration as it moves from inner leaflet, to outer leaflet with $t_{1/2} = 22 min.$ Similar studies with BODIPY-DG analog give a $t_{1/2} = 70 msec.$

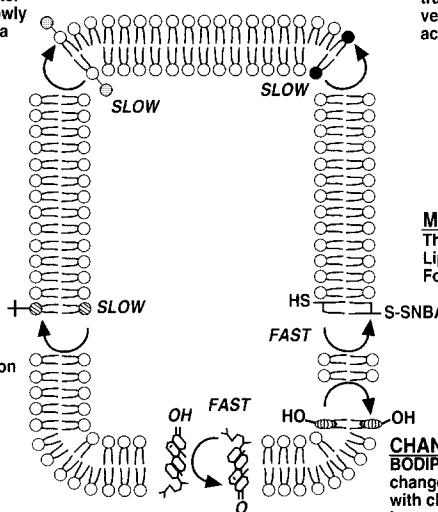


Fig. 6. Summary of key experiments examining transbilayer lipid movement in liposomes. Abbreviations: IAA, isethionylacetimidate; TNBS, trinitrobenzenesulfonate; DTNBA, dithiobisnitrobenzoic acid (Ellman's reagent). Reaction of DTNBA with RSH gives R-S-SNBA. Each of the experiments was specifically designed to initially sample only the outer leaflet of the bilayer and then at subsequent periods detect the movement of lipid from the inner to the outer leaflet of the bilayer.

bisnitrobenzoic acid (Ellman's reagent). When liposomes containing the thiol analog of DG were reacted with Ellman's reagent, the $t_{1/2}$ for transmembrane movement was determined to be 15 s. Concentration-dependent changes in the fluorescent properties of BODIPY lipids have also been used to estimate transbilayer movement of DG and ceramide (J. Bai, 1997). By the fluorescence method the $t_{1/2}$ for transbilayer movement of BODIPY-DG is 70 ms and that of BODIPY-Cer is in 22 min (see Fig. 6). These results strongly suggest that the polar moiety of phospholipids is the portion of the molecule that greatly retards transbilayer movement of these molecules in model membranes. Cholesterol is another non-polar lipid whose transmembrane movement has been examined. Treatment of PC/cholesterol liposomes with cholesterol oxidase demonstrated that the entire cholesterol pool could be readily oxidized with a $t_{1/2}$ of 1 min at 37°C (J.M. Backer, 1981) (see Fig. 6).

Thus, studies with model membranes provide clear evidence that the transbilayer movement of phospholipids is a very slow process in this system, whereas the process appears to be rapid for non-polar lipids. The results imply that if transbilayer movement of phospholipids does occur in biological membranes, it must be a facilitated process.

4.2. Intramembrane lipid translocation and biological membranes

4.2.1. Prokaryotes

The primary consideration in the genesis of any biological membrane is the location of the synthetic apparatus that manufactures the subunits of the membrane and its relationship to the final distribution of its products. In *Escherichia coli*, substantial evidence indicates that the synthesis of phospholipids occurs at the inner (cytoplasmic) membrane by enzymes that have their active sites on the cytoplasmic surface of the inner membrane [13]. Such an orientation allows free access of water-soluble substrates and reaction products to the cytosol.

In experiments performed with *B. megaterium*, Rothman and Kennedy [15] used chemical modification with TNBS, under conditions where the probe did not enter the cell, to distinguish between PE molecules located on the outer and inner sides of the cell membrane. This technique was coupled with pulse-chase experiments with [32 P]inorganic phosphate and [3 H]glycerol and demonstrated that newly synthesized PE is initially found on the cytoplasmic surface of the cell membrane and is rapidly translocated to the outer leaflet of the membrane with a $t_{1/2}$ of 3 min at 37°C. Although the translocation is rapid, it does not occur coincident with synthesis, but rather, with a significant delay after the molecule is synthesized. In addition, the translocation can continue in the absence of PE synthesis. These findings indicate that lipid synthesis and translocation are two distinct events.

The energetic requirements for transmembrane movement of phospholipids have been investigated (K.E. Langley, 1979). Using *B. megaterium* and a TNBS probe, these studies demonstrated that the transbilayer movement of newly synthesized PE was unaffected by inhibitors of ATP synthesis and protein synthesis. Thus, the driving force for phospholipid translocation in *B. megaterium* is independent of metabolic energy, lipid synthesis, and protein assembly into cell membranes.

More recent work with closed vesicles derived from *B. megaterium* membranes demonstrates that NBD analogs of PE, PG, and PC can translocate across the membrane with a $t_{1/2}$ of 30 s at 37°C (S. Hrafnisdottir, 1997). Similar types of experiments conducted with closed vesicles isolated from *E. coli* inner membrane reveal that NBD phospholipids traverse the bilayer with a $t_{1/2}$ of 7 min at 37°C (R. Huijbregts, 1996). This latter process is insensitive to protease and *N*-ethylmaleimide treatments and does not require ATP. Collectively, the data indicate that transbilayer lipid movement is rapid and does not require metabolic energy in bacterial membranes. The basic characteristics of the lipid translocation in the intact cell appear to be retained in isolated membranes.

4.2.2. Eukaryotes

4.2.2.1. Transbilayer movement of lipid at the endoplasmic reticulum. In eukaryotic systems a detailed pattern of synthetic asymmetry has emerged with respect to the topology of the enzymes of phospholipid synthesis in rat liver microsomal membranes. Protease mapping experiments (R. Bell, 1981) have indicated that the active sites of the phospholipid synthetic enzymes are located on the cytosolic face of the ER. Thus, in both prokaryotic and eukaryotic systems it appears that the site of synthesis of the

bulk of cellular phospholipid is the cytosolic side of the membrane. This asymmetric localization of synthetic enzymes strongly implicates transbilayer movement of phospholipids as a necessary and important event in membrane assembly that is required for the equal expansion of both leaflets of the bilayer (reviewed in A. Zachowski, 1993).

The transbilayer movement of phospholipids in microsomal membranes has been measured using several different approaches. In preparations of liver microsomes that were first radiolabeled with lipid precursors *in vivo*, the transbilayer movement of lipids was examined using phospholipid transfer proteins (D.B. Zilversmit, 1977). The results from these experiments provided evidence that PC, PE, PS, and PI from both membrane leaflets were exchanged between labeled microsomes and excess acceptor membranes with a maximal $t_{1/2}$ of ~ 45 min. This value, was set as an upper limit because the amount of lipid transfer protein used could not exchange out the labeled phospholipid with a $t_{1/2}$ of less than 45 min.

In a different approach, a water-soluble, short-chain (dibutyryl) analog of PC was used to measure the rate of uptake and luminal sequestration by isolated liver microsomes (W. Bishop, 1985). This PC analog was taken up in a time- and temperature-dependent manner. The kinetics of uptake were saturable with respect to substrate concentration and the transport activity was protease sensitive. The transporter was also shown to be stereospecific in its action and it was unaffected by the addition of ATP. Virtually identical properties have also been described for a microsomal transporter that utilizes butyryl-lyso-PC (Y. Kawashima, 1987).

Additional studies utilized spin-labeled analogs of PC, PE, PS and SM, and the $t_{1/2}$ for the translocation of these lipid analogs from the cytosolic face to the luminal face of the microsomes was calculated to be 20 min. The transport process did not require ATP and the translocation of each class of lipid showed identical sensitivity to inhibition by *N*-ethylmaleimide (Fig. 7). Furthermore, different species of lipid showed transport kinetics that were consistent with mutual competition for a single transporter. These

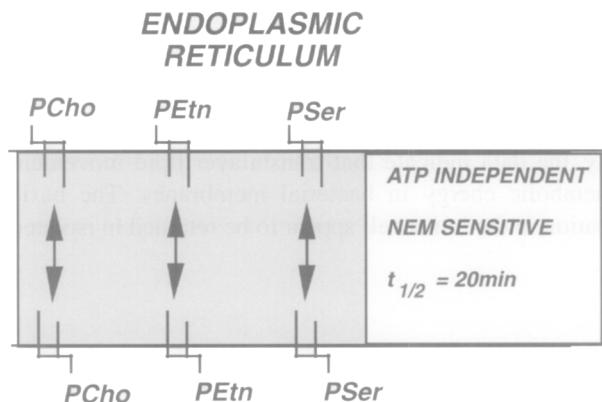


Fig. 7. Transbilayer movement of phospholipids in eukaryotic membranes. The general features of the transmembrane transporters of the ER are shown. The ER transporter does not require ATP and is inhibited by *N*-ethylmaleimide (NEM). The structure (┐) represents the DG portion of the lipids and PCho, PEtn and PSer are abbreviations for phosphocholine, phosphoethanolamine and phosphoserine, respectively.

results indicate that the ER has a relatively nonspecific, ATP-independent transporter that is capable of translocating multiple species of lipid across the bilayer.

Thus, the data from both bacteria and animal cells demonstrate that transbilayer movement of phospholipid occurs on a time scale of minutes, in an ATP-independent fashion in membranes that contain the majority of the enzymes involved in their biosynthesis. These intramembrane transport properties observed in the major biosynthetic membranes, however, are not generally true for other membrane systems. This is especially true of the plasma membrane.

4.2.2.2. Transbilayer lipid movement at the eukaryotic plasma membrane. The transbilayer movement of lipids at the cell surface of eukaryotes is being understood with increasing molecular and biochemical detail. Three fundamental classes of transport are now recognized and consist of the aminophospholipid translocases (also called flippases) the scramblases and the ATP binding cassette (ABC) pumps. A schematic summary of some of the properties of these proteins is shown in Fig. 8.

Aminophospholipid translocase. Several studies using either short-chain versions of PS, or spin-labeled or fluorescent analogs of PS and PE have established the properties of the translocases on the erythrocyte and nucleated cells [10,11]. The aminophospholipid translocase recognizes PS and PE at the external surface of the plasma membrane and translocates the lipids to the cytoplasmic side. This flipping process requires ATP on the cytosolic side of the membrane. The affinity for PS is about 30-fold higher than PE, and the equilibrium distribution is 95% PS and 90% PE in the inner membrane. The aminophospholipid translocases are susceptible to protease digestion and inactivation with *N*-ethylmaleimide. Additional susceptibilities to inhibition by AlF_4^- and Na_2VO_4

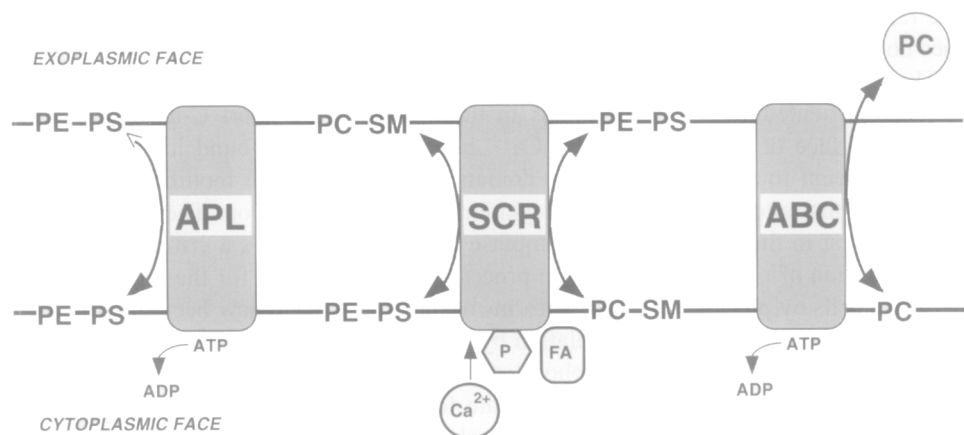


Fig. 8. Summary of the general features of transbilayer transporters found in plasma membranes. APL, amino phospholipid translocase; SCR, bidirectional transporter or scramblase; ABC, ATP binding cassette protein. The APL and ABC transporters utilize cytoplasmic ATP to drive transport. The APL is specific for PS and PE and the reaction greatly favors internalization. The ABC transporter pumps PC out of the plasma membrane. The scramblase is activated by high intracellular Ca^{2+} levels and regulated by phosphorylation and acylation. The scramblase is nonspecific and randomizes the lipid distribution across the bilayer.

suggest that the translocases belong to the P-type ATPase family. The stoichiometry of ATP utilization per phospholipid translocation event is approximately one. All of the characteristics are consistent with the amino phospholipid translocase translocase being a specific plasma membrane ATPase that is activated by aminophospholipids.

An erythrocyte ATPase that is Mg^{2+} and PS dependent, has been partially purified and reconstituted into vesicles. The reconstituted protein preparation is capable of transporting PS across the bilayer, albeit with low efficiency [11]. A second P-type ATPase, designated ATPase II, has been purified to homogeneity and its cDNA cloned. The deduced protein sequence reveals that it contains 3 P-type ATPase consensus sequences (X. Tang, 1996). Definitive reconstitution of the ATPase II into vesicles and demonstration of its aminophospholipid translocase activity has not yet been accomplished.

The cDNA sequence of the ATPase II was used to identify the yeast ortholog, which is the *DRS2* gene, that encodes the Drs2p protein, and for which *drs2* mutants are known, that have defects in ribosomal assembly. Some *drs2* mutant strains appear to have defects in their NBD-PS transport (X. Tang, 1996). However, *drs2Δ* mutants lacking any of the gene product have normal NBD-PS and NBD-PE transport (A. Siegmund, 1998). Thus, the aminophospholipid translocase has yet to be definitively identified in either mammalian or yeast systems.

Bidirectional transporters. The bidirectional transporters at the plasma membrane function to randomize the lipid distribution across the plane of the bilayer, and are commonly referred to as scramblases [10]. The action of the scramblase is similar to that of the previously described transbilayer transporter present in the ER. The protein was first functionally identified in erythrocytes, but is also present in nucleated cells. The scramblase shows no lipid specificity and essentially collapses the asymmetry of lipids at the cell surface. Phospholipids, SM and glycosphingolipids all serve as substrates. The randomizing function of this plasma membrane protein is activated by Ca^{2+} and does not require ATP.

The scramblase protein was purified to homogeneity and its cDNA cloned [16]. The protein is oriented with its N-terminus in the cytosol and a short C-terminus at the external surface of the membrane. A Ca^{2+} binding domain is found in the cytosolic region adjacent to the transmembrane domain. Post-translational modifications to the scramblase that alter activity include acylation and phosphorylation. There is currently much interest in the regulation of scramblase function, as it plays a critical role in the externalization of phosphatidylserine, a process that is important for the recognition of apoptotic cells by phagocytes. Four isoforms of the protein have now been identified (T. Wiedmer, 2000). In addition to regulation by Ca^{2+} and oligomerization, the scramblase activity can be enhanced by phosphorylation directed by protein kinase C- δ (S. Frasch, 2000). The protein is also a substrate for the protein tyrosine kinase c-Abl (J. Sun, 2001). Detailed understanding of the mechanism of scramblase action is now likely to provide important fundamental information about the energetics and maintenance of lipid asymmetry.

ABC transporters. The ABC transporters are a large family of proteins involved in moving molecules across membranes in ATP-dependent reactions (I. Klein, 1999). A subset of this family transports molecules that include xenobiotics, bile acids, and

hydrophobic compounds. In 1993, Smit and coworkers described a transgenic mouse with null alleles for the ABC transporter, *mdr2*, that exhibited a profound defect in transporting PC into bile. These findings led Ruetz and Gros [17] to examine the activity of the *mdr2* protein as a PC translocase. The heterologous expression of the *mdr2* cDNA in yeast leads to incorporation of *mdr2* protein into yeast secretory vesicles and acquisition of the ability to translocate NBD-PC across the bilayer. The translocation process is time-, temperature- and ATP-dependent. These findings indicate that the *mdr2* protein acts as a PC transporter. Unequivocal experiments in transgenic mice establish that the human MDR3 protein and mouse *mdr2* have identical function [18].

Further implication of ABC transporters in lipid translocation comes from work with human ABC A1, ABC R and ABC G5 plus ABC G8. The ABC A1 protein acts on cholesterol export and is described in Chapter 20. The ABC R protein is defective in Stargardt's macular dystrophy (R. Allikmets, 1997). In mice with null alleles for ABC R, retinylidene PE (a conjugate of retinaldehyde and PE) accumulates in the inner membrane of rod outer segment discs, due to failure of translocation by ABC R (J. Weng, 1999). These findings implicate ABC R as a translocase for retinylidene PE. The dimeric ABC G5 and ABC G8 transporter is defective in individuals with sitosterolemia (M. Lee, 2001). Under normal conditions, intestinal absorption of the plant sterol, sitosterol, appears to be minimized by an efflux pumping mechanism that continually translocates the sterol back to the lumen of the gut. In individuals with ABC G5 or ABC G8 defects, this pumping mechanism is lost, and blood and tissue levels of this deleterious sterol increase dramatically.

Data implicating ABC family transporters are growing, and it is likely that additional members will be added to the list. It is important to highlight one fundamental difference observed between the ABC transporters and other translocases (Fig. 8). In each case of the ABC family, the transported substrate either enters an environment that is a different phase from the membrane or the substrate is rapidly moved into another membrane or metabolized. In contrast, the substrate for the aminophospholipid translocases and scramblases remains within the bilayer across which it is transported.

4.3. Intermembrane lipid transport

From a theoretical perspective a number of processes could contribute to the intermembrane transport of lipids. These are outlined in Fig. 9 and include monomer solubility and diffusion (A), soluble carriers such as lipid transfer proteins (B), carrier vesicles (C), membrane apposition and transfer (D) and membrane fusion processes (E). Lipids such as free fatty acids, lysophosphatidic acid and CDP-DG may have sufficient solubility to allow for some monomeric transport but most other lipids are likely to require one of the other potential mechanisms due to their extremely low solubility.

4.3.1. Transport in prokaryotes

The presence of multiple membrane systems in organisms, such as Gram-negative bacteria, photosynthetic bacteria, and the eukaryotes, raise significant questions about the mechanisms of membrane biogenesis. In a 'simple' organism such as *E. coli* there are two membrane systems: the inner or cytoplasmic membrane, and the outer

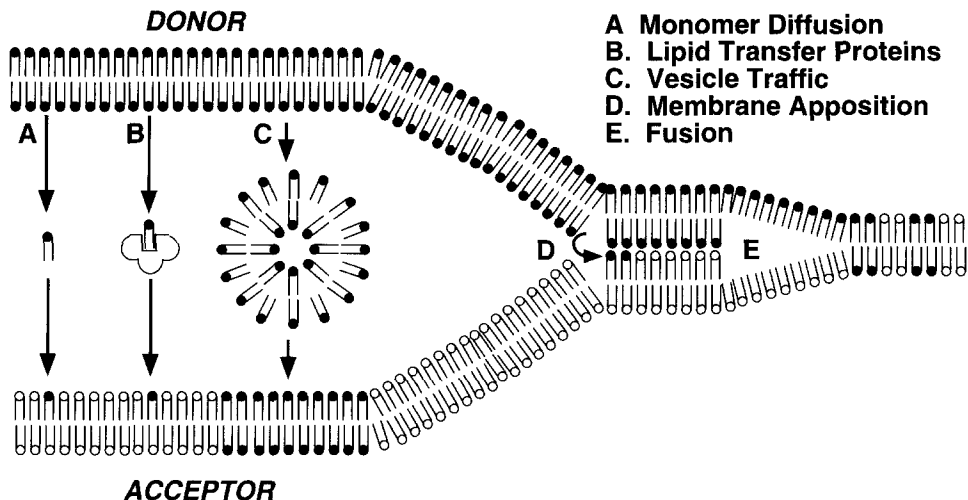


Fig. 9. Theoretical mechanisms for transporting lipids and altering membrane composition.

membrane (Chapter 3) [13]. The entire apparatus for phospholipid synthesis is located at the inner membrane. Consequently, there must exist a mechanism for exporting phospholipids from the inner membrane to the outer membrane.

In some of the earliest studies of phospholipid transport between the inner and outer membranes of *E. coli*, labeling of PE revealed that the specific activity of this lipid was fivefold higher in the inner membrane than the outer membrane, immediately following a 30-s pulse with [^3H]glycerol (A.M. Donohue-Rolfe, 1980). During the chase period the specific activity of the outer membrane increased, while that of the inner membrane decreased. After several minutes the specific activities of both membranes asymptotically approached the same value, which indicated radioequilibration between the membranes. The $t_{1/2}$ for the translocation of PE was determined to be 2.8 min. The transport in these studies was independent of protein synthesis, lipid synthesis, and ATP synthesis. It appeared, however, to be dependent upon the cell's protonmotive force.

In recent work, Doerrler et al. [19] have examined the role of an ABC transporter in *E. coli* named *msbA*, in the phenomena of lipid transport between the inner and outer membrane. The *msbA* is an essential gene whose overexpression suppresses growth arrest in *E. coli* strains defective in the final steps of lipid A synthesis. Examination of the mode of rescue effected by high copy *msbA*, in the lipid A synthesis mutants, suggested that the protein acted to transport toxic lipid A precursors from the inner to the outer membrane. The function of *msbA* was examined further by creating a temperature-sensitive allele (*msbA^{ts}*). When cultures of strains with *msbA^{ts}* are shifted to the non-permissive temperature, cell growth arrests after about 60 min. In wild-type cells, subjected to the same treatments, cell growth is unaffected and 40–60% of the phospholipid, lipid A and protein, moved from the inner to the outer membrane. In the *msbA^{ts}* strain, <10% of the phospholipid and lipid A moved to the outer membrane, whereas protein traffic was unaffected. Furthermore, electron micrographs

reveal significant invagination of the inner membrane consistent with excess lipid accumulation at this site. These experiments provide striking evidence for the role of *msbA* in the export of both phospholipid and lipid A from the inner membrane to the outer membrane of *E. coli*. These findings raise the possibility of other ABC transporters acting at sites of membrane contact or apposition, as a general mechanism for intermembrane lipid transport.

4.3.2. Transport in eukaryotes

Currently, there is a broad understanding of the elements of interorganelle transport for several different lipid classes. In many studies the questions have been narrowly focused to the movement of one lipid class between a donor and an acceptor compartment that are temporally, metabolically and geographically segregated within the cell. The discussion of these processes is organized by class of lipid and then by membrane systems examined.

4.3.2.1. Phosphatidylcholine.

Transport of newly synthesized PC from the ER to the plasma membrane. The principal site of PC synthesis is the ER (Chapter 8). The transport of PC from the ER to the plasma membrane has been examined using pulse-chase labeling with a [^3H]choline precursor and rapid plasma membrane isolation with cationic beads (M. Kaplan, 1985). These studies reveal that PC transport is an extremely rapid process occurring with a $t_{1/2} \approx 1$ min (Fig. 10). This transport is unaffected by metabolic poisons that deplete cellular ATP levels, disrupt vesicle transport, or alter cytoskeletal arrangement. The mechanism of this transport is presently unknown, but the results are consistent with a soluble carrier mechanism such as PC transfer protein, or zones of apposition that facilitate rapid intermembrane transfer. Recent work by Pichler and colleagues has identified a subfraction of the ER that associates closely with the plasma membrane in yeast (H. Pichler, 2001). Future studies examining the effects of agents or mutations that disrupt these intracellular membrane associations will be critical for determining their role in lipid traffic.

Transport of newly synthesized PC from the ER to the mitochondria. Using conventional subcellular fractionation techniques, the transport of nascent PC to the mitochondria of BHK cells was examined by pulse-chase experiments with a [^3H]choline precursor (M.P. Yaffe, 1983). These experiments show that the newly made PC pool equilibrates between the outer mitochondrial membrane and the ER in approximately 5 min (Fig. 10). Similar studies performed in yeast (G. Daum, 1986) revealed that the PC pool rapidly equilibrates between the ER and mitochondria. Addition of metabolic poisons did not eliminate the PC radioequilibration in yeast. Studies with isolated mitochondria demonstrate that PC loaded into the outer mitochondrial membrane can be transported to the inner membrane in an energy-independent manner (M. Lampl, 1994). Consistent with this finding is the observation that PC rapidly moves across the membrane of vesicles derived from mitochondrial outer membranes prepared from either mammalian cells or yeast (D. Dolis, 1996; M. Janssen, 1999).

Transport of exogenous PC analogs from the cell surface to intracellular organelles. Clear evidence for the movement of PC from the plasma membrane to intracellular

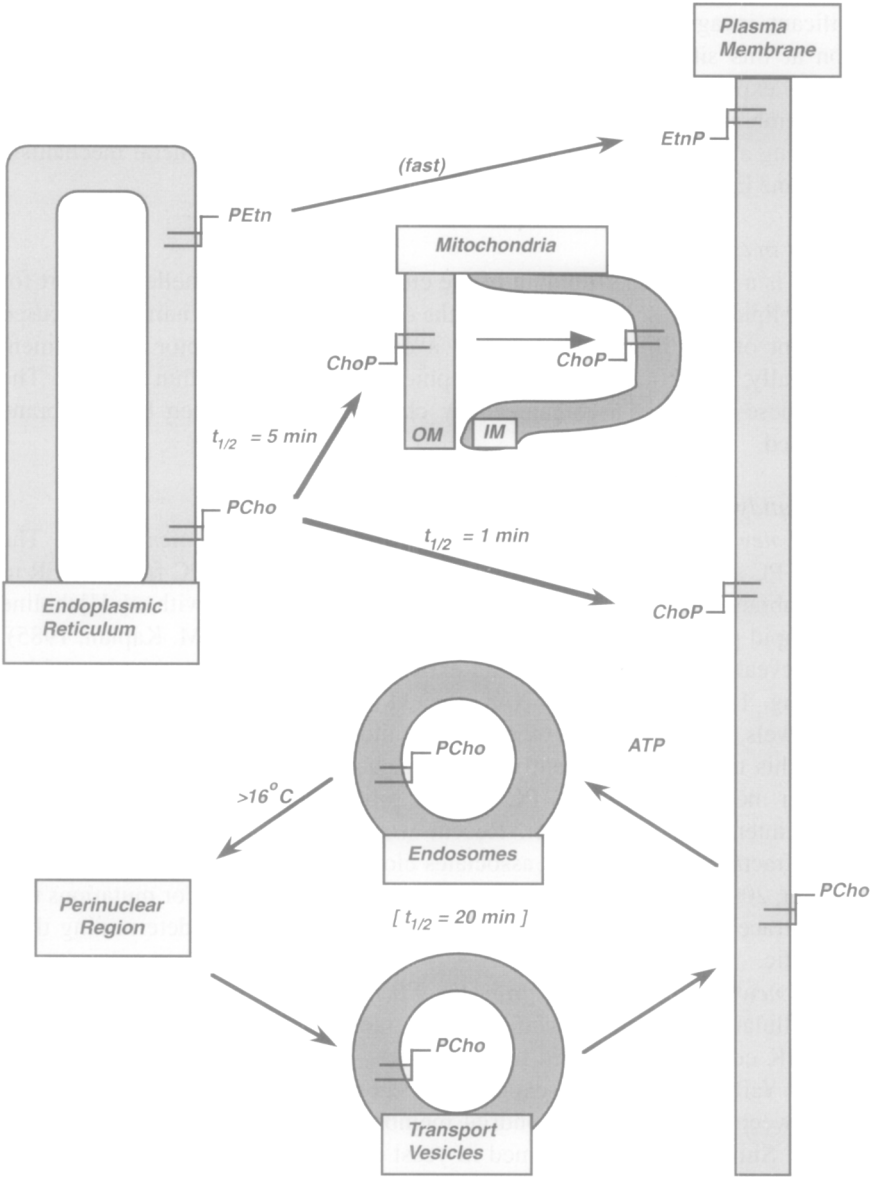


Fig. 10. Interorganelle transport of PC and PE within eukaryotic cells. The structure (F) represents the DG portion of the phospholipid, and PCho and PEtn are the abbreviations for phosphocholine and phosphoethanolamine respectively. OM and IM are abbreviations for the outer and inner membranes of the mitochondria. The $t_{1/2}$ for PC transport from the perinuclear region of the cell to the plasma membrane is shown in brackets and estimated to be 20 min.

organelles has been obtained using the fluorescent lipid analog NBD-PC (R. Sleight, 1984). The fluorescent lipid can be pulse-labeled into the outer leaflet of the plasma membrane at 2°C . Upon warming the cells, the fluorescent lipid is transported from the

plasma membrane to the perinuclear region of the cell in the proximity of the Golgi apparatus and the centrioles, via an ATP-dependent process (Fig. 10). The lipid transport occurs by endocytosis and the process can be disrupted by reducing the temperature to 16°C which causes the PC to accumulate in endosomal vesicles. The kinetics for endocytosed NBD-PC transport from intracellular membranes back to the plasma membrane occurs with $t_{1/2} = 20$ min. During the transit cycle the NBD-PC remains restricted to the non-cytosolic face of the respective membranes. The kinetics of this vesicle-based recycling of PC between the cell interior and the plasma membrane are markedly different from those for transport of newly synthesized PC to the cell surface. The disparity between the two transport processes suggests that there is restricted intermixing of nascent and recycling pools of PC.

4.3.2.2. *Phosphatidylethanolamine.*

Transport of newly synthesized PE to the plasma membrane. When an ethanolamine precursor is used, the primary site of PE synthesis is the ER (Chapter 8). The appearance of newly synthesized PE at the external leaflet of plasma membrane has been determined using chemical modification of the cell surface with TNBS at reduced temperature (R. Sleight, 1983). The results indicate that the initial rate of transport of PE is rapid and proceeds without a lag (Fig. 10). The transport process is insensitive to metabolic poisons that disrupt vesicle transport and cytoskeletal structure. The rapid transport kinetics occur at rates consistent with a soluble carrier-mediated process or transfer at zones of apposition between membranes. Analysis of the kinetics of the process is complicated since only PE at the outer leaflet of the plasma membrane is measured, and the ATP-dependent aminophospholipid transporter activity within the plasma membrane (P. Devaux, 1988; O. Martin, 1987) may be a required step for the lipid to arrive at this location. Despite these complications the results clearly indicate that the initial rate of arrival of PE at the plasma membrane occurs on a time scale that clearly distinguishes it from well characterized vesicle transport phenomena, and is independent of processes involved in protein transport to the cell surface.

PE derived from a PS precursor that is decarboxylated at the mitochondria is also transported to the plasma membrane (J. Vance, 1991) (Fig. 11). This mitochondrial PE is transported to the plasma membrane, with greater efficiency than PE synthesized from an ethanolamine precursor. The mechanism of this translocation remains to be elucidated but the process is unaffected by brefeldin A, a fungal metabolite that alters the structure and function of the Golgi apparatus.

Transport of newly synthesized PE to the mitochondria. Early studies examining the movement of newly synthesized PE from the ER to the mitochondria of hepatocytes demonstrated that the process was markedly slower ($t_{1/2} \approx 2$ h) than that observed for PC (M.P. Yaffe, 1983). These experiments used classical rate sedimentation to isolate the organelles. More recent studies indicate that such mitochondrial fractions are likely to contain another resolvable compartment, the mitochondria-associated membrane (MAM) (J. Vance, 1990). Evidence obtained using CHO-K1 cells [20] indicates that nascent PE (made via CDP-ethanolamine) is transported to the MAM but not to the inner mitochondrial membrane. It remains unclear whether some of this PE is transported to the outer mitochondrial membrane. The results are consistent with little import of PE

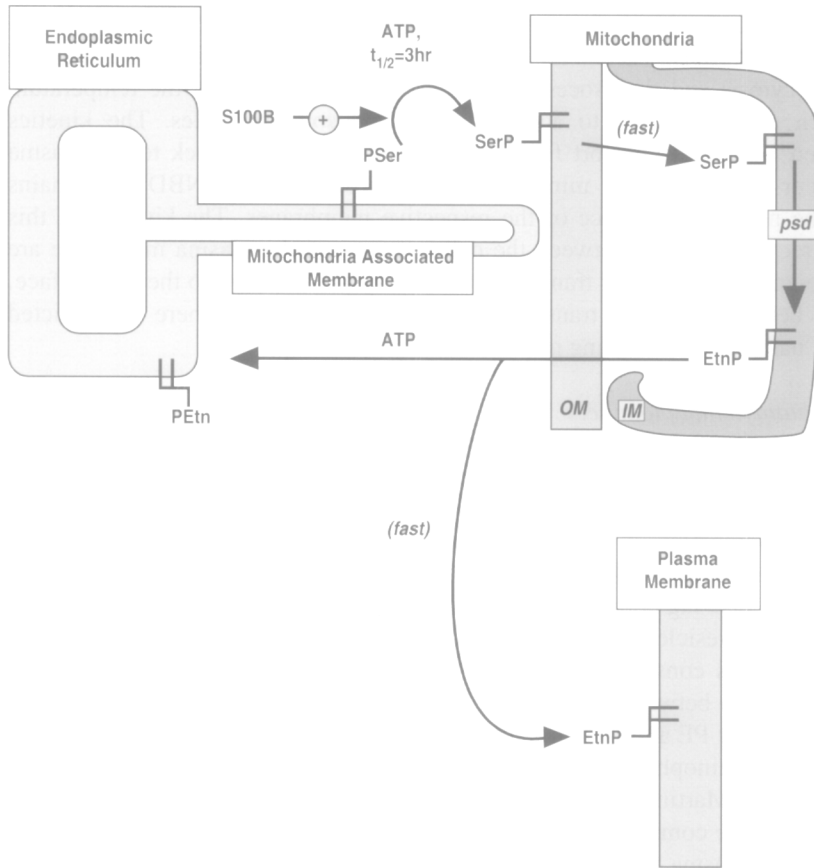


Fig. 11. Interorganelle transport of PS in eukaryotic cells. The structure (F) represents the DG portion of the phospholipid. Pser, PEtn and PCho are abbreviations for phosphoserine, phosphoethanolamine and phosphocholine. The term *psd* stands for PS decarboxylase. The rate for the transport of PS between the outer (OM) and inner (IM) mitochondrial membrane has not been determined but appears to be on the order of minutes.

derived from ethanolamine, into the mitochondria. Furthermore, yeast mutants lacking a functional allele for PS decarboxylase 1 are markedly deficient in mitochondrial PE (P.J. Trotter, 1995). The reduced PE in mitochondria cannot be fully restored by PE synthesized in the ER from an ethanolamine precursor, or that made in the Golgi or vacuole by PS decarboxylase 2 (R. Birner, 2001; M. Storey, 2001). These latter findings clearly demonstrate that there is compartmentation and restricted transport of different pools of PE within cells.

4.3.2.3. Phosphatidylserine.

Transport of newly synthesized PS to the mitochondria. The location of PS decarboxylase at the inner mitochondrial membrane [8] provides a convenient method for determining the arrival of PS at this cellular location. The extremely low steady

state level of PS at the mitochondrial inner membrane (Table 1) coupled with kinetic considerations, indicates that PS is rapidly decarboxylated upon its arrival at the inner membrane. The general features of nascent PS transport are outlined in Fig. 11. The initial studies with intact mammalian cells that used PS decarboxylation as an indicator for lipid transport identified a clear ATP requirement for the transport process. Subsequent reconstitution studies, with isolated organelles, established that mitochondria could take up PS in an ATP-independent process. These findings indicated a requirement for ATP at a stage earlier than the presence of PS at the outer mitochondrial membrane. Additional studies with isolated organelles provided evidence for a tight association between specialized elements of the ER and the mitochondria. These *in vitro* associations were also shown to have *in vivo* counterparts evidenced by electron microscopy (D. Ardail, 1993). Successful isolation of these specialized ER structures, now called the mitochondria-associated membrane (MAM), indicated that they are selectively enriched in a subset of lipid synthetic enzymes especially PS synthase (J. Vance, 1990). A MAM structure has also been identified and isolated from yeast cells [21]. Pulse-chase experiments coupled with subcellular fractionation have now established the PS destined for the mitochondria must transit through the MAM and that exit from the MAM requires ATP [20].

The synthesis of PtdSer and transport to the mitochondria have been successfully reconstituted using permeabilized cells [8]. Permeabilized cells retain cellular organelles and cytoarchitecture but are depleted of soluble cellular components. The compromised plasma membrane enables the addition of defined soluble components under controlled conditions to reconstitute the transport processes. The transport of PS to the mitochondria in permeabilized cells occurs in the absence of cytosol, displays an absolute requirement for ATP and occurs with a $t_{1/2}$ of approximately 3 h at 37°C. This transport does not require ongoing synthesis of PS, and 45 fold dilution of the permeabilized cells does not alter the rate or extent of transport. These results are consistent with a membrane bound transport intermediate that utilizes zones of close membrane apposition or fusion between the ER and mitochondria. Although there is not an absolute requirement for cytosol in the transport reaction, a soluble 9 kDa Ca^{2+} binding protein, named S100B, that is highly conserved across mammalian species can enhance the transport several fold (O. Kuge, 2001). Permeabilized yeast have also been used to examine PS transport (G. Achleitner, 1995). Unlike mammalian cells, the transport of PS to yeast mitochondria does not require ATP.

Genetic approaches to identifying components involved in interorganelle aminophospholipid transport. Genetic tools constitute a powerful approach for identifying the components involved in lipid transport. The ability to isolate mutant strains defective in a given transport step, and then clone the genes by complementation of the mutation, can lead to clear molecular and mechanistic resolution of complex processes. The genetic aspects of PS and PE transport in eukaryotic systems are shown in Fig. 12. Understanding the genetic approach to examining aminophospholipid transport requires an appreciation that the synthesis and decarboxylation of PS and the methylation of PE are all geographically separate events within a cell. A basic hypothesis of the genetic approach is that there are specific genes, designated *PST* for (*PS* transport), and *PEE* (for *PE* export), that either regulate or directly participate in the transfer process. Prominent methods for identifying mutant strains defective in these processes rely on

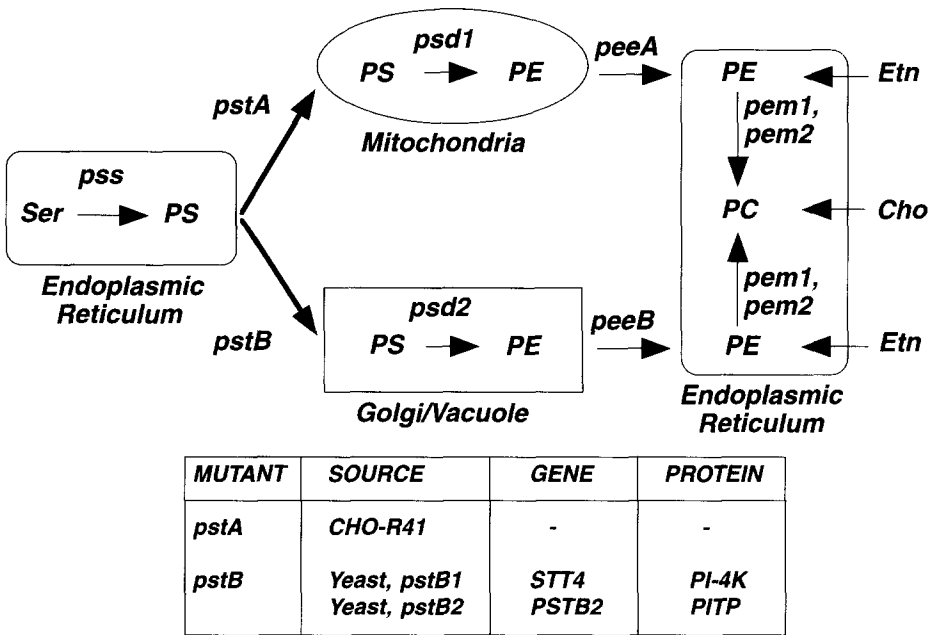


Fig. 12. Genetic analysis of aminoglycerophospholipid transport in eukaryotes. The transport of PS synthesized in the ER is hypothesized to be regulated by *PSTA* and *PSTB* genes. The acronym stands for *PS* transport (either A or B pathways). Likewise, the transport of PE synthesized in the mitochondria or Golgi/vacuole is proposed to be regulated by *PEEA* and *PEEB* genes. The acronyms stand for *PE* export (either A or B pathways). Both known and proposed mutations along the metabolic and transport pathways appear in lower case italics. The table summarizes the mutants, genes and proteins that have been identified. Other abbreviations: *pss*, PS synthase; *psd*, PS decarboxylase, *pem*, PE methyltransferase; PI-4K, PI-4-kinase; PITP, PI transfer protein; Cho, choline; Etn, ethanolamine; Ser, serine.

the identification of drug-resistant mammalian cells [22] or isolation of Etn auxotrophs (described in more detail in [8]) in suitable genetic backgrounds in yeast.

Studies with mammalian systems have identified a mutant line of CHO cells, designated R-41, that is resistant to an antibiotic that recognizes PE and causes cytolysis [22]. The cells have normal enzyme activity for PS synthases and decarboxylase, but labeling with [¹⁴C]serine reveals a defect in PE formation. Further analyses with isolated mitochondria demonstrate that the rate of import of PS from the outer to the inner mitochondrial membrane occurs at approximately 40% of the rate found for wild-type cells. By the scheme shown in Fig. 12 this cell line belongs to the *pst A* class of mutants. Although the mutant line has a lesion in PS import into mitochondria, protein import into the inner membrane is unaffected. This cell line now provides an important tool for cloning the complementing cDNA that encodes an element required for PS import into the mitochondria.

In the yeast system, selection for Etn auxotrophs has yielded a number of mutant strains and genes that are involved in aminophospholipid transport [8]. Currently, the majority of the work has provided information about the transport of nascent PS from the ER to the Golgi/vacuole compartment in the *pstB* arm of the pathway. One

mutant strain, *pstB1*, accumulates PS and shows reduced formation of PE. The gene complementing the defect, *STT4*, encodes a PI-4-kinase (P. Trotter, 1998). The results suggest that PI-4P or perhaps PI4,5P₂ may serve as a recognition motif on either the donor or the acceptor membranes participating in PS transport. Polyphosphoinositides are known to form recognition sites for the assembly of multiple protein components involved in membrane fusion machines [2].

A second mutant strain identified in the yeast system, *pstB2*, displays a 50% accumulation of PS and a 70% decrease in PE formation relative to wild-type strains when labeled with [³H]serine. The gene that complements the defect, named *PSTB2*, is related to the PI transfer protein in yeast encoded by the *SEC14* gene. The protein encoded by the *PSTB2* gene, denoted PstB2p, is a PI transfer protein, but does not transfer PS (W. Wu, 2000). PstB2p is not a cofactor for the PS decarboxylase 2 but a protein that appears to regulate the access of PS to the enzyme. The site of action of PstB2p has now been shown to be at the acceptor membranes (W. Wu, 2001). The mechanism of action of PstB2p may be to dock donor and acceptor membranes, or it may act to regulate PI kinase activity in donor membranes.

4.3.2.4. Sphingolipids.

Ceramide transport from ER to Golgi. Ceramide is synthesized in the ER and the majority of this lipid is subsequently transported to the Golgi apparatus where it is metabolized to sphingomyelin and glycosphingolipids such as GlcCer and LacCer (Chapter 14) [14]. Measurement of conversion of ceramide to either sphingomyelin or glycosphingolipids can serve as an indicator of ceramide transport (Fig. 13). Hanada and coworkers have isolated mutant strains of CHO cells that are resistant to a toxin,

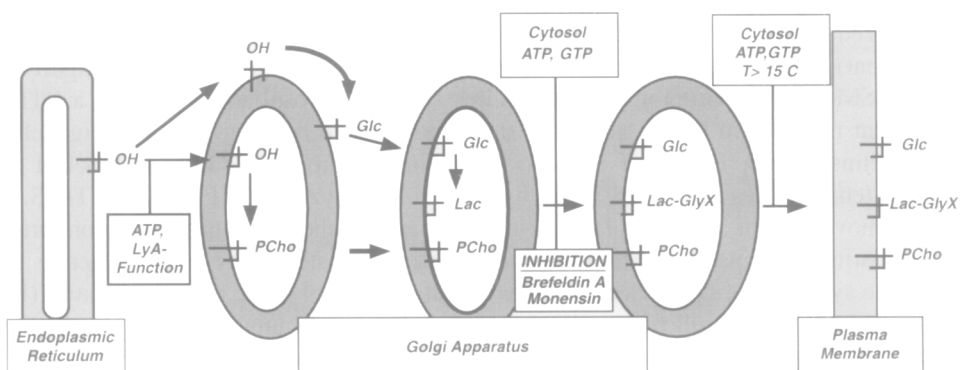


Fig. 13. Interorganelle transport of sphingolipids in eukaryotic cells from ER to plasma membrane. The structure (⊥) represents the Cer portion of sphingolipids. PCho, Glc and Lac are the abbreviations for phosphocholine, glucose, and lactose. Cer is transported to the Golgi in an ATP-dependent reaction. GlcCer is synthesized on the cytosolic face of the Golgi. SM is synthesized on the luminal face. The LyA mutation selectively affects the access of Cer to the site of SM synthesis. GlcCer must reach the luminal face of the Golgi for conversion to LacCer and more complex glycosphingolipids, denoted by GlyX moieties. Movement of the sphingolipids through the Golgi requires cytosol, ATP, and GTP and is inhibited by brefeldin A, monensin, GTPγS and reduced temperature. The $t_{1/2}$ for sphingolipid transport from the Golgi to the plasma membrane is 20 min.

lysenin, that binds cell surface SM and causes cytolysis [23]. One class of mutants (LyA) is selectively defective in SM synthesis, but not glycosphingolipid synthesis, despite normal activity of SM synthase. These findings suggest that the routing of Cer to SM synthase and GlcCer synthase is different and regulated by different gene products.

Ceramide transport to the locus of SM synthase demonstrates that the process can be reconstituted in permeabilized cells (T. Funakoshi, 2000) and requires the addition of ATP and cytosol. Depletion of ATP in intact cells also yields arrest of ceramide (and BODIPY-ceramide) trafficking to SM synthase. When wild-type permeabilized cells are reconstituted with cytosol from the LyA cells, ceramide transport-dependent SM synthesis does not occur. Thus, the LyA lesion resides in a soluble protein that participates in transport of nascent Cer to the locus of SM synthase. In contrast to these findings the GlcCer synthesis failed to exhibit a clear requirement for cytosol for ceramide transport to the enzyme.

Transport of newly synthesized sphingolipids from the Golgi to the plasma membrane. The synthesis and intracellular trafficking of SM and glucosylceramide (GlcCer) has been examined using several different fluorescent ceramides and short-chain radiolabeled ceramides [14]. When fibroblasts are incubated with NBD-Cer at 2°C, it is rapidly taken up and distributed randomly among all cell membranes (N. Lipsky, 1985). Upon warming the cells to 37°C, the fluorescent lipid concentrates in the Golgi apparatus as it is converted to NBD-SM and NBD-GlcCer. These sphingolipids are subsequently exported from the Golgi apparatus to the plasma membrane by a process that is partially monensin-sensitive and brefeldin-A-sensitive in most cells and occurs with a $t_{1/2}$ of 20 min, a time similar to that required for the transport of proteins from the Golgi to the plasma membrane (Fig. 13). However, there appears to be a pool of Glc-Cer that can be transported by routes insensitive to inhibitors of vesicle trafficking [14]. Vesicle-based protein transport is arrested in mitotic cells as is the transport of newly synthesized NBD-SM and NBD-GlcCer (Kobayashi, 1989). Experiments using (non-fluorescent) short-chain analogs of Cer in permeabilized cells indicate that the export of nascent SM from the Golgi apparatus requires ATP and cytosol and occurs via a GTP-dependent mechanism that is also consistent with vesicle budding from the organelle (J.B. Helms, 1990). Export of nascent sphingomyelin from the Golgi is blocked at reduced temperatures such as 15°C and by the non-hydrolyzable GTP analog, GTP γ S.

The movement of sphingolipids between elements of the Golgi has been monitored in reconstituted preparations from mutant Chinese hamster ovary cells defective in either the synthesis of lactosylceramide or the attachment of sialic acid to the latter (B. Wattenberg, 1990). In cell free systems, donor Golgi that accumulate lactosylceramide transfer this lipid to acceptor Golgi that are devoid of the substrate. The acceptor Golgi add sialic acid to the lactosyl ceramide to make GM₃. The lipid transfer reaction between Golgi compartments requires ATP and cytosol and is inhibited by GTP γ S. The properties of glycosphingolipid transport between Golgi compartments are thus identical to those found for vesicular protein transport.

Import of exogenous sphingolipids. The NBD and BODIPY analogs of SM, GlcCer, and LacCer can be readily inserted into the outer leaflet of the plasma membrane of fibroblasts at reduced temperature (see Fig. 3). When fibroblasts treated in such a manner are warmed to 37°C the fluorescent sphingolipids are internalized and accumulate in the

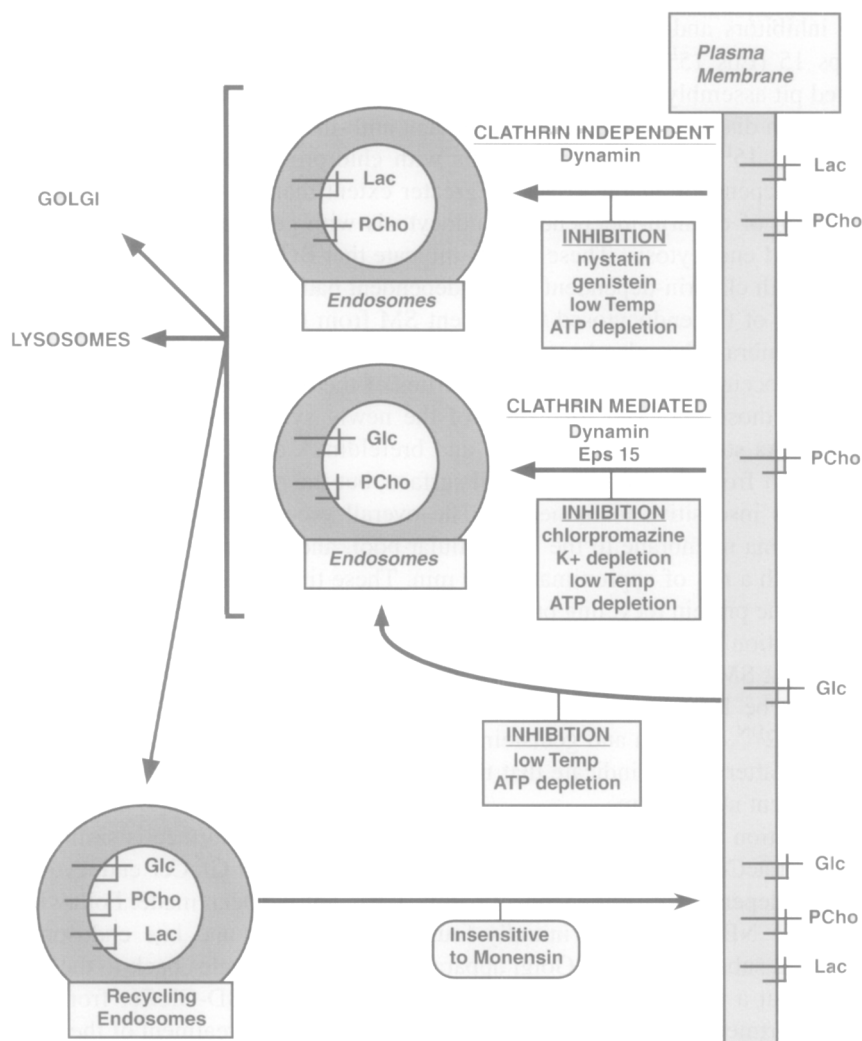


Fig. 14. Interorganelle transport of exogenously supplied sphingolipids. The structure ($\perp$) represents the Cer portion of sphingolipids. PCho, Lac and Glc are the abbreviations for phosphocholine lactose and glucose, respectively. Following insertion into the plasma membrane at reduced temperature, SM is internalized to the endosomal compartments by both clathrin-dependent and -independent pathways. LacCer is endocytosed primarily by the clathrin-independent pathways. The endocytic process can be generally inhibited by ATP depletion and reduced temperature. The endocytosed lipid can recycle back to the plasma membrane and this recycling is insensitive to monensin. The endocytosed lipid can also be transported to the Golgi apparatus or lysosomes.

endosomal compartments of the cell. General inhibition of endocytosis by ATP depletion, or maintenance at low temperature, effectively prevents any internalization of the polar sphingolipids. Internalized NBD-SM subsequently accumulates in the perinuclear region of the cell containing the centrioles (M. Koval, 1989) and the Golgi apparatus (see Fig. 14). The initial steps of BODIPY-SM internalization have been dissected using

a variety of inhibitors and dominant negative structural variants of dynamin 2 (Dyn 2^{DN}) and Eps 15 (Eps 15^{DN}) [24]. The Eps 15 protein has a regulatory function in clathrin coated pit assembly. The internalization of BODIPY-SM is completely arrested by Dyn 2^{DN} which disrupts both clathrin-dependent and -independent endocytosis. The expression of Eps 15^{DN}, or treatment of cells with chlorpromazine or K⁺ depletion, inhibits clathrin-dependent endocytosis to a greater extent than SM endocytosis. Conversely, inhibition of clathrin-independent endocytosis with genistein or nystatin does not fully block SM endocytosis. These results indicate that BODIPY-SM is likely to be internalized by both clathrin-dependent and -independent pathways.

The movement of the endocytosed fluorescent SM from the internalized pool, back to the plasma membrane has also been examined in fibroblasts (M. Koval, 1989). This transport process occurs via vesicles. The properties of the recycling pool of NBD-SM are distinct from those observed for export of the newly synthesized SM out of the Golgi (Fig. 13). As stated above, monensin and brefeldin A arrest newly synthesized NBD-SM transport from the Golgi to the cell surface, but the recycling of endocytosed fluorescent SM is insensitive to monensin. The overall process of internalization of SM from the plasma membrane to the intracellular pool, and transport back to the cell surface occurs with a $t_{1/2}$ of approximately 40 min. These time constants are similar to those for membrane protein recycling processes from the plasma membrane.

The internalization of BODIPY-LacCer follows a route that partially overlaps with that for fluorescent SM [24]. The BODIPY-LacCer is internalized into endosomes and subsequently can be localized to the Golgi. The endocytosis of BODIPY-LacCer is inhibited by Dyn 2^{DN}, nystatin and genistein but not Eps 15^{DN}, chlorpromazine, or K⁺ depletion. These latter results indicate that the fluorescent LacCer is endocytosed by a clathrin-independent mechanism.

The internalization and recycling of NBD-GlcCer from the cell surface is similar to that for SM and LacCer analogs [14,25]. The assignment of NBD-GlcCer endocytosis to either clathrin-dependent or -independent pathway has not yet been made. Following internalization the NBD-GlcCer is initially found in both early and late endosomal compartments and subsequently the Golgi apparatus. The lipid recycles back to the cell surface (Fig. 14) at a rate similar to that for SM. Transport of NBD-GlcCer from the endosomal compartment to the plasma membrane is insensitive to treatment of the cells with either monensin or brefeldin A.

4.3.2.5. Cholesterol.

Transport of cholesterol to and from the plasma membrane. Following its synthesis at the ER, cholesterol is transported throughout the cell and becomes enriched in the plasma membrane [9]. The transport of newly synthesized cholesterol to the plasma membrane has been examined in tissue culture cells using pulse-chase experiments with either the rapid plasma membrane isolation procedure (M. Kaplan, 1985), caveolae isolation (A. Uittenbogaard, 1998), oxidation of accessible cholesterol by cholesterol oxidase (Y. Lange, 1985), or desorption of newly labeled cholesterol with methyl- β -cyclodextrin (S. Heino, 2000). These lines of experimentation have revealed that the minimum transport time for cholesterol to the plasma membrane is 10 min at 37°C (Fig. 15). The transport process can be completely blocked by reducing the temperature

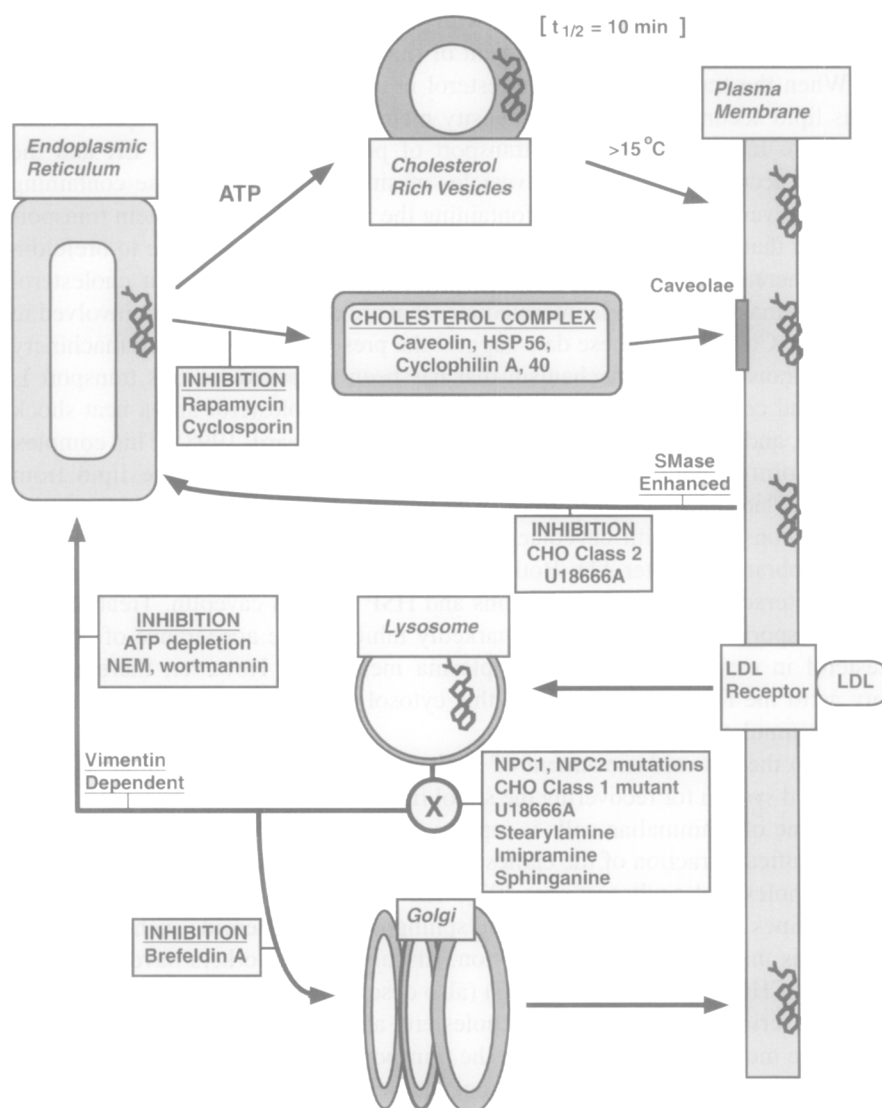


Fig. 15. Interorganelle transport of cholesterol. Newly synthesized cholesterol is transported from the ER to the plasma membrane in an ATP- and temperature-dependent process. One intermediate identified in this transport is a low-density cholesterol-rich fraction believed to be comprised of vesicles. A second proposed intermediate consists of a soluble cholesterol/protein complex. It is not clear if the vesicle fraction and soluble complex are the same. Cholesterol present in the plasma membrane can be induced to move to the ER by SMase treatment of the cell surface. This latter process is inhibited by hydrophobic amines and class 2 mutations in CHO cell lines. Low-density lipoprotein (LDL) derived cholesteryl ester enters the lysosome and is cleaved to form free cholesterol. The lysosomal cholesterol is exported from the lysosomes by a process regulated by NPC1 and NPC2 gene products that is susceptible to inhibition by hydrophobic amines. The cholesterol exported from the lysosome also traverses the Golgi en route to the plasma membrane, or travels directly to the ER via a process that exhibits partial dependence on intermediate filaments and requires ATP.

to 15°C or depleting cellular ATP levels with metabolic poisons. The transport of nascent cholesterol is unaffected by treatment of the cells with cytoskeletal poisons or monensin. When the translocation of cholesterol is inhibited by maintaining the cells at 15°C, this lipid accumulates in a low-density membrane fraction (M. Kaplan, 1985; Y. Lange, 1985). Intermediates in the transport of proteins between the ER and the Golgi apparatus accumulate at 15°C in vesicles of similar density to those containing cholesterol. However, the compartment containing the intermediates in protein transport is different from that containing cholesterol because the former is sensitive to brefeldin A treatment, whereas the latter is not [26]. This result demonstrates that cholesterol travels to the plasma membrane via intermediates that are distinct from those involved in protein transport. Collectively, these data suggest the presence of specialized machinery for cholesterol transport. One mechanism that has been proposed for this transport is non-vesicular and consists of a complex of caveolin with cholesterol and a heat shock protein, HSP 56, and the cyclophilins A and 40 (A. Uittenbogaard, 1998). This complex is believed to form a cytosolic cholesterol carrier that can transport the lipid from the ER to a caveolae-rich fraction of plasma membrane. Pulse-chase experiments with [³H]acetate are consistent with caveolae serving as an entry point for cholesterol at the plasma membrane (Chapter 11). Both cyclosporin A and rapamycin are predicted to disrupt the interactions of the cyclophilins and HSP 56 with caveolin. Treatment of cells with cyclosporin A and rapamycin markedly inhibited the appearance of nascent [³H]cholesterol in caveolae and the total plasma membrane. However, there is still uncertainty as to the relationship between this cytosolic complex and the low-density fraction that accumulates at 15°C.

In addition to the outward movement of cholesterol to the plasma membrane, cells display a retrieval system for recovering the sterol from the plasma membrane. When the plasma membrane of mammalian cells is rapidly depleted of SM by sphingomyelinase treatment, a significant fraction of the cholesterol is transported to the ER and esterified by acyl CoA:cholesterol acyltransferase [27]. The cholesterol retrieval is blocked by hydrophobic amines including U18666A, and sphingosine, and steroids such as progesterone [27] but is insensitive to ATP depletion. Liscum and coworkers have isolated a cell line denoted CHO 3–6 (N. Jacobs, 1997) (also described as a Class 2 mutant) that is defective in recovering plasma membrane cholesterol after sphingomyelinase treatment, that should prove most useful for dissecting the transport mechanism.

Recycling of exogenous cholesterol. Exogenous cholesterol imported into the cell via the low-density lipoprotein (LDL) receptor can be utilized for membrane biogenesis and regulation of sterol metabolism (Chapter 15). The mechanisms whereby lipoprotein-derived cholesterol (generated from cholesteryl esters within lysosomes) is disseminated throughout the cell is being understood with increasing detail (Liscum, 1999; E. Blanchette-Mackie, 2000). The current view indicates that approximately 70% of the lysosomal cholesterol pool is directed to the plasma membrane, whereas 30% is directed to the ER by a separate pathway. One set of inhibitors or mutations appears to affect the export of cholesterol from the lysosomes before the bifurcation of the transport between the plasma membrane and the ER. These early acting conditions include U18666A, imipramine, Niemann–Pick C mutations and CHO class 1 mutants. Subsequent to the bifurcation in the pathway the routing to the plasma membrane is sensitive to brefeldin

A disruption of the Golgi. After the bifurcation the transport to the ER is sensitive to disruption of intermediate filaments, ATP depletion, *N*-ethylmaleimide treatment, and (weakly) wortmannin intoxication.

Important insights into the mechanism of cholesterol transport have come from LDL metabolism in cells from individuals with Niemann–Pick type C (NPC) disease. In NPC fibroblasts, cholesterol transport from the lysosomal compartment to the plasma membrane is markedly retarded compared to that in normal fibroblasts (Liscum, 1999; E. Blanchette-Mackie, 2000). The NPC cells also have impaired regulation of acyl CoA : cholesterol acyl transferase, 3 hydroxy-3-methyl glutaryl CoA reductase and LDL receptor levels, in response to LDL. In addition, free cholesterol accumulates in the lysosomal compartment. In contrast, the transport of newly synthesized cholesterol from the endoplasmic reticulum to the plasma membrane of NPC fibroblasts is essentially identical to that found for normal cells. These findings localize one abnormality of NPC disease to cholesterol export from the lysosomes to other organelles.

Two genes, NPC1 and NPC2 are now known to be responsible for the NPC phenotype. The NPC1 gene product shows significant homology to the morphogen receptor, *Patched*, to a protein of unknown function, NPCL1, and to members of the RND (resistance–nodulation–division) family of prokaryotic permeases. NPC1 has sequence homology to proteins containing sterol sensing domains (3-hydroxy-3-methylglutaryl CoA reductase and SREBP cleavage activating protein, Chapter 15). NPC1 is also closely related to bacterial permeases that transport hydrophobic compounds including acriflavine [28,29]. Consistent with the proposed permease function, NPC1-defective cells cannot efficiently export acriflavine out of their lysosomes. The characteristics of NPC1 protein structure most closely resemble those of fatty acid permeases, and expression of human NPC1 protein in *E. coli* markedly enhances the uptake of oleic acid by the bacteria. Thus a specific biochemical function assignable to NPC1 is modulation of fatty acid transport. The relationship between this defined function and the export of cholesterol remains to be clarified.

The NPC2 gene has also been identified [29]. The gene product is a soluble lysosomal protein that was identified in a global proteomics screen of lysosomal constituents. The protein is found in secreted and intralysosomal forms, with retrieval of the secreted protein mediated by the mannose-6-P receptor. Incubation of NPC2-defective cells with medium containing the secreted form of wild-type NPC2 leads to protein uptake and rectification of the cholesterol accumulation seen in the mutant cells. In addition, individuals with the NPC2 phenotype (which is identical to that for NPC1) show specific mutations in the NPC2 gene, thereby confirming the gene-mutation–disease relationship. The emerging picture is that soluble NPC2 and membrane-bound NPC1 must cooperate in the recognition and translocation of cholesterol out of the lysosome.

Cholesterol import into mitochondria. In steroidogenic tissues cholesteryl esters are hydrolyzed in response to hormonal stimuli, and cholesterol is imported into mitochondria for the synthesis of pregnenolone, the precursor of all steroid hormones. The transit of cholesterol between the outer and inner mitochondrial membrane is regulated by steroidogenic acute regulatory protein (StAR) [30]. StAR is rapidly synthesized in response to hormonal stimuli and targeted to the mitochondria by N-terminal sequences. The StAR protein is imported into the mitochondria and becomes

associated with the matrix and inner membrane. Initially, transit intermediates of StAR were proposed to be the cholesterol carriers between the outer and inner membranes. However, current data demonstrate that import of StAR is not essential and that the association of the C-terminus of StAR with the outer mitochondrial membrane may be the critical interaction required for promoting cholesterol transfer. The mechanism of StAR action is not understood, but interaction of the C-terminus with cholesterol and outer membrane proteins may play a role in assembling a transport complex that moves the lipid to the inner membrane. Individuals with lipoid congenital adrenal hyperplasia lack functional StAR and are unable to make pregnenolone. Analysis of the StAR sequences in these individuals reveals that the mutations accumulate in the carboxy terminal region of the protein. The carboxy terminus of StAR and a structural homolog MLN-64 that binds cholesterol are now recognized to define a large protein family capable of binding hydrophobic molecules (L.M. Iyer, 2001).

4.3.2.6. Phospholipid transfer proteins and membrane biogenesis. Since their discovery in the early 1960s, the PLTPs have been attractive candidates for soluble lipid carriers between membranes *in vivo* (K. Wirtz, 1968). However, there have been two major points of debate about a lipid trafficking role for the PLTPs. The first point has remained that the proteins effect exchange of lipid between model membranes and do not yield a net transfer of mass. Such an action by the proteins *in vivo* might be able to change lipid composition of membranes but would not be able to cause net synthesis and accumulation of new lipid mass. However, to be circumspect one cannot rule out the possibility that either protein modification or protein–protein interactions *in vivo*, may enable the proteins to accomplish net transfer. A second point of concern has been whether the lipid binding and exchange of the proteins simply reflects a lipid binding property that has other functions.

Genetic approaches addressing the role of PLTPs have provided significant insight into their function [31]. The identification of the *SEC14* gene and its corresponding Sec14p protein, as the major PI/PC transfer protein in yeast enabled critical tests of *in vivo* function. The *SEC14* gene was shown to be essential and play an important role in regulating protein traffic through the Golgi. Tests of the required function of Sec14p demonstrate that the ability of the protein to effect transfer of PI is completely dispensable (S. Phillips, 1999). The PC bound form of Sec14p is now thought to be crucial to its function. In the PC bound form, Sec14p is proposed to be a negative regulator of phosphocholine cytidyltransferase (Chapter 8) (H.B. Skinner, 1995). The consequences of this regulation of the cytidyltransferase are thought to be steady state maintenance of a Golgi diacylglycerol pool required for secretory vesicle formation. Other functions of Sec14p have also been proposed including the regulation of polyphosphoinositide pools that are required for operation of the secretory pathway. This latter idea finds support from data in other systems that show a requirement for PI transfer proteins in either exocytosis or signal transduction that appear related to modulation of polyphosphoinositide pools [32].

Another prominent test of *in vivo* lipid transfer protein function has focused upon the nonspecific lipid transfer protein known as SCP2. Early studies suggested that SCP2 played an important role in intracellular cholesterol traffic. However, antibodies

raised to the protein localized the antigen primarily in peroxisomes, rather than the predicted cytosolic compartment. More detailed analysis revealed that SCP2 arose as a proteolytic fragment of the peroxisomal enzyme 3-ketoacyl-CoA thiolase (Chapter 5). These findings were more consistent with a role for SCP2 in peroxisomal β -oxidation than cholesterol traffic. The development of gene-targeted mice with defects in SCP2 expression revealed a lesion in pristanic acid oxidation and in cholesterol side-chain oxidation [33] but not in intracellular lipid traffic. Collectively, these data fail to define any role for SCP2 in cholesterol transport.

The *in vivo* role of PC transfer protein has also been examined by genetic deletion in mice (A. van Helvoort, 1999). Animals with homozygous null alleles for the PC transfer protein had no identifiable phenotype. Examination of tissues with high rates of synthesis and secretion of PC, such as liver and lung, also failed to show any perturbation in lipid traffic or metabolism, indicating that the PC transfer protein cannot be essential for these processes.

Thus far, critical biochemical and genetic studies of lipid transfer proteins do not demonstrate the proteins act as lipid carriers that directly function in the transport reactions for membrane biogenesis. However, the issue of a role for such proteins remains open, in light of the observations that some lipids (e.g., PE) move with surprising rapidity between membranes in reactions that are not demonstrably ATP-dependent. Certainly, other mechanisms are possible for such rapid lipid transport reactions, but additional data are required before a role lipid for transfer proteins can be ruled out.

5. Future directions

Lipid transport is a fundamental process essential to all cell growth, division and differentiation. Our understanding of lipid transport has changed markedly in the last five years, and the pace of change is now increasing. Most notably, the identification of mammalian cell lines, yeast, and bacterial strains with defects in lipid traffic is a major advance. The identification of human diseases with lesions in lipid traffic makes additional important tools, and in some cases cell lines, available. Advances in reconstitution of lipid traffic in permeabilized cells now allow for more precise and critical tests of protein function in the processes. The application of fluorescent probes continues to provide new insights and real time images of selected aspects of lipid transport. As the examination of these processes now begins to enter the realm of the manipulation of mutant cells, genes, and gene products, there remains much to be accomplished. Future studies need continued focus on the development of new genetic tools. For many of the lipid trafficking processes described in this chapter there are still no mutants available, and a concerted effort must be made to develop novel selections and screens that attack the voids in our understanding. The current expansion in genomic information and the ease of manipulating genes in heterologous systems now also allows for approaches in which educated guesses can be used for targeting candidate sequences.

Any candidate sequence that appears in yeast can now be obtained in a hemizygous null strain from commercial sources. Straightforward manipulation allows for the

recovery of strains harboring null alleles that are covered by a plasmid-borne copy of the wild-type gene under inducible or repressible promoters. Such tools allow for rapid critical testing of gene product function in lipid traffic. In addition, phylogenetic jumping among databases can also permit the rapid identification, isolation, and testing of specific gene products in suitable *in vivo* and *in vitro* assay systems. The mechanisms of intracellular lipid traffic in membrane assembly have thus far been difficult to elucidate, but recent advances are grounds for much optimism. The current molecular tools, combined with new genetic strategies are likely to provide both new research opportunities and rewards for those who tackle this long-standing problem of cell biology.

Abbreviations

BODIPY	boron dipyrromethene difluoride
Cer	ceramide
DG	diacylglycerol
diC8-	dioctanoyl-
ER	endoplasmic reticulum
ESR	electron spin resonance
GlcCer	glucosylceramide
GTP	guanosine triphosphate
GTP γ S	guanosine 5'- <i>O</i> -(γ -thiotriphosphate)
IAA	isethionylacetimidate
LDL	low-density lipoprotein
PC	phosphatidylcholine
PE	phosphatidylethanolamine
PG	phosphatidylglycerol
PI	phosphatidylinositol
PS	phosphatidylserine
PSD	phosphatidylserine decarboxylase
MAM	mitochondria-associated membrane
mdr	multidrug resistance
NBD	<i>N</i> -[7-(4-nitrobenzo-2-oxa-1,3-diazole)]-6-aminocaproyl
NPC	Niemann–Pick type C
SM	sphingomyelin
TNBS	trinitrobenzenesulfonate

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Lipoprotein structure

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1. Introduction

Lipoproteins are soluble complexes of proteins (apolipoproteins) and lipids that transport lipids in the circulation of all vertebrates and even insects. Lipoproteins are synthesized in the liver, in the intestines, arise from metabolic changes of precursor lipoproteins, or are assembled at the cell membranes from cellular lipids and exogenous lipoproteins or apolipoproteins. In the circulation, lipoproteins are highly dynamic. They undergo enzymatic reactions of their lipid components, facilitated and spontaneous lipid transfers, transfers of soluble apolipoproteins, and conformational changes of the apolipoproteins in response to the compositional changes. Finally, lipoproteins are taken up and catabolized in the liver, kidney, and peripheral tissues via receptor-mediated and other mechanisms. This chapter deals almost exclusively with the human lipoproteins.

1.1. Main lipoprotein classes

Although the assembly, structure, metabolism, and receptor interactions of lipoproteins are determined by their apolipoprotein components, the most common classifications of lipoproteins are based on their hydrated density or mobility on agarose gel-electrophoresis.

The classification into chylomicrons (CM), very low-density (VLDL), low-density (LDL), and high-density (HDL) lipoproteins is based on their relative contents of protein and lipid that determine the densities of these lipoprotein classes. Chylomicrons have only 1–2% protein while HDL have about 50% protein by weight. The diameters of lipoproteins are inversely correlated with their densities and range from about 6000 Å for CM down to 70 Å for the smallest HDL (Fig. 1).

The general structural organization is similar for all the lipoprotein classes: the apolipoproteins and amphipathic lipids (mostly phospholipids and unesterified cholesterol) form a 20-Å shell on the surface of spherical particles. This shell encloses a core of neutral lipids (triacylglycerols, cholesteryl esters, and small amounts of unesterified cholesterol and other dissolved lipids, e.g., lipid-soluble vitamins). The main protein components are characteristic of each lipoprotein class; they are indicated in Fig. 1, and will be described in detail in Section 3 of this chapter.

The principal functions of the lipoprotein classes are determined by their apolipoprotein (apo) and lipid components. The CM are synthesized in the intestines for the transport of dietary triacylglycerols to various tissues. VLDL are synthesized in the

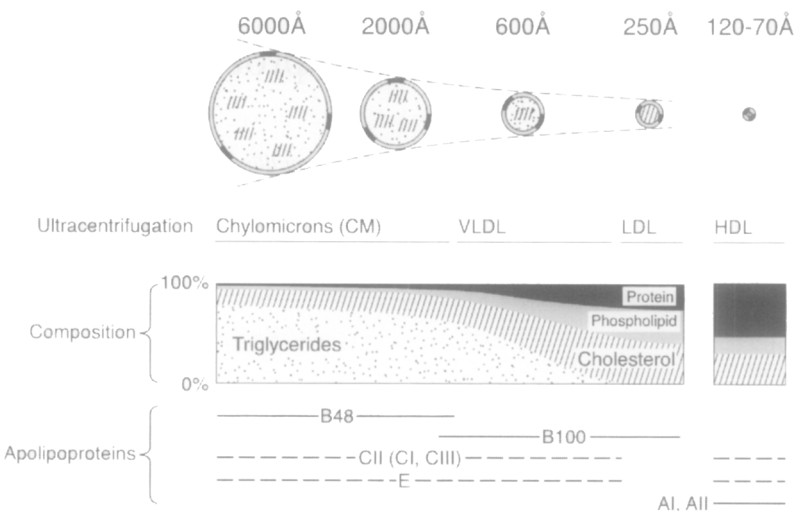


Fig. 1. Major lipoprotein classes (CM, VLDL, LDL, HDL) based on their density. Lipoprotein diameters range from about 6000 Å for CM to 70 Å for HDL. The outer shell (~20 Å) of all lipoproteins consists of apolipoproteins, unesterified cholesterol, and phospholipids; the spherical core contains triacylglycerols and cholesteryl esters. CM and VLDL have the highest contents of triacylglycerols, and 1–10% apolipoproteins by weight; LDL and HDL contain mostly cholesteryl esters in their cores, and 20–50% of apolipoproteins. The major apolipoprotein components of the various classes of lipoproteins are indicated with the solid lines; secondary or minor apolipoprotein components are indicated with the dashed lines. In this figure, 'cholesterol' refers to both esterified and unesterified cholesterol; triglycerides = triacylglycerols.

liver for the export of endogenous triacylglycerols, while LDL arise from the metabolic transformation of VLDL in circulation. The function of LDL is to deliver cholesteryl esters to peripheral tissues and to the liver. HDL are synthesized and assembled in the liver and intestine or are formed from metabolic transformations of other lipoproteins in circulation, and from cellular lipids at the cell membranes (see Chapter 20). HDL remove excess cholesterol from cells and transport it to liver and steroidogenic tissues for metabolism and excretion.

Lipoproteins are also classified by their electrophoretic mobility on agarose gels into α , pre β , and β lipoproteins, corresponding to HDL, VLDL, and LDL density classes respectively; CM, when present, remain at the electrophoretic origin.

Although lipoprotein concentrations in blood plasma are highly variable, depending on age, sex, feeding state, metabolic/hormonal state, and disease state of individuals, a representative lipoprotein distribution for a fasting, healthy, adult male in plasma is approximately 0 mg/dl for CM, 150 mg/dl for VLDL, 410 mg/dl for LDL and 280 mg/dl for HDL [1].

1.2. Lipoprotein subclasses

The lipoproteins within each class are heterogeneous in terms of their density, size, and lipid and apolipoprotein contents and compositions, as well as in their functional

properties. They can be separated into subclasses by ultracentrifugation, gel filtration, electrophoresis, or affinity chromatography methods.

Based on ultracentrifugal and gel-filtration separations, HDL have been subdivided into HDL₁, HDL₂, and HDL₃ subclasses, from the largest and least dense to the smallest and most dense particles. The HDL₁ subclass is enriched in apo E and is least abundant and often disregarded. Non-denaturing gel electrophoresis has further separated the main HDL₂ and HDL₃ subclasses into HDL_{2a}, HDL_{2b}, HDL_{3a}, HDL_{3b}, and HDL_{3c} species spanning a density range from 1.085 to 1.171 g/ml and size (diameter) range from 106 to 76 Å, respectively [2].

Using anti-apolipoprotein immunoaffinity columns two major subclasses of HDL have been separated: one containing apo A1 but no apo A2 (LpA-I) and another containing both apo A1 and apo A2 (LpA-I/A-II). Minor proteins (apo E, apo Cs) may or may not be present in significant amounts in these HDL subclasses [3]. On average, human HDL contain about 70% by weight of apo A1, 20% of apo A2, and 10% of the minor apolipoproteins.

Two-dimensional separations of HDL (agarose gel-electrophoresis in one dimension and non-denaturing polyacrylamide gel-electrophoresis in the second dimension) have yielded α -migrating, pre β -migrating, and γ -migrating HDL subclasses of various sizes and compositions. The pre β -migrating subclasses are present in low concentrations in plasma (in contrast to the abundant α species), and in somewhat higher concentrations in interstitial fluid, but are metabolically very important as they represent the nascent forms of HDL that are especially active in cholesterol uptake from cells (pre β ₁-HDL) and cholesterol esterification by lecithin cholesterol acyltransferase (LCAT) (pre β ₂- and pre β ₃-HDL) [4]. Because of their key role in cholesterol uptake from cells, the pre β ₁-HDL have been studied quite intensively (J.P. Kane, 1985). They contain 2 molecules of apo A1 per particle, and no apo A2 nor other proteins. Their mass ranges from 60 to 80 kDa, with a content of about 90% protein, 7% phospholipids, 0.3% unesterified cholesterol, and 1.8% esterified cholesterol. The pre β ₁-HDL represent only 7% of total apo A1 in plasma and have an apo A1 conformation distinct from apo A1 in α -migrating HDL. Much less is known about the other pre β -HDL subclasses, except that they are discoidal in shape, contain large proportions of phospholipid, and have masses around 300 kDa.

Other lipoprotein classes can also be separated into subclasses of varying density and size by the same separation methods. Subclasses of LDL in the density range from 1.027 to 1.060 g/ml and size range from 270 to 210 Å have been obtained and shown to have different metabolic properties [5]. Small dense LDL, containing high amounts of triacylglycerols, appear to be the most proatherogenic LDL species. Chylomicrons and VLDL undergo continuous density, size, and composition changes due to the hydrolysis of their triacylglycerols by lipoprotein lipase and exchanges of soluble apolipoproteins; therefore, these lipoprotein classes consist of a continuous spectrum of particles.

2. Lipid components

2.1. Lipid composition

The lipid content and composition of the major lipoprotein classes are listed in Table 1 [6], which shows that the total lipid content is inversely correlated with the density of the lipoproteins. Glycerolipids, mainly triacylglycerols, are the major lipid components of CM and VLDL, but constitute less than 11% of the lipids of LDL and HDL, which are enriched in cholesteryl esters (24–51%). Unesterified cholesterol is found in all the lipoprotein classes in relatively low proportions because it is actively esterified by LCAT on HDL and then redistributed to LDL and VLDL by cholesterol ester transfer protein. The total phospholipid content of lipoproteins increases with increasing density and is directly related to the surface area of the lipoprotein particles, as the surface of lipoproteins is covered by a monolayer of phospholipids (PL) and apolipoproteins. By far, the main phospholipid constituents are phosphatidylcholines (PC) (57–81% wt PL) followed by sphingomyelins (12–26% wt PL). Lyso PC and other phospholipids (phosphatidylethanolamine, phosphatidylserine, and phosphatidylinositol) constitute 5–15% wt PL. Glycolipids are present only in trace amounts and free fatty acids contribute less than 3% to the total mass of lipids.

2.2. Fatty acid composition

In the fasted state, the fatty acid composition of lipoprotein lipids reflects their biosynthetic origins and metabolic transformations in circulation (Table 2) [6].

The glycerolipids (predominantly triacylglycerols) have high proportions of 18 : 1 and 16 : 0 fatty acids, reflecting synthesis in the liver. The bulk of cholesteryl esters of human lipoproteins is formed in circulation by the action of LCAT. This enzyme acts on HDL

Table 1
Lipid composition of lipoprotein classes ^a

	CM ^b	VLDL ^c	LDL ^c	HDL ^c
Density (g/ml)	<0.94	0.94–1.006	1.006–1.063	1.063–1.210
Total lipid (% wt)	98–99	90–92	75–80	40–48
Glycerolipids (% wt lipid) ^d	81–89	50–58	7–11	6–7
Cholesteryl esters (% wt lipid)	2–4	15–23	47–51	24–45
Unesterified cholesterol (% wt lipid)	1–3	4–9	10–12	6–8
Phospholipids (% wt lipid) ^c	7–9	19–21	28–30	42–51
PC (% wt PL)	57–80	60–74	64–69	70–81
SM (% wt PL)	12–26	15–23	25–26	12–14
Lyso PC (% wt PL)	4–10	~5	3–4	~3
Other (% wt PL)	6–7	6–10	2–10	5–10

^a Adapted from Skipski [6].

^b Chylomicrons (CM) were isolated during absorption of fat meals from plasma or lymph.

^c VLDL, LDL, and HDL were isolated from fasting plasma or serum.

^d Most glycerolipids are triacylglycerols, only about 4% are diacylglycerols or monoacylglycerols.

^c Phospholipid (PL), phosphatidylcholine (PC), sphingomyelin (SM).

Table 2

Fatty acid composition of glycerolipids, cholesteryl esters and phospholipids present in lipoprotein classes^a

Fatty acid	VLDL	LDL	HDL
Glycerolipids:			
16:0	26.9	23.1	23.4
18:0	2.9	3.3	3.5
18:1	45.4	46.6	43.8
18:2	15.7	15.7	15.9
20:4	2.5	5.4	7.6
Cholesteryl esters:			
16:0	11.5	11.3	11.4
18:0	1.4	1.0	1.1
18:1	25.6	22.4	22.4
18:2	51.8	59.9	54.8
20:4	6.1	6.9	6.3
Phospholipids:			
16:0	33.8	36.0	31.8
18:0	14.7	14.3	14.5
18:1	12.2	11.6	12.3
18:2	20.3	18.9	20.6
20:4	13.6	13.2	15.7

^a Adapted from Skipski [6]. Lipoproteins were isolated from fasting plasma. Composition is in wt% of total fatty acids.

phosphatidylcholines and unesterified cholesterol to form lyso PC and the cholesteryl ester products. In this reaction, LCAT uses preferentially PC species with 18:2 or 18:1 fatty acids in the *sn*-2 position, thus enriching cholesteryl esters in these fatty acids. In contrast, PC containing 18:0 or 20:4 fatty acids is a poor substrate for LCAT, explaining the decreased contents of these fatty acids in the cholesteryl esters. The fatty acid composition of the phospholipids found in lipoproteins is similar to the phospholipid fatty acid composition of cell membranes, especially of hepatic and intestinal cells, including relatively high contents of the essential fatty acids 18:2 and 20:4.

In general, in the fasting state, the fatty acid compositions across the lipoprotein classes for specific types of lipids are fairly similar due to the transfers of lipids among all the lipoproteins by cholesterol ester transfer protein and the phospholipid transfer protein. Postprandially the fatty acid composition of VLDL glycerolipids reflects to some extent the fatty acid composition of dietary fat, but the fatty acid compositions of LDL and HDL are hardly affected. In chylomicrons, the fatty acid compositions do reflect the fatty acid composition of the meal, especially 8–10 h after the meal when CM concentrations in lymph and plasma are maximal.

2.3. Lipid organization

The surface of all lipoproteins consists of a lipid monolayer containing all the phospholipids, and about 2/3 of all unesterified cholesterol, plus the corresponding apolipoproteins.

The dynamic properties of the lipid monolayers depend largely on the nature of the constituent lipids. For example, the surface lipids of LDL are more condensed and rigid than those of HDL due to the presence of more saturated fatty acids in the phospholipids of LDL, a higher sphingomyelin-to-PC ratio, and a higher unesterified cholesterol-to-phospholipid ratio in LDL (M.C. Phillips, 1989). The surface monolayer of VLDL is even more fluid than that of HDL due to differences in the monolayer lipids. Apolipoproteins exert little or no effect on the average dynamic properties of the surface lipids especially in LDL and VLDL. In fact, isolated surface lipids, reconstituted into vesicles or microemulsions, have similar fluidity, diffusion rates, and mobility as the lipid monolayer components in the intact lipoproteins. The core lipids (triacylglycerols and cholesteryl esters) partition poorly into the surface lipid monolayer, accounting for approximately 3 and 1 mol%, respectively, of the surface lipids [7].

Under physiological conditions, the interior of lipoproteins is a fluid spherical droplet of the neutral lipids, including small amounts of dissolved unesterified cholesterol (about 1/3 of the total) and other lipophilic molecules. In HDL and VLDL, the core lipids do not appear to be organized because of the small volume available in HDL, and because of the high content of fluid triacylglycerols in VLDL. However, the core of LDL particles is known to have the cholesteryl esters organized into two interdigitated concentric layers that may undergo cooperative phase transitions at temperatures between 19–32°C. This organization is optimal at a specific cholesteryl ester/triacylglycerol ratio of 7/1 where a separate fluid phase of triacylglycerol is present in the center. At different core lipid ratios there is mixing of the neutral lipids [8]. Phase transitions in the core lipids of LDL apparently result in changes in the secondary structures of apolipoprotein B100 on the surface. This indicates some coupling of hydrophobic regions of the apolipoprotein with adjacent neutral lipids.

3. Apolipoproteins

3.1. Classes and general properties

The apolipoproteins found in plasma are classified into two broad types: the non-exchangeable and the exchangeable (or soluble) apolipoproteins. Apolipoprotein B100 (apo B100) and apo B48, the principal protein components of LDL, VLDL, Lp(a) and CM are non-exchangeable apolipoproteins. They are very large and water-insoluble proteins that are assembled with lipids at their site of synthesis in the endoplasmic reticulum of liver or intestinal cells (Chapter 19). These non-exchangeable apolipoproteins circulate bound to the same lipoprotein particle through various metabolic transformations in plasma, until they are cleared, as lipoproteins, via specific receptors (Chapter 21). In contrast, the exchangeable apolipoproteins (e.g., apo A1, apo A2, apo Cs, apo E) have much smaller molecular masses than apo B100 or apo B48, are more or less soluble in water in their delipidated states, can transfer between lipoprotein particles, and can acquire lipids while in circulation (see Table 3) [9].

The common function of all apolipoproteins is to help solubilize neutral lipids in the circulation. The apolipoproteins bind readily to phospholipid/water interfaces

Table 3
Major human apolipoproteins

Apolipoprotein ^a	Molecular weight ^b	Lipoprotein class ^c	Concentration in plasma (mg/dl)
Apo A1	28,100	HDL , <i>CM</i>	130
Apo A2 ^d	17,400	HDL	40
Apo A4 ^e	44,500	CM	15
Apo(a) ^f	3–8 × 10 ⁵	Lp(a)	0.1–40
Apo B100 ^g	512,000	LDL , VLDL	250
Apo B48 ^g	242,000	CM	
Apo C1	6,600	VLDL , CM , <i>HDL</i>	3
Apo C2	9,000	VLDL , CM , <i>HDL</i>	12
Apo C3	9,000	VLDL , CM , <i>HDL</i>	12
Apo D ^h	22,000	HDL	12
Apo E ⁱ	34,200	VLDL , CM , HDL	7

^a Other, minor apolipoproteins isolated from lipoprotein fractions include apo F, apo H, apo J, apo L, and apo M. They are present in plasma in low concentrations and do not have well-defined functions in lipoprotein metabolism.

^b Polypeptide molecular weights do not include carbohydrate contributions.

^c In bold are the lipoprotein classes containing the highest proportion of the apolipoprotein; in italics are secondary lipoprotein classes.

^d Human apo A2 is a disulfide-linked dimer of two identical monomers of 8.7 kDa; other mammalian apo A2s are monomeric.

^e Apo A4 is a glycoprotein found in lymph CM and is mostly (90%) lipid-free in plasma. Exists in several polymorphic states.

^f Apo(a) is a highly polymorphic protein, containing variable numbers of kringle structural units and glycan chains. Homologous to plasminogen, apo(a) is bound to apo B100 in LDL by a disulfide linkage forming Lp(a).

^g Apo B48 contains 48% of the N-terminal sequence of apo B100. Both are glycoproteins. Apo B48 is present in variable amounts depending on feeding state and CM concentration.

^h Apo D belongs to the lipocalin protein family, it may bind progesterone, but its role in lipoprotein metabolism is unknown.

ⁱ Apo E is glycosylated and consists of three polymorphic forms (apo E2, E3, and E4).

and, under appropriate conditions, can spontaneously form discrete particles with phospholipids. In vivo, the assembly of apolipoproteins with lipids to form lipoproteins may require the assistance of cell proteins such as the microsomal lipid transfer protein or the ABC1 transporter.

Intimately related to their ability to bind phospholipids, and to solubilize neutral lipids within lipoprotein particles, the apolipoproteins have the ability to change conformation to adjust to changing lipid contents, compositions, and metabolic states of the lipoproteins. In apo A1, the adaptable structural regions have been called the mobile or hinge domains. Analogous domains likely exist in other apolipoproteins, as epitope recognition by specific antibodies and exposure to proteolytic digestion change with changing lipid contents and compositions of various lipoproteins.

Several of the exchangeable apolipoproteins in their respective lipoproteins are known to activate or inhibit plasma enzymes. Apo A1, and to a lesser extent apo E, apo A4, and apo C1, activate LCAT, apo C2 activates lipoprotein lipase, and apo C3 and apo A2 may act as inhibitors of hepatic lipase.

Apolipoproteins also have roles in receptor recognition. The best characterized interaction is the recognition of the LDL receptor by apo B100- and apo E-containing lipoproteins. Recently, a receptor for HDL (SR-BI) has been described which binds various apolipoproteins, including apo A1 and apo A2.

In addition to the well-known functions of apolipoproteins in lipid binding and solubilization, modulation of enzymatic activities, and receptor recognition, other functions have been described for apolipoproteins. For example, apo E has been implicated in in situ nerve repair and regeneration, as well as in plaque formation in Alzheimer's disease. Apo(a) may have a function in the clotting process, while apo A4, produced in the intestine and the hypothalamus, may have a role in signaling satiety in the fed state.

Evidently, the multiple functions of apolipoproteins are determined by their unique, modular structures encoded by families of genes.

3.2. Gene organization

The gene structures of the major exchangeable apolipoproteins are very similar to each other, whereas the apo B gene is distinct, as are the gene structures of apo(a) and apo D. Most of the genes of the exchangeable apolipoproteins contain four exons and three introns, with similar locations of intron/exon boundaries, and similar intron and exon lengths for the first three exons (see Fig. 2) [10]. The differences in the total length of the mRNA are due to the differences in the length of the 4th exon. The mRNAs encompass a 5'-untranslated region, a region encoding the signal sequence, a short pro-segment, the mature sequence of the protein, and a 3'-untranslated region. Exons 3 and 4 of the genes encode the entire mature sequence. For the apo A4 gene, the only difference from the other exchangeable apolipoprotein genes is the absence of the first exon and intron in the 5'-untranslated region. The homologies in the gene and protein sequences of the exchangeable apolipoproteins indicate that these genes evolved by gene duplication from a primordial gene resembling the gene of apo C1.

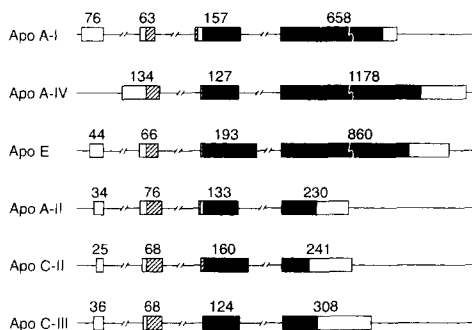


Fig. 2. Organization of the genes of the exchangeable apolipoproteins [10]. The boxes represent the exons joined by introns (broken lines). The open portions of the boxes correspond to the 5'- and 3'-untranslated regions, the hatched portions are the signal peptide regions, and the filled parts represent the regions that code the mature protein sequences. The narrow open portions in exon 3 of the apo A1 and apo A2 genes represent their pro-segments. Numbers above the exons indicate the number of nucleotides in each exon.

In contrast to the genes of the exchangeable apolipoproteins, the apo B gene is short with respect to the length of its 29 exons. It is also quite asymmetric: 19 introns are concentrated within the first 1000 codons, and two very long exons (exons 26 and 29) occur in the 3'-third of the gene sequence. No homology is found between the apo B gene and other known genes [9]. It encodes one of the longest known polypeptide chains containing 4536 amino acid residues. Apo B48 is encoded by the same gene as apo B100, but the mRNA is edited by a specific enzyme in human intestine. The enzyme changes codon 2153 from a Gln to a stop codon, resulting in the production of apo B48, a protein of 242 kDa, lacking the LDL receptor-binding region of apo B100 (D. Driscoll, 1990).

3.3. Primary sequences

The exchangeable apolipoproteins, in addition to having similar genes, have similar primary amino acid sequences. Their sequences contain 11 and 22 amino acid homologous repeats, the latter consisting of two 11-mers, as shown in Fig. 3 [11]. The last 33 amino acids encoded by exon 3 can be aligned into three 11-mers, while the sequences encoded by exon 4 fit, in general, into 22-mer segments that often start with

A	A											B											C												
	1	2	3	4	5	6	7	8	9	10	11	1	2	3	4	5	6	7	8	9	10	11	1	2	3	4	5	6	7	8	9	10	11		
A-I	11	V	K	D	L	A	T	V	V	D	V	L	K	D	S	G	R	D	Y	V	S	Q	F	E	G	S	A	L	V	K	Q	L	N	43	
E	29	A	L	G	R	F	W	D	Y	L	V	W	V	T	T	Y	G	K	D	L	M	E	K	V	L	S	S	Q	V	L	Q	E	L	R	61
A-II	7	V	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	S	V	Q	K	S	P	E	V	L	Q	A	K	39
C-III	8	L	L	S	S	L	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	S	V	Q	K	S	P	E	V	L	Q	A	K	40
C-II	18	V	K	E	S	L	S	S	Y	W	E	S	L	E	K	T	A	R	E	L	I	S	R	I	K	Q	S	E	L	S	A	K	M	R	39
C-I	7	A	L	D	K	L	K	E	F	G	N	T	E	D	K	A	R	E	L	I	S	R	I	K	Q	S	E	L	S	A	K	M	R	39	

B	A											B																						
	1	2	3	4	5	6	7	8	9	10	11	1	2	3	4	5	6	7	8	9	10	11												
A-I	44	K	L	V	L	D	E	N	W	D	S	T	E	S	K	G	L	R	Q	E	K	L	65											
	66	P	V	T	Q	F	W	D	N	L	T	E	D	K	E	L	V	R	K	Q	E	K	87											
	88	L	L	S	S	L	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	98											
	99	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	120											
	121	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	142											
	143	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	164											
	165	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	186											
	187	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	208											
	209	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	219											
	220	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	241											
E	62	A	V	L	M	E	S	E	T	R	K	S	A	Y	K	E	E	L	A	Q	C	G	83											
	84	P	V	A	E	E	T	R	K	S	A	Y	K	E	E	L	A	Q	C	G	R	L	105											
	106	Q	Y	R	G	K	L	V	Q	R	K	G	L	E	A	Q	R	D	S	D	A	L	116											
	117	Q	Y	R	G	K	L	V	Q	R	K	G	L	E	A	Q	R	D	S	D	A	L	138											
	139	S	H	L	Q	E	E	T	R	K	S	A	Y	K	E	E	L	A	Q	C	G	R	160											
	161	V	Y	L	Q	E	E	T	R	K	S	A	Y	K	E	E	L	A	Q	C	G	R	182											
	183	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	201											
	202	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	217											
	218	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	237											
	241	A	K	L	V	E	E	D	Q	Q	Q	W	L	S	N	F	V	L	T	G	50													
	267	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	289											
A-II	40	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	50											
C-III	41	P	L	S	F	M	Q	S	G	Y	M	K	H	A	K	T	A	K	D	A	L	E	71											
C-II	51	D	L	Y	W	S	T	V	K	D	F	S	T	Y	T	W	G	D	I	F	Q	V	51											
C-I	51	D	L	Y	W	S	T	V	K	D	F	S	T	Y	T	W	G	D	I	F	Q	V	72											
	40	E	W	S	E	T	Q	K	V	K													50											

Fig. 3. Amino acid sequences of human apolipoproteins arranged in 11-mer and 22-mer repeats [11]. (A) The last 33 amino acids encoded by exon 3 are divided into three (A, B, C) 11-mer repeats. (B) The sequences encoded by exon 4 contain 22-mer repeats each consisting of two 11-mers (A and B), and some 11-mer repeats. If there are 10 or more amino acids in the same column with equivalent physical properties (i.e., hydrophobic, basic, acidic), they are boxed. Numbers at the beginning and end of a sequence give the amino acid locations in the mature apolipoprotein.

a Pro (J.I. Gordon, 1986). The significance of these repeated sequences is that they are predicted to form amphipathic α -helices. These helices in apolipoproteins and synthetic peptides have been shown experimentally to bind to phospholipid surfaces and to be effective in solubilizing lipids. Thus, the 22 amino acid repeats in α -helical organization are the lipid-binding units of the exchangeable apolipoproteins. Furthermore, specific repeats in the sequences encoded by exon 4 have other functional roles such as LCAT activation by the 143–165 repeat of apo A1, LDL receptor-binding by the 136–150 region of apo E, and lipoprotein lipase activation by residues 44–79 of apo C2.

In contrast, the apo B100 sequence contains many internally repeated sequences that bear little resemblance to the sequences of the exchangeable apolipoproteins [12]. There are unique, Pro-rich repeats of 25 and 52 residues that contain high proportions of hydrophobic amino acids. These regions could be in contact with the core lipids in LDL and VLDL. Other features of the apo B100 sequence include (1) 25 cysteine residues (16 in disulfide linkages), (2) at least 4 heparin-binding regions, and (3) 19 potential glycosylation sites (16 of which are occupied by glycan chains, contributing about 9% of the molecular mass of apo B100). Most of the disulfide linkages and free Cys residues cluster in the amino-terminal region of the apolipoprotein. The receptor-binding domain of apo B100, between residues 3353 and 3371, does have some homology with the corresponding receptor-binding region of apo E.

Apo(a) is a distinct, highly polymorphic glycoprotein that is covalently linked to apo B100 of LDL by a disulfide linkage, to form Lp(a). Apo(a) is homologous to plasminogen, containing variable kinds and numbers of kringle sequences. Each kringle repeat contains about 80 amino acids and three internal disulfide bridges. While apo(a) has sequences homologous to those forming the plasminogen catalytic, serine protease site, the peptide sequence corresponding to the cleavage site required for activation of plasminogen is modified in apo(a), so that the serine protease activity is not expressed. In fact, the function of apo(a) in plasma is not known in the 30% of the population that expresses this apolipoprotein in significant levels (A.M. Scanu, 1988).

3.4. Secondary structures

The exchangeable apolipoproteins, devoid of lipids, have substantial amounts of α -helical structure in physiological aqueous solutions and their α -helix content increases markedly upon lipid binding. For example, apo A1 is about 50% helical in the lipid-free state, and becomes 60–85% helical when bound to lipids, as measured by circular dichroism spectroscopy.

Various computer algorithms have predicted the existence of amphipathic α -helix segments, coinciding quite well with the 22-mer repeats of the exchangeable apolipoproteins. The predicted helical structures have a non-polar face, and a larger polar face. Depending on the relative distribution of the charged residues on the polar face, the amphipathic helices of the exchangeable apolipoproteins fall into three classes: A, G*, and Y helices [13].

In the A-type helices (see Fig. 4) the basic residues are found at the boundary between the polar and non-polar helix faces. The hydrophobic chains of the basic amino acids, in fact, contribute to the hydrophobicity of the non-polar face of the helix. The

(2), an amphipathic β -strand region (residues 827–1961); (3) a region with α -helices resembling A- and Y-type amphipathic helices and including other kinds of helices (residues 2103–2560); (4) a second β -strand region (residues 2611–3867); and (5) a C-terminal, third region of amphipathic α -helices (residues 4061–4338) [14]. The N-terminal region of apo B100, containing the G-helices and high contents of Cys, consists of one or more globular domains (residues 22–303 and 512–721) that are involved in interactions with the microsomal triglyceride transfer protein, a protein required for the assembly of nascent apo B100 with lipids. The central and C-terminal regions of α -helices may be involved in reversible lipid binding by apo B100, and the β -strand regions may participate in irreversible interactions with lipids, possibly involving core lipids.

3.5. Three-dimensional structures in solution

While most of the known functions of apolipoproteins are associated with their lipid-bound states, lipid-free or lipid-poor exchangeable apolipoproteins do exist in plasma and interstitial fluid, and have important metabolic roles in lipid uptake from cells, transfers between lipoproteins, structural remodeling of lipoproteins, and apolipoprotein catabolism.

Fragments of two apolipoproteins, apo E (22 kDa N-terminal fragment) and apo A1 (C-terminal fragment missing 43 N-terminal amino acid residues) have been crystallized and their structures determined by X-ray methods (Fig. 5). The apo E structure [15] (Fig. 5A) consists of an elongated (65 Å) four-helix bundle and a connecting short helix. The structure is stabilized by hydrophobic interactions, salt-bridges, and leucine-zipper

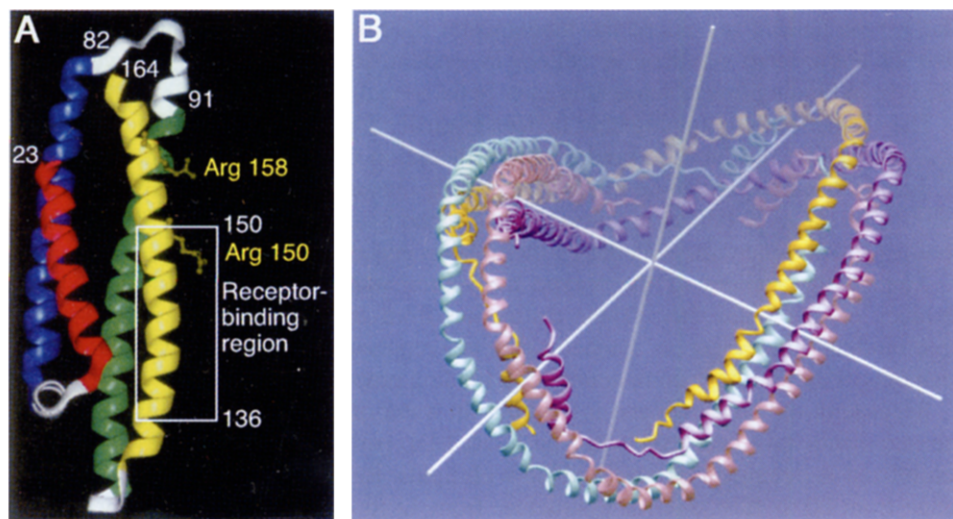


Fig. 5. (A) Ribbon X-ray structure of the apo E 22 kDa N-terminal fragment, showing the elongated 4-helix bundle of the lipid-free protein monomer [15]. The LDL receptor-binding region is boxed. (B) X-ray structure of a tetramer of the apo A-I fragment lacking the N-terminal 43 amino acids [16].

interactions. The receptor-binding region contains a cluster of basic residues on the surface of one long helix. This structure confirms the importance of amphipathic α -helices as a fundamental structural motif of apolipoproteins, even in the lipid-free state. Indeed, the four main helices contain 19, 28, 36, and 35 amino acids and encompass several of the predicted 11-mer and 22-mer repeats. The first two helices and their connecting short helix consist of the three 11-mer repeats encoded by exon 3 plus the first 22-mer repeat of exon 4, while the last two helices each consist of two 22-mer repeats. However, as important as this partial structure of apo E is for our understanding of apolipoprotein folding in solution, it does not represent accurately the configuration of the receptor-binding region because the lipid-free form of this apo E fragment binds to the LDL receptor with very poor affinity. Only in the lipidated form does apo E adopt the correct structure for high-affinity binding to the receptor.

It is hypothesized that when apo E binds to lipid surfaces the four-helix bundle opens at a hinge region to expose the hydrophobic faces of the helices to the lipid. In addition, the C-terminal 10-kDa fragment, missing from the crystal structure, is a major lipid-binding domain of apo E, but its contribution to the intact structure of apo E is not known.

A fragment of apo A1 lacking 43 amino acids from the N-terminus has been crystallized into 2 or 3 distinct conformations. The best studied one is a tetramer where each monomer polypeptide chain forms continuous amphipathic α -helices that curve into a horseshoe with a diameter of about 100 Å [16]. The monomers are organized into dimer pairs of antiparallel overlapping sequences giving a closed, distorted circular structure (Fig. 5B). This structure again illustrates the importance of amphipathic α -helices, shows that they can form long continuous structures, and suggests models for interaction with lipids; however, it does not agree with known hydrodynamic and spectroscopic properties of lipid-free apo A1 monomers under physiologic conditions. The best current information for intact monomeric apo A1 in solution indicates that it has an elongated and rather rigid structure (125×25 Å) about two helices thick. The N-terminal Trp residues cluster near each other, and the compact and structured N- and C-terminal regions of the protein are within 35 Å of each other (A. Jonas, 2002). In fact, one of the crystallized forms of the apo A1 fragment suggests a monomeric protein in a hair-pin fold of extended helices, that is quite compatible with the proposed fold of the intact monomeric apo A1.

The three-dimensional structures of two insect apolipoproteins, apolipophorin-III from locust and Sphinx moth, have been solved by X-ray and NMR methods (M.A. Wells, 1991; R.O. Ryan, 1997). Both of these proteins bind reversibly to insect lipoprotein (lipophorin) surfaces and undergo dramatic structural changes in the process. The structures of the insect apolipophorins, similar to that of the apo E fragment, consist of elongated helix bundles of 5 amphipathic helices (Fig. 6).

It is important to note that exchangeable apolipoproteins have an extraordinary capacity for structural adaptation in response to solution conditions or to different lipid environments. In solution, high salt conditions, pH changes, or inclusion of organic solvents can lead to distinct structural states, with abnormally high α -helix contents, and specific degrees of oligomerization, as seen for the crystallized tetramer of apo A1. Under physiologic conditions, the transformation from a lipid-free to a lipid-bound state often involves a marked increase in α -helix content, and transfer of hydrophobic amino

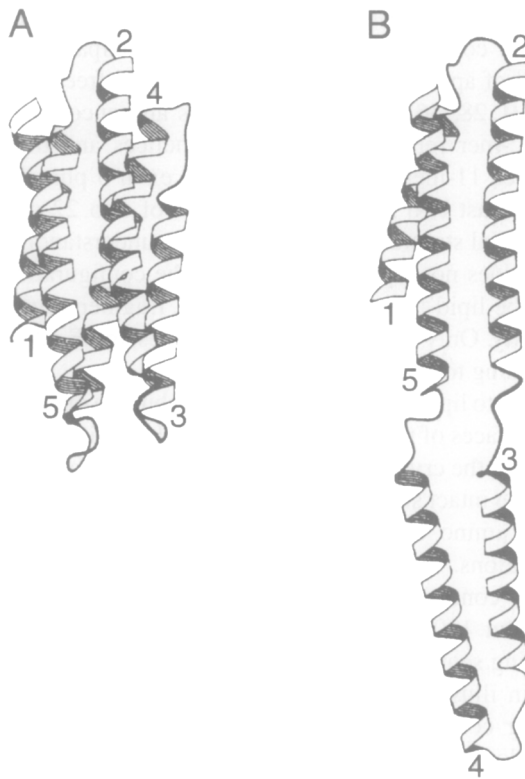


Fig. 6. Ribbon diagrams of the structure of moth apolipophorin III: (A) the helix-bundle configuration determined by X-ray analysis of the lipid-free protein; (B) proposed 'open' configuration of the apolipophorin as it binds to the surface of the insect lipoproteins. The helices are numbered 1 through 5, starting at the N-terminus (D.R. Breiter, 1991).

acid residues from a buried protein environment to a lipid environment. In lipid-bound states, changes in the content and composition of the lipid — surface phospholipids, cholesterol, or core lipids — can induce structural changes in the apolipoproteins. Also, binding of other apolipoproteins to the lipoprotein surface can affect the conformation of the resident apolipoproteins. Thus, there are potentially many distinct conformational states accessible to each exchangeable apolipoprotein. This is likely the case also for lipid-bound apo B100 and apo B48.

4. Complexes of apolipoproteins with lipids

4.1. Binding of apolipoproteins to phospholipid surfaces

Interactions of apolipoproteins with phospholipids are essential for the assembly of lipoproteins, stabilization of lipoprotein structures, and expression and modulation of

apolipoprotein functions. The main experimental approaches for the study of apolipoprotein interactions with phospholipids have used isolated, exchangeable apolipoproteins in conjunction with aggregated lipids dispersed in water or spread at the air/water interface. The aggregated lipid states include lipid monolayers, various types of liposomes (small unilamellar vesicles, large unilamellar vesicles, multilamellar vesicles), and emulsions. All these lipid systems consist of or include phospholipids, especially phosphatidylcholines.

Apolipoproteins bind or adsorb readily to the phospholipid surfaces [17]. For fluid egg PC surfaces, in vesicle or emulsion form, the binding affinities (K_{ds}) for exchangeable apolipoproteins range from 0.2 to 9 μM , depending on the apolipoprotein, presence or absence of cholesterol in the surface, and curvature of the surface. Apo A4 has the lowest affinity for egg PC surfaces, in agreement with the observation that most of the apo A4 in plasma is lipid-free. The other exchangeable apolipoproteins have affinities comparable to one another, and similar calculated stoichiometries of amino acids bound per PC, averaging 0.70 ± 0.35 . The percent of phospholipid surface occupied by the apolipoproteins, at saturation, is in the range from 8 to 12%. In the absence of cholesterol, this corresponds to the maximal compression of the surface PC, required to accommodate the amphipathic helices of the apolipoproteins. Cholesterol added to the surface initially increases the available surface area for apolipoprotein binding, but then decreases it at higher cholesterol contents (e.g., 30 mol%) [17].

The rates of interaction with phospholipid surfaces are highest for lower molecular weight apolipoproteins or peptide analogs that are monomeric and unfolded [18]. However, upon binding to phospholipid surfaces most apolipoproteins and peptides become highly α -helical. In fact, helix formation largely accounts for the negative enthalpy of binding. At the phospholipid surface, the amphipathic helices lie with their axes parallel to the surface. The polar and charged residues of the α -helices are at the level of the polar head groups of the phospholipids, but do not appear to interact electrostatically with them. The prevalence and importance of intra or intermolecular helix-to-helix interactions is not yet elucidated.

Desorption of exchangeable apolipoproteins from phospholipid surfaces occurs in two steps: unfolding and loss of α -helical structure, followed by a slow desorption step whose rate depends on the length of the apolipoprotein and its affinity constant for the surface. Generally, the smaller molecular weight apolipoproteins desorb more rapidly and exchange between phospholipid surfaces more readily than larger apolipoproteins. One exception is apo A2, which has a higher affinity for PC than apo A1, and can displace apo A1 from model PC surfaces and from native lipoproteins.

As already noted, delipidated apo B100 and apo B48 are not soluble in water; therefore, their investigation has been limited to the proteins solubilized with detergent or reconstituted with selected lipids.

The properties of lipoprotein-like particles reconstituted with defined lipids will be described in the next section.

4.2. Lipoprotein-like complexes

Although phospholipid liposomes are favored systems for the study of apolipoprotein binding to phospholipid surfaces, vesicle–apolipoprotein complexes are not ideal models for lipoproteins. Vesicles have an interior water compartment not present in lipoproteins, are incapable of solubilizing large amounts of neutral lipids within the phospholipid bilayer, and are too large to mimic the surface curvature of HDL. Thus, methods have been developed to prepare small, micellar complexes of exchangeable apolipoproteins (in particular apo A1) with lipids that mimic discoidal and spherical HDL in shape, composition, and functional properties. For LDL and VLDL, microemulsions and emulsions of lipids of selected diameter and composition, with added apo B100, make good models of the native lipoproteins.

Several methods are known for the reconstitution of HDL-like complexes from pure components: (1) spontaneous formation of HDL discs from dimyristoylphosphatidylcholine liposomes; (2) detergent mediated reconstitution of HDL discs with various phospholipids; and (3) co-sonication of apolipoproteins and lipids to form either discoidal or spherical HDL analogs [19].

Dimyristoyl-PC liposomes can bind apolipoproteins reversibly, as described in the preceding section; however, at the transition temperature (T_m) of the lipid (24°C) and at sufficiently high proportions of apolipoprotein to dimyristoyl-PC (1/3 or greater, wt/wt), the apolipoproteins can lyse the liposomes to give rise to small discs analogous to nascent HDL. The rate of the liposome disruption and solubilization depends on the temperature of the reaction. It is highest at the onset of the main phase transition of dimyristoyl-PC and decreases a thousand-fold on either side of T_m . While the same reaction does occur with dipalmitoyl-PC liposomes at the T_m of 41°C, the rate is much slower than for dimyristoyl-PC. For long-chain, unsaturated PC, such as palmitoyl-oleoyl-PC, the rates of reaction are too slow to be measured at accessible temperatures. Therefore, for practical purposes, under ordinary conditions, only dimyristoyl-PC can be used effectively to reconstitute HDL discs by this method. In this system, smaller apolipoproteins and peptide analogs disrupt dimyristoyl-PC liposomes at higher rates than larger apolipoproteins. Thus under some circumstances synthetic peptide mimics of A-type amphipathic helices may lyse even egg PC or palmitoyl-oleoyl-PC vesicles.

The mechanism of the disruption of dimyristoyl-PC liposomes into discoidal reconstituted HDL particles requires binding of the apolipoprotein to the liposome surface, followed by protein–protein interactions that lead to penetration of the bilayer by apolipoprotein, and breakdown of the liposome into bilayer discs. The discs are surrounded and stabilized on the periphery by the amphipathic helices of the apolipoproteins [20].

A more universal method for producing discoidal reconstituted HDL particles uses detergent (usually Na cholate) to solubilize the phospholipid into mixed micelles, followed by addition of exchangeable apolipoprotein and removal of detergent by dialysis or column chromatography. Depending on the proportions of PC to apolipoprotein, in the range from 1/1 to 4/1 (wt/wt), reconstituted HDL discs of different diameters containing different numbers of apolipoprotein molecules with varied conformations

can be produced. Cholesterol (up to 15 mol%) or small amounts of other lipids can be readily incorporated during the lipid solubilization step.

The third method for making HDL-like particles is extensive co-sonication of apolipoproteins with phospholipids, in the absence or presence of neutral lipids. Usually, the components are mixed in the proportions found in native HDL and yield discoidal or spherical analogs of HDL depending on the absence or presence of neutral lipids. While the ability to synthesize spherical reconstituted HDL particles by this method is very attractive, controlling particle homogeneity and yield is difficult.

To produce models of LDL, VLDL, and CM, lipid mixtures of PC and a neutral lipid are first sonicated to give metastable emulsion particles of the general size of the desired lipoproteins, and then the apo B component is added from a detergent dispersion, or exchangeable apolipoproteins are added in solution [17].

The advantages of reconstituted lipoproteins over the native lipoproteins are that the model particles can be made with a single apolipoprotein and one or a few defined lipid components to study each component individually. In addition, particles of uniform size can be isolated where high-resolution structural analysis is potentially possible. Also the reconstituted lipoproteins lend themselves for the study of structure–function relationships. In fact, reconstituted lipoproteins display all the known functions ascribed to native lipoproteins, including enzyme activation, receptor binding, and uptake and transfer of lipids.

4.3. Reconstituted HDL

The best studied reconstituted HDL are the discoidal particles containing apo A1 and a single type of PC, with or without added small amounts (<20 mol%) of other phospholipids or cholesterol [20]. The discoidal shape of the particles has been confirmed by negative-stain electron microscopy, small-angle X-ray scattering, and atomic force microscopy methods. All methods indicate a disc thickness of 45–55 Å, that corresponds to the phospholipid bilayer thickness, and diameters ranging from 70 to 180 Å. The main phase transition of the PC is preserved in the particles, but T_m is shifted about 3°C to higher temperatures, indicating a greater ordering and restriction of the PC in the particles than in corresponding liposomes.

Analyzed by non-denaturing gradient gel electrophoresis the particles display discrete diameters and size distributions that vary depending on the initial PC/apo A1 ratios of the preparations. Isolated particles of a specific size have reproducible, characteristic physical and functional properties. For example, some particles containing palmitoyl-oleoyl-PC (78 and 109 Å) are poor substrates for LCAT while others, particularly the 96 Å particles, are the best known LCAT substrates (A. Jonas, 1989). In addition, there is evidence of different binding affinities of the particles for SR-BI receptors (D.R. van der Westhuyzen, 2001). Thus reconstituted HDL subclasses, with different lipid contents, have different apolipoprotein conformations that result in dramatically altered functional properties. In vivo, this conformational adaptability of apolipoproteins probably leads to metabolic switching for the diverse functions of HDL subclasses. Apo A1 conformation is also regulated by the saturation or unsaturation of the PC acyl chains and by high contents of cholesterol (>15 mol %) or polyun-

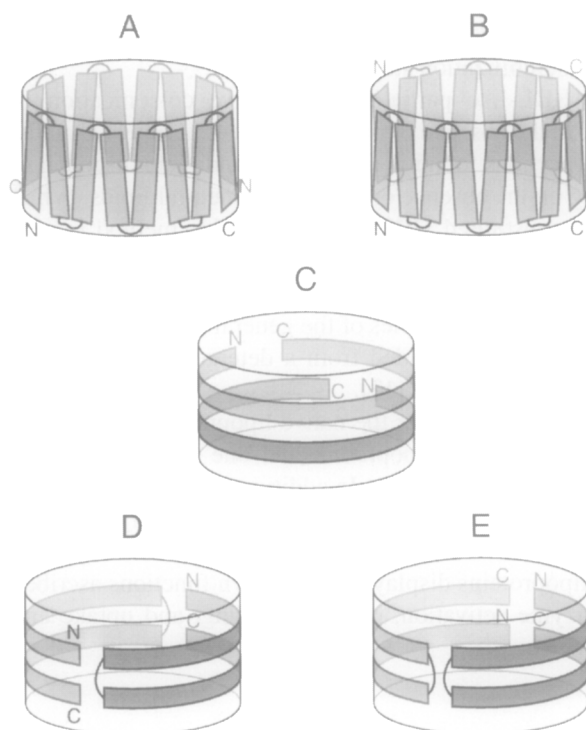


Fig. 7. Diagrams of the proposed arrangement of two apo A1 molecules (residues 44–243) on the periphery of a discoidal reconstituted HDL particle (M.A. Tricerri, 2001). N, denotes the N-terminus and C, the C-terminus of each protein molecule. (A) 'Picket-fence' model in head-to-tail arrangement. (B) 'Picket-fence' model in a head-to-head arrangement. (C) 'Belt' model. (D) 'Hair-pin' fold in a head-to-tail arrangement. (E) 'Hair-pin' fold in a head-to-head arrangement.

saturated PC. Other exchangeable apolipoproteins (apo A2, apo E, apo A4) also form discoidal reconstituted HDL with diameters that roughly correspond to their contents of amphipathic α -helices and molecular weights.

Reconstituted spherical HDL can be made by co-sonication of selected HDL components (e.g., apo A1, PC, cholesteryl ester) or by extensively reacting discoidal reconstituted HDL with LCAT in the presence of an exogenous source of cholesterol. The products are spheroidal and contain 2, 3, or 4 apo A1 molecules per particle, PC, cholesterol, and a cholesteryl ester core. The diameters of the particles range from 80 to 120 Å. Although not well studied, the conformation of apo A1 appears distinct from that in the discoidal particles and variable depending on the particle diameter (A. Jonas, 1990).

In reconstituted discoidal HDL, the apolipoproteins form a protective shell, one helix thick, around the periphery of the discs. According to current experimental evidence and models [21] (Segrest, 1999; Jonas, 2001), the helix axes are predominantly perpendicular to the acyl chains of the PC bilayer (see Fig. 7C,D). Earlier models, however, oriented the helical repeats almost parallel to the acyl chains, with the

antiparallel helical segments joined by tight β -turns (Fig. 7A,B) (Rosseneu, 1990; Schulten, 1997). The spatial organization of the amphipathic helices in spherical reconstituted HDL is not known and has not been modeled.

4.4. Structures of native lipoproteins

Electron microscopic images of lipoproteins show predominantly spherical shapes. Only nascent HDL appear as stacks of discs by negative-stain transmission electron microscopy. The stacks are artifacts of the method, since in solution nascent HDL and their reconstituted HDL analogs are free-standing discs. Regarding the structure of native HDL, their electron microscopic images do not have enough resolution to show any significant surface features, and other methods have not yet been successful in providing dependable structural information. The computer models generated for the reconstituted HDL discs are probably applicable for the larger nascent HDL particles, but no credible computer models exist for the spherical HDL. Based on the dimensions of HDL₃ particles, and the molecular volumes and surface areas of its component phospholipids, cholesterol, cholesteryl esters and amino acids, a space-filling model was constructed by Edelstein et al. in 1979 [22]. Their model is still valid today, with the exception that more apolipoprotein helix contacts are probably present, and the space under the helices is probably filled with cholesteryl ester (or triacylglycerol) molecules rather than with unesterified cholesterol, as shown in Fig. 8.

The N-terminal domain of apo B may have structural homology with lipovitellin. Lipovitellin is a soluble oocyte lipoprotein that contains 16% by weight of lipid and has sequence homology with the N-terminal 670 residues of apo B and with the microsomal lipid transfer protein. The X-ray structure [23] of lipovitellin contains a funnel-shaped cavity formed by two β -sheets lined with hydrophobic residues that are presumed to bind

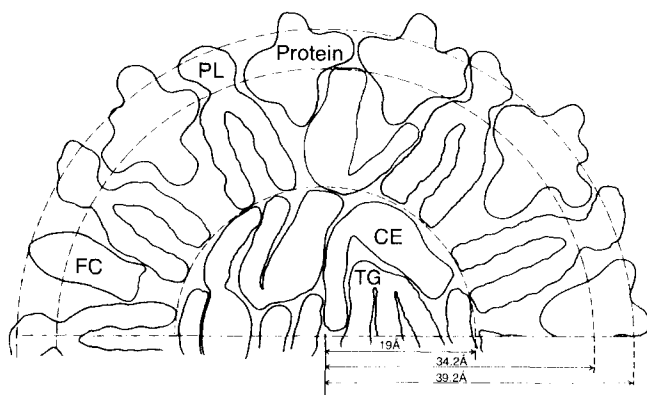


Fig. 8. Structural model of human HDL₃ based on the composition and dimensions of the particles, and the molecular volumes and surface areas of the individual components. PL, phospholipid; FC, unesterified cholesterol; CE, cholesteryl ester; TG, triacylglycerol. Modified form of the model presented in Edelstein et al. [22] showing a cholesteryl ester molecule in the space under the apolipoprotein helices, and unesterified cholesterol (FC) in the surface monolayer.

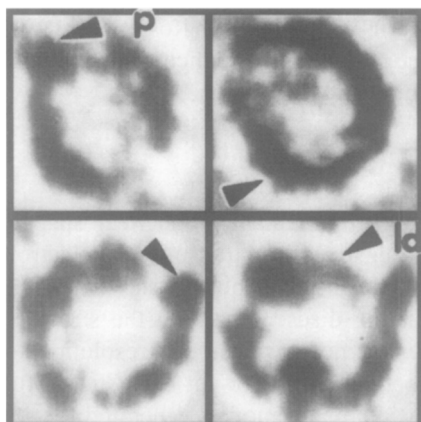


Fig. 9. Individual images of LDL obtained by cryoelectron microscopy [22]. The box sizes are 270×270 Å. The unlabeled arrow indicates a region of high density that corresponds to a domain of apo B100; arrow with a 'p' indicates the pointed, N-terminal domain of apo B100; and an arrow with 'ld' points to a region of low density in apo B100.

phospholipids. A similar structure may be present in apo B and in the microsomal lipid transfer protein. Higher-resolution cryoelectron microscopy and negative-stain electron microscopy combined with monoclonal antibody tagging have revealed some structural features of apo B100 on LDL particles of relatively homogeneous sizes [24]. Images of LDL appear slightly ovoid and with average diameters of 233 Å. Individual LDL images show four or five high-density regions that correspond to the single molecule of apo B100. The dense regions are joined by lower-density connections. Although the entire protein appears in many images as a ring on the surface of the LDL (Fig. 9), the actual distribution of the protein domains is probably more dispersed and complex. The N-terminal domain of apo B100 can be identified as a high-density pointed end that protrudes more from the particle surface than any of the other domains. Some of the putative domains have approximately circular projections but the largest one, the one corresponding to residues 1700–3070, is elongated and is located at 50 – 140° latitude from the N-terminus.

5. Future directions

In spite of the great advances made during the last two decades in the elucidation of the structures and structure–function relationships of lipoproteins, much remains to be accomplished in this field of research.

Key areas of future research:

- (1) Structural and functional studies of the roles of minor lipid and apolipoprotein components of lipoproteins, including the products of oxidative reactions that occur in vivo and in vitro.
- (2) Investigation of the conformational adaptability of apolipoproteins during assembly